

RRB MATHS MINIMUM FORMULAS

Sawaal dekho $\rightarrow$ value uthao $\rightarrow$ formula mein daalo $\rightarrow$ answer

Sirf 53 Master Formulas (pehle 185)

28 Chapters • har formula ke saath **VALUE DAALO** table

Dono books ke 8,463 sawaal (English 5,213 + Hindi 3,250) — har ek kisi formula ki ek row se ✓

1,235 solved examples + 159 random “unseen” test sawaal — book ki answer key se computer-verified ✓

Books: *RRB Mathematics – Chapterwise Solved Papers* (English) • *RRB Group D Maths – Chapterwise Solved Papers* (हिन्दी)

Kaise Use Karein — 3 Step, Dimaag Kam

1. **PEHCHAN**: sawaal ke shabd (keywords) dekho → chapter ka formula mil gaya (jaise **PC-2**).
2. **VALUE DAALO table**: us formula ki table mein wo **row** dhoondo jo sawaal jaisi ho (“Sawaal mein ye ho”).
3. Row ke saamne likhe formula mein **sawaal ke numbers rakh do** → answer. Jo poocha hai use x maano, baaki sab master formula mein daalo.

5 GOLDEN TRICKS — naya (unseen) sawaal bhi in se khulta hai

1. **100 maan lo** — percent, profit, discount, income, population: shuru wali value 100 rakh do, phir badlav lagao.
2. **LCM = kul kaam / kul doori** — time & work, pipe, alternate days: din / ghante ka LCM = poora kaam, phir rate jodo-ghatao.
3. **Ratio ⇒ 1 hissa (k)** — ratio, age, partnership, mixture, coins: 1 hissa = di hui value ÷ uske parts.
4. **Value daalo** — algebra / trigonometry ke “simplify / equals” wale: $x = 1$ ya 2 , $\theta = 45^\circ$ ya 30° daal ke sawaal aur options ka maan milao.
5. **Options daalo** — equation, umar, digits, puzzle: har option ko shart mein rakh ke check karo (5–10 sec).

Notation

- **EN 12** = English book ke usi chapter ka Q. 12 • **HI 30** = Hindi book ka Q. 30 • **(2)** = VALUE DAALO table ki row 2.
- → **PC-2** = ye sawaal doosre chapter ke formula se hota hai.

Guarantee kya hai (sach-sach)

- Pehle wali PDF ke 185 formulas ko 53 master formulas mein jodaa: har purana formula ab kisi naye formula ki **ek row** hai — isliye dono books ka koi sawaal chhoota nahi (8,463/8,463, har chapter ke end mein **COVERAGE CHECK**).
- **1,235** solved examples dobara computer se book ki answer key par check kiye — 1,235/1,235 ✓ (har card mein unme se kuch row-wise chhappe hain).
- **Unseen test**: har formula ke 3 aise random sawaal liye jo pehle kabhi hal nahi kiye gaye the, aur sirf is PDF ke card se hal kiye — **159/159** ✓ (2 sawaal mein book ne options hi nahi chhappe the, unki jagah naya random sawaal liya). Poori list **Appendix A** mein.
- Baaki sawaal type (row) ke hisaab se jode gaye hain, har ek ka alag se computer-calculation nahi hua. Bilkul naye type ka sawaal aaye to 5 Golden Tricks lagao — 100% ki guarantee koi bhi formula nahi de sakta, par is exam level ke sawaal inhi patterns mein aate hain.

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Ek Nazar Mein — 53 Master Formulas

Sirf ye yaad karo. Har formula ki VALUE DAALO table chapter mein hai.

1. Number System संख्या पद्धति

NS-1 BHAAG MACHINE (divisible · remainder · unit digit)

$$N = D \times Q + R \quad (0 \leq R < D)$$

NS-2 GINTI + FACTOR MACHINE (kitne numbers · kitne factors · prime)

$$\# \text{multiples of } d \text{ in } [A, B] = \left\lfloor \frac{B}{d} \right\rfloor - \left\lfloor \frac{A-1}{d} \right\rfloor \quad N = p^a q^b r^c \Rightarrow \text{Factors} = (a+1)(b+1)(c+1)$$

NS-3 DO-NUMBER + SERIES MACHINE

$$\text{Bada} = \frac{S+D}{2}, \quad \text{Chhota} = \frac{S-D}{2} \quad 1+2+\dots+n = \frac{n(n+1)}{2}$$

2. Decimal Fractions दशमलव भिन्न

DF-1 CONVERT MACHINE (decimal $\leftrightarrow$ fraction · terminating · shift)

$$0.\overline{ab} = \frac{ab}{99} \quad 0.a\overline{bc} = \frac{abc-a}{990} \quad \text{Terminating} \iff \text{neeche sirf } 2^m \times 5^n$$

DF-2 COMPARE + OPERATE + “HISSA SE POORA” MACHINE

$$\frac{a}{b} \text{ vs } \frac{c}{d} : a \cdot d \text{ vs } b \cdot c \quad \frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd} \quad \text{Poora} = \frac{\text{Hissa ki value}}{\text{Hissa ki fraction}}$$

3. Indices and Surds घातांक तथा करणी

IS-1 POWER MACHINE (same base $\Rightarrow$ powers barabar)

$$a^m a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n} \quad (a^m)^n = a^{mn} \quad a^{-n} = \frac{1}{a^n} \quad a^0 = 1 \quad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

IS-2 ROOT + SURD MACHINE ($\sqrt{\quad}$ nikaalo · simplify · rationalise)

$$\sqrt{a^2 b} = a\sqrt{b} \quad (a + \sqrt{b})(a - \sqrt{b}) = a^2 - b \quad \sqrt{A \pm 2\sqrt{B}} = \sqrt{m} \pm \sqrt{n} \quad (m+n=A, mn=B)$$

4. Simplification सरलीकरण

SIM-1 VBODMAS + “?” + APPROX MACHINE

$$\mathbf{V} \rightarrow \mathbf{O} \rightarrow \mathbf{D} \rightarrow \mathbf{M} \rightarrow \mathbf{A} \rightarrow \mathbf{S} \quad a \square ? = b \Rightarrow ? = b \text{ (ulta } \square \text{) } a$$

SIM-2 IDENTITY + SHORTCUT MACHINE (bada calculation kabhi nahi)

$$a^2 - b^2 = (a+b)(a-b) \quad (a \pm b)^2 = a^2 \pm 2ab + b^2 \quad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

5. LCM & HCF लघुत्तम समापवर्त्य एवं महत्तम समापवर्तक

LH-1 HCF-LCM NIKAALO + LINK MACHINE

$$\text{HCF} = \text{common primes ki MIN power} \quad \text{LCM} = \text{saare primes ki MAX power} \quad \text{HCF} \times \text{LCM} = a \times b$$

LH-2 SAATH KAB / SABSE BADA NAAP MACHINE (application + remainder)

$$\text{Saath / sabse chhota number} \Rightarrow \text{LCM} \quad \text{Sabse bada naap / sabse bada divisor} \Rightarrow \text{HCF}$$

6. Percentage प्रतिशतता

PC-1 PERCENT = HISSA MACHINE (of · marks · income · votes)

$$x\% \text{ of } Y = \frac{x \times Y}{100} \quad Y = \frac{\text{Value} \times 100}{x} \quad A \text{ is } \left(\frac{A}{B} \times 100 \right) \% \text{ of } B$$

PC-2 MULTIPLIER MACHINE (badha · ghata · do baar · population)

$$\text{Naya} = \text{Purana} \times \frac{100 \pm x}{100} \quad \text{Net}\% = a + b + \frac{a \times b}{100}$$

7. Profit & Loss लाभ और हानि

PL-1 CP $\rightarrow$ SP MULTIPLIER MACHINE (profit · loss · MP · “100 maan lo”)

$$SP = CP \times \frac{100 \pm P}{100} \quad P\% = \frac{SP - CP}{CP} \times 100 \quad \frac{MP}{CP} = \frac{100 + P}{100 - D}$$

PL-2 GINTI · RATE · NAKLI BAAT MACHINE

$$P\% = \left(\frac{\text{Jitna milna chahiye}}{\text{Jitna diya}} - 1 \right) \times 100 \quad \Rightarrow \quad \text{SP of } x = \text{CP of } y : P\% = \frac{y - x}{x} \times 100$$

8. Discount बट्टा / छूट

DS-1 DISCOUNT MULTIPLIER MACHINE

$$SP = MP \times \frac{100 - d}{100} \quad \text{Do discount} = a + b - \frac{a \times b}{100} \quad \frac{MP}{CP} = \frac{100 + P}{100 - D}$$

9. Ratio & Proportion अनुपात और समानुपात**RP-1** “1 HISSA” MACHINE (share · jodna · coins · bachat)

$$1 \text{ hissa} = \frac{\text{Di hui value}}{\text{uske parts}} \Rightarrow \text{Hissa} = \text{Total} \times \frac{a}{a+b+c}$$

RP-2 CROSS-GUNA MACHINE (proportion · ratio badlo)

$$\frac{a}{b} = \frac{c}{d} \iff a \times d = b \times c$$

10. Partnership साझेदारी**PT-1** PROFIT $\propto$ PAISA $\times$ SAMAY MACHINE

$$P_A : P_B : P_C = C_A t_A : C_B t_B : C_C t_C \quad \text{Hissa} = \text{Profit} \times \frac{C_i t_i}{\sum C t}$$

11. Work & Time समय और कार्य**WT-1** LCM UNIT MACHINE (Kaam = Rate $\times$ Din)

$$\text{Kul kaam} = \text{LCM}(\text{din}) \quad \text{Rate} = \frac{\text{Kaam}}{\text{Din}} \quad \text{Din} = \frac{\text{Kaam}}{\text{Rates ka jod}} \quad (\text{do log: } \frac{ab}{a+b})$$

WT-2 M·D·H MACHINE (log · din · ghante)

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

12. Alligation / Mixture मिश्रण / एलीगेशन**MX-1** ALLIGATION CROSS MACHINE (milaane ka ratio)

$$\frac{\text{Sasta}}{\text{Mehenga}} = \frac{d-m}{m-c} \quad m = \frac{q_1 c + q_2 d}{q_1 + q_2}$$

MX-2 “JO NAHI BADLA” + REPLACEMENT MACHINE

$$C_1 V_1 = C_2 V_2 \quad \text{Bacha shuddh} = \text{Shuru} \times \left(1 - \frac{x}{V}\right)^n$$

13. Pipe & Cistern नल और टंकी**PP-1** NET RATE MACHINE (WT-1 hi hai: bharo +, khaali -)

$$\frac{1}{T} = \sum \frac{1}{\text{bharne wale}} - \sum \frac{1}{\text{khaali karne wale}} \quad (\text{bharo } a, \text{ khaali } b : T = \frac{ab}{b-a})$$

14. Simple Interest साधारण ब्याज**SI-1** P·R·T MACHINE (simple interest)

$$SI = \frac{P \times R \times T}{100} \quad A = P + SI \quad (\text{jo poocha use akela karo: } P = \frac{100 SI}{RT}, R = \frac{100 SI}{PT})$$

15. Compound Interest चक्रवृद्धि ब्याज**CI-1** CI AMOUNT MACHINE ($A = P \times \text{multiplier}^n$)

$$A = P \left(1 + \frac{R}{100}\right)^n \quad CI = A - P \quad \begin{array}{l} \text{half-yearly} \\ \text{quarterly} \end{array} \left| \begin{array}{l} R/2 \\ R/4 \end{array} \right| \begin{array}{l} 2n \\ 4n \end{array}$$

CI-2 CI – SI FARAK MACHINE

$$\mathbf{2 \text{ saal: } CI - SI = P \left(\frac{R}{100}\right)^2} \quad \mathbf{3 \text{ saal: } CI - SI = P \left(\frac{R}{100}\right)^2 \left(3 + \frac{R}{100}\right)}$$

16. Problems on Age आयु संबंधी प्रश्न**AG-1** UMAR MACHINE (ratio + “farak kabhi nahi badalta”)

$$\text{Umar} = ax, bx \quad \frac{ax+t}{bx+t} = \frac{c}{d} \Rightarrow x = \frac{t(c-d)}{ad-bc} \quad (t \text{ saal pehle} \Rightarrow t \text{ negative})$$

17. Average औसत**AV-1** TOTAL MACHINE (Total = Average $\times$ n)

$$\mathbf{Total} = \text{Average} \times n \quad \Rightarrow \quad \text{Naya aadmi} = \text{Naya total} - \text{Purana total}$$

18. Speed, Time & Distance चाल, समय और दूरी**ST-1** D = S $\times$ T MACHINE (units · late-early · ratio · do sawari)

$$D = S \times T \quad \text{km/h} \xrightarrow{\times \frac{5}{18}} \text{m/s} \quad \text{Late-Early: } D = \frac{S_1 S_2}{S_2 - S_1} \times \Delta t$$

ST-2 AVERAGE + RELATIVE SPEED MACHINE

$$\text{Avg speed} = \frac{\text{Kul doori}}{\text{Kul samay}} \left(\text{barabar doori: } \frac{2xy}{x+y} \right) \quad \text{Relative} = S_1 + S_2 \text{ (aamne-saamne), } S_1 - S_2 \text{ (peeche)}$$

19. Train रेलगाड़ी**TR-1** TRAIN MACHINE (ST-1 hi hai: D = dono lambai, S = relative)

$$\text{Samay} = \frac{L_{\text{train}} + L_{\text{cheez}}}{\text{Relative speed}} \quad (\text{pole / aadmi / ped: } L_{\text{cheez}} = 0; \text{ km/h} \times \frac{5}{18})$$

20. Boat & Stream नाव और धारा**BS-1** D / U MACHINE (dhaara ke saath / khilaaf)

$$D = B + S \quad U = B - S \quad \Rightarrow \quad B = \frac{D + U}{2}, \quad S = \frac{D - U}{2}$$

21. Mensuration क्षेत्रमिति**MN-1** 2-D SHAPE TABLE (area · perimeter)

Shape	Area	Perimeter / aur
Triangle	$\frac{1}{2}bh, \sqrt{s(s-a)(s-b)(s-c)}$	$a + b + c, s = \frac{a+b+c}{2}$
Equilateral	$\frac{\sqrt{3}}{4}a^2$	$h = \frac{\sqrt{3}}{2}a$
Rectangle	lb	$2(l + b), d = \sqrt{l^2 + b^2}$
Square	$a^2 = \frac{d^2}{2}$	$4a, d = a\sqrt{2}$
Rhombus	$\frac{1}{2}d_1d_2$	$a^2 = (\frac{d_1}{2})^2 + (\frac{d_2}{2})^2$
Parallelogram	$b \times h$	$2(a + b)$
Trapezium	$\frac{1}{2}(a + b)h$	
Circle	πr^2	$2\pi r$
Sector (θ)	$\frac{\theta}{360}\pi r^2$	$\text{arc} = \frac{\theta}{360}2\pi r$

MN-2 3-D SHAPE TABLE (volume · surface · pighlao)

Shape	Volume	Surface
Cube	a^3	$6a^2$ (4 deewar $4a^2$), $d = a\sqrt{3}$
Cuboid	lbh	$2(lb + bh + hl), d = \sqrt{l^2 + b^2 + h^2}$
Cylinder	$\pi r^2 h$	$2\pi r h, 2\pi r(r + h)$
Cone	$\frac{1}{3}\pi r^2 h$	$\pi r l, \pi r(r + l), l = \sqrt{r^2 + h^2}$
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$
Hemisphere	$\frac{2}{3}\pi r^3$	$2\pi r^2, 3\pi r^2$

MN-3 SIZE-CHANGE MACHINE (power dekho, ratio guna)

$$\text{Lambai} \times k \Rightarrow \text{Area} \times k^2, \quad \text{Volume} \times k^3 \quad \% \text{ change (area)} = 2x + \frac{x^2}{100}$$

22. Algebra बीजगणित**AL-1** “VALUE DAALO” IDENTITY MACHINE

$$x + \frac{1}{x} = k \Rightarrow x^2 + \frac{1}{x^2} = k^2 - 2, \quad x^3 + \frac{1}{x^3} = k^3 - 3k \quad a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$$

AL-2 EQUATION MACHINE (linear · quadratic · remainder)

$$ax^2 + bx + c = 0: \quad \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad D = b^2 - 4ac \quad P(x) \div (x - a) \Rightarrow \text{shesh} = P(a)$$

AL-3 SERIES MACHINE (AP · GP)

$$\text{AP: } T_n = a + (n - 1)d, \quad S_n = \frac{n}{2}(a + l) \quad \text{GP: } T_n = ar^{n-1}, \quad S_n = a \frac{r^n - 1}{r - 1}$$

AL-4 VENN MACHINE (sets)

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

23. Trigonometry त्रिकोणमिति**TG-1** “θ DAALO” + VALUE TABLE MACHINE

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \sec^2 \theta - \tan^2 \theta = 1, \quad \csc^2 \theta - \cot^2 \theta = 1$$
TG-2 ANGLE BADLO MACHINE (90° jodi · 180° · A ± B)

$$A + B = 90^\circ \Rightarrow \sin A = \cos B, \tan A \tan B = 1 \quad \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

TG-3 UNCHAI-DOORI MACHINE (heights & distances)

$$\tan \theta = \frac{\text{Unchai}}{\text{Doori}} \quad 30^\circ: h = \frac{d}{\sqrt{3}} \quad 45^\circ: h = d \quad 60^\circ: h = d\sqrt{3}$$

24. Co-ordinate Geometry निर्देशांक ज्यामिति**CG-1** POINT MACHINE (doori · section · area)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad P = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right) \quad \text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

CG-2 LINE + CIRCLE MACHINE (slope)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = -\frac{a}{b} \quad y = mx + c \quad \parallel: m_1 = m_2, \quad \perp: m_1 m_2 = -1$$

25. Geometry ज्यामिति**GE-1** KON (ANGLE) MACHINE (line · triangle · polygon · quadrilateral)

$$\text{Seedhi rekha} = 180^\circ \quad \text{Triangle} = 180^\circ \quad \text{Bahari kon} = \text{door ke do andar ke kon ka jod} \quad n\text{-gon ka jod} = (n - 2) \times 180^\circ$$

GE-2 LAMBAI MACHINE (Pythagoras + similar triangle)

$$c^2 = a^2 + b^2 \quad \text{Similar: side ratio } k \Rightarrow \text{area ratio } k^2$$

GE-3 CIRCLE MACHINE (chord · tangent · kon)

$$r^2 = d^2 + \left(\frac{\text{chord}}{2}\right)^2 \quad \text{Tangent}^2 = d^2 - r^2 \quad \angle \text{centre} = 2 \times \angle \text{paridhi}$$

26. Statistics & Probability सांख्यिकी और प्रायिकता**SP-1** MEAN-MEDIAN-MODE MACHINE

$$\bar{x} = \frac{\sum f x}{\sum f} \quad \text{Median} = \text{sort karke beech wala} \quad \text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

SP-2 SPREAD MACHINE (range · variance · SD)

$$\text{Range} = \max - \min \quad \sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 \quad \sigma = \sqrt{\sigma^2} \quad CV = \frac{\sigma}{\bar{x}} \times 100$$

SP-3 PROBABILITY MACHINE

$$P = \frac{\text{Favourable}}{\text{Total}} \quad P(\text{nahi}) = 1 - P \quad {}^nC_r = \frac{n!}{r!(n-r)!}$$

27. Data Interpretation आंकड़ों का विश्लेषण**DI-1** PADHO-UTHAO-DAALO MACHINE (pie · % · ratio · average)

$$1\% = 3.6^\circ \quad \% \text{ change} = \frac{\text{Naya} - \text{Purana}}{\text{Purana}} \times 100 \quad \text{Ratio} = \frac{\sum \text{pehla}}{\sum \text{doosra}} \quad \text{Avg} = \frac{\text{Jod}}{\text{Ginti}}$$

28. Miscellaneous विविध**MS-1** POORA-HISSA + EK-KA-KITNA + ULTA CHALO MACHINE

$$\text{Poora} = \frac{\text{Bacha hua}}{(1-a)(1-b)(1-c)\dots} \quad 1 \text{ ka} = \frac{\text{Kul}}{\text{Ginti}} \quad \text{Puzzle} \Rightarrow \text{aakhri se ulta chalo / options daalo}$$

Chapter 1

Number System (Sankhya Paddhati) संख्या पद्धति

3 formulas (pehle 10) · English book 345 sawaal · Hindi book 161 sawaal

NS-1 — BHAAG MACHINE (divisible · remainder · unit digit)

PEHCHAN – Question mein ye dikhe

divisible by, remainder / leaves, least number to be added or subtracted, greatest / smallest n-digit number divisible by, unit digit / last digit, $a^n \pm b^n$ divisible; विभाज्य, शेषफल, इकाई अंक, जोड़ा / घटाया जाए

MASTER FORMULA – sirf ye yaad karo

$$N = D \times Q + R \quad (0 \leq R < D)$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Divisible hai? / missing digit x	$R = 0$ chahiye. 2, 5, 10 : last digit · 3, 9 : digits ka jod · 4, 8 : last 2 / 3 digits · 11 : (odd jagah ka jod – even jagah ka jod) = 0, 11 ... · bada D $\Rightarrow$ co-prime tukde ($12 = 3 \times 4$, $72 = 8 \times 9$)
② Remainder of $a \times b$, a^n , chain	Har number ki jagah uska remainder rakho: $R(ab) = R(a) \cdot R(b)$ · $a \equiv \pm 1 \Rightarrow (\pm 1)^n$ · chain: naya $R = R_1 \bmod D_2$
③ Kitna jodein / ghatayein; sabse bada / chhota n-digit	Jodo = $D - R$ · Ghatao = R · Sabse bada = $99 \dots 9 - R$ · Sabse chhota = $10 \dots 0 + (D - R)$ · perfect square: agla square – N
④ Unit digit of a^n	Power $\div 4 \Rightarrow$ bacha r (0 ho to 4) $\Rightarrow$ (a ka last digit) ^r ka last digit · 0, 1, 5, 6 $\Rightarrow$ wahi digit
⑤ $a^n \pm b^n$ / lagataar numbers ka product kis se kate	$a^n - b^n \div (a - b)$ hamesha, n even $\Rightarrow \div (a + b)$ bhi · $a^n + b^n$, n odd $\Rightarrow \div (a + b)$ · k lagataar numbers ka product $\div k! \cdot \overline{pq} - \overline{qp} = 9(p - q)$

TERMS – Har letter ka matlab

N = number (bhajya) · D = divisor (bhajak) · Q = quotient (bhagphal) · R = remainder (shesh).

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 1**: 9 aur 11 dono se divisible?
10098: digit-sum 18 ✓(9); $(8 + 0 + 1) - (9 + 0) = 0$ ✓(11) ✓ **book ans (b)**
- ① **EN 44**: x4461 divisible by 11, x = ?
 $(1 + 4 + x) - (6 + 4) = 0 \Rightarrow x = 5$ ✓ **book ans (d)**
- ② **EN 43**: $14528 \div d$: quotient 83, remainder 3; d = ?
 $d = \frac{14528 - 3}{83} = 175$ ✓ **book ans (d)**
- ② **EN 6**: $171 \times 172 \times 173 \div 17$ ka remainder
rem: $1 \times 2 \times 3 = 6$ ✓ **book ans (c)**
- ③ **EN 26**: Largest 4-digit divisible by 83
 $9999 \bmod 83 = 39 \Rightarrow 9999 - 39 = 9960$ ✓ **book ans (c)**
- ③ **EN 37**: Smallest 4-digit divisible by 47
 $1000 \bmod 47 = 13 \Rightarrow 1000 + (47 - 13) = 1034$ ✓ **book ans (c)**
- ④ **EN 292**: 213^6 ka last digit
 $6 \bmod 4 = 2 \Rightarrow 3^2 = 9$ ✓ **book ans (d)**
- ④ **EN 285**: $1373^{36} - 1442^{20}$
 $3^{4k} \rightarrow 1, 2^{4k} \rightarrow 6 \Rightarrow 1 - 6 \rightarrow 11 - 6 = 5$ ✓ **book ans (c)**
- ⑤ **EN 17**: $41^{43} + 43^{43}$ divisible by?
odd power $\Rightarrow \div (41 + 43) = 84$ ✓ **book ans (d)**
- ⑤ **EN 32**: $3^{71} + \dots + 3^{75}$ divisible by?
 $3^{71}(1 + 3 + 9 + 27 + 81) = 3^{71} \cdot 121 \Rightarrow 11$ ✓ **book ans (c)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (116) EN 1–11, 14–17, 19–21, 24, 26–30, 32–55, 67, 68, 71–73, 152, 182, 253, 262, 265, 268, 271, 275, 279, 280, 283–285, 288, 292, 294, 297, 298, 331 · **HI** 1–3, 5–27, 29, 30, 32, 33, 35–40, 57, 119, 126, 142, 147, 158, 159 | *Ch 5 (L.S. – M.S.):* EN 222

NS-2 — GINTI + FACTOR MACHINE (kitne numbers · kitne factors · prime)

PEHCHAN – Question mein ye dikhe

how many numbers between A and B divisible by, number of factors / divisors, odd / even factors, prime / co-prime / twin prime, how many primes; कितनी संख्याएँ, गुणनखंड, अभाज्य

MASTER FORMULA – sirf ye yaad karo

$$\# \text{multiples of } d \text{ in } [A, B] = \left\lfloor \frac{B}{d} \right\rfloor - \left\lfloor \frac{A-1}{d} \right\rfloor \quad N = p^a q^b r^c \Rightarrow \text{Factors} = (a+1)(b+1)(c+1)$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① A se B ke beech kitne d se kate	Master formula · “p ya q” = $n(p) + n(q) - n(\text{LCM})$ · “p aur q” = $n(\text{LCM})$ · squares = $\lfloor \sqrt{B} \rfloor - \lfloor \sqrt{A} \rfloor + 1$
② Kitne factors (odd / even / square)	Total = $(a+1)(b+1) \dots$ · Odd = 2 ki power hatake · Even = Total – Odd · odd factors $\Leftrightarrow$ N perfect square · square banana: odd power wale prime se $\times / \div$
③ Prime hai? / kitne prime	$\sqrt{N}$ tak ke primes se bhaag — koi na kaate $\Rightarrow$ prime · 1–100 mein 25 primes · co-prime $\Leftrightarrow$ $\text{HCF} = 1$ · chain $pq = A, qr = B \Rightarrow q = \text{HCF}(A, B)$

TERMS – Har letter ka matlab

$\lfloor x \rfloor$ = x ka poora (integer) hissa · p, q, r = prime numbers · a, b, c = unki powers.

SOLVED – Asli book questions (row ke saath, verified)

① **EN 13:** 1–700 mein 17 se divisible

$$\lfloor 700/17 \rfloor = 41 \checkmark \text{ book ans (b)}$$

① **EN 12:** 1–100: 3 ya 4 se divisible

$$33 + 25 - 8 = 50 \checkmark \text{ book ans (a)}$$

② **EN 281:** Factors of 4200

$$2^3 \cdot 3 \cdot 5^2 \cdot 7 \Rightarrow 4 \cdot 2 \cdot 3 \cdot 2 = 48 \checkmark \text{ book ans (a)}$$

② **EN 276:** 120 ke odd, even divisors

$$2^3 \cdot 3 \cdot 5: \text{odd} = 2 \cdot 2 = 4, \text{even} = 16 - 4 = 12 \checkmark \text{ book ans (c)}$$

③ **EN 78:** 50 se kam primes

table: 15 $\checkmark$ book ans (b)

③ **EN 86:** 50 aur 60 ke beech primes ka sum

$$53 + 59 = 112 \checkmark \text{ book ans (d)}$$

IS FORMULA SE HONE WALE BOOK KE SAWAAL (99) EN 12, 13, 22, 23, 25, 31, 65, 74, 76–93, 95–109, 111, 257, 264, 266, 274, 276, 278, 281, 282, 286, 287, 289–291, 293, 295, 296, 299–301, 329, 332, 339 · **HI** 4, 28, 31, 34, 52–56, 58–64, 66–74, 116, 143–146, 148, 157, 160, 161 | *Ch 5 (L.S. – M.S.):* EN 174

NS-3 — DO-NUMBER + SERIES MACHINE

PEHCHAN – Question mein ye dikhe

sum and difference / product of two numbers, consecutive numbers, digits reversed, fraction of a number, sum of squares / cubes, place value / face value, rational / irrational; योग, अंतर, गुणनफल, क्रमागत, स्थानीय मान

MASTER FORMULA – sirf ye yaad karo

$$\text{Bada} = \frac{S+D}{2}, \quad \text{Chhota} = \frac{S-D}{2} \quad 1+2+\dots+n = \frac{n(n+1)}{2}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Do numbers: jod S, farak D, product P; lagataar numbers; digits ulte	Master formula $\cdot x^2 - y^2 = S \cdot D \cdot x^2 + y^2 = S^2 - 2P \cdot n$ lagataar ka jod $S \Rightarrow$ beech wala $= S/n \cdot \frac{a}{b}x - \frac{c}{d}x = k \Rightarrow x = k \div (\frac{a}{b} - \frac{c}{d}) \cdot 2$ -digit ulta: farak $= 9(x - y)$, jod $= 11(x + y)$
② Squares / cubes ka jod, place value, rational	$\sum n^2 = \frac{n(n+1)(2n+1)}{6} \cdot \sum n^3 = \left[\frac{n(n+1)}{2}\right]^2$ • pehle n odd $= n^2 \cdot$ handshakes $= \frac{n(n-1)}{2} \cdot$ place value $= \text{digit} \times \text{jagah (1, 10, 100)} \cdot$ face value $= \text{digit} \cdot \sqrt{(\text{non-square})}$, $\pi \Rightarrow$ irrational

TERMS – Har letter ka matlab

S = jod (sum) $\cdot$ D = farak (difference) $\cdot$ P = guna (product) $\cdot$ n = kitne numbers.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 124:** Sum 45, difference 19
 $(45 + 19)/2 = 32$, 13 ✓ **book ans (b)**
- ① **EN 131:** S = 25, P = 136 $\Rightarrow x^3 + y^3$
 $25^3 - 3(136)(25) = 15625 - 10200 = 5425$ ✓ **book ans (a)**
- ① **EN 142:** S = 20, P = 96 $\Rightarrow$ difference
 $\sqrt{400 - 384} = 4$ ✓ **book ans (a)**
- ② **EN 141:** $3^2 + \dots + 18^2$
 $S(18) - S(2) = 2109 - 5 = 2104$ ✓ **book ans (d)**
- ② **EN 138:** $5^3 + \dots + 14^3$
 $105^2 - 10^2 = 11025 - 100 = 10925$ ✓ **book ans (b)**
- ② **EN 328:** 40 log handshakes
 $40 \cdot 39/2 = 780$ ✓ **book ans (b)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (290) EN 18, 56–64, 66, 70, 75, 94, 110, 112–151, 153, 155–172, 175–180, 184–192, 195, 197–252, 254–256, 258–261, 263, 267, 269, 270, 272, 273, 277, 302–328, 330, 333, 335, 336, 338, 340, 343, 344 • **HI** 41–51, 65, 75–115, 117, 118, 120–122, 125, 127–141, 149–152, 154, 156 | Ch 3 (Ghatank – Karani): EN 123 | Ch 5 (L.S. – M.S.): EN 172 • **HI** 105 | Ch 7 (Labh aur Hani): EN 41, 108, 118, 154 | Ch 9 (Anupat aur Samanupat): EN 149 • **HI** 4 | Ch 10 (Saajhedaari): EN 20–22 • **HI** 5, 18 | Ch 28 (Vividh): EN 42

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
NS-1	BHAAG MACHINE (divisible $\cdot$ remainder $\cdot$ unit digit)	72	43
NS-2	GINTI + FACTOR MACHINE (kitne numbers $\cdot$ kitne factors $\cdot$ prime)	64	34
NS-3	DO-NUMBER + SERIES MACHINE	195	80
$\rightarrow$ DF-1	CONVERT MACHINE (decimal $\leftrightarrow$ fraction $\cdot$ terminating (Ch 2)	1	0
$\rightarrow$ DF-2	COMPARE + OPERATE + “HISSA SE POORA” MACHINE (Ch 2)	5	2
$\rightarrow$ SIM-1	VBODMAS + “?” + APPROX MACHINE (Ch 4)	3	0
$\rightarrow$ LH-2	SAATH KAB / SABSE BADA NAAP MACHINE (application + (Ch 5)	1	2
$\rightarrow$ PC-1	PERCENT = HISSA MACHINE (of $\cdot$ marks $\cdot$ income $\cdot$ vot (Ch 6)	2	0
$\rightarrow$ PC-2	MULTIPLIER MACHINE (badha $\cdot$ ghata $\cdot$ do baar $\cdot$ popu (Ch 6)	2	0
Total (book ke sawaal)		345/345 ✓	161/161 ✓
$\rightarrow$ DF-1: EN 69 $\cdot$ $\rightarrow$ DF-2: EN 173, 174, 181, 183, 341 HI 123, 124 $\cdot$ $\rightarrow$ LH-2: EN 154 HI 153, 155 $\cdot$ $\rightarrow$ PC-1: EN 194, 196 $\cdot$ $\rightarrow$ PC-2: EN 193, 334 $\cdot$ $\rightarrow$ SIM-1: EN 337, 342, 345			

Chapter 2

Decimal Fractions (Dashamlav Bhinn) दशमलव भिन्न

2 formulas (pehle 9) · English book 232 sawaal · Hindi book 133 sawaal

DF-1 — CONVERT MACHINE (decimal ↔ fraction · terminating · shift)

PEHCHAN – Question mein ye dikhe

p/q form of recurring decimal (bar), terminating / non-terminating, reciprocal, “if $23 \times 19 = 437$ then $0.0437 \div 1.9$ ”, rounding / correct to n places, equivalent fraction, lowest form; आवर्ती दशमलव, सांत, व्युत्क्रम, तुल्य भिन्न

MASTER FORMULA – sirf ye yaad karo

$$0.\overline{ab} = \frac{ab}{99} \quad 0.\overline{abc} = \frac{abc - a}{990} \quad \text{Terminating} \iff \text{neeche sirf } 2^m \times 5^n$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo ⇒ answer
① Bar wala decimal → p/q	Jitne digit repeat ⇒ utne 9 , jo repeat nahi ⇒ unke baad utne 0 ; upar = (saare digits) – (non-repeat digits) · poora number alag jodo: $4.\overline{5} = 4 + \frac{5}{9}$
② Terminating hai ya recurring	Pehle lowest form; neeche sirf 2 aur 5 ⇒ terminating (decimal places = $\max(m, n)$), warna recurring
③ Decimal guna / bhaag, diya hua product, rounding	Digits wahi, sirf point khisko: (upar ke decimal places) – (neeche ke) · rounding: agla digit $\geq 5 \Rightarrow$ upar
④ Reciprocal	$\frac{a}{b} \rightarrow \frac{b}{a} \cdot \frac{a}{b} + \frac{b}{a} = \frac{a^2+b^2}{ab} \cdot \frac{1}{\frac{1}{x}+\frac{1}{y}} = \frac{xy}{x+y}$
⑤ Equivalent / lowest form / part of whole	$\frac{a}{b} = \frac{c}{d} \iff ad = bc$ · lowest form: dono ÷ HCF · hissa ÷ poora (same unit)

TERMS – Har letter ka matlab

Bar (̄) = repeat hone wale digits · m, n = 2 aur 5 ki powers.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 73: $0.\overline{12} = ?$
 $\frac{12}{99} = \frac{4}{33}$ ✓ book ans (b)
- ① EN 72: $0.27 = ?$
 $\frac{27-2}{90} = \frac{25}{90} = \frac{5}{18}$ ✓ book ans (c)
- ② EN 52: 57/120, 47/150, 61/110, 43/140: terminating?
 $\frac{57}{120} = \frac{19}{40}$, $40 = 2^3 \cdot 5$ ✓ (baaki mein 3, 11, 7) ✓ book ans (a)
- ② EN 50: 3/8 kitne decimal places ke baad khatam
 $8 = 2^3 \Rightarrow 3$ ✓ book ans (c)
- ③ EN 160: $0.1404 \div 0.06$
 $\frac{14.04}{6} = 2.34$ ✓ book ans (b)
- ③ EN 178: $23 \times 19 = 437 \Rightarrow 0.0437 \div 1.9$
 0.0437 (4 places) $\div 1.9$ (1 place) $\Rightarrow 23$ ke saath 3 places: 0.023 ✓ book ans (c)
- ④ EN 117: $5/7$ aur $9/5$ ke reciprocals ke sum ka reciprocal
 $\frac{1}{\frac{1}{5}+\frac{1}{9}} = \frac{45}{88}$ ✓ book ans (c)
- ④ EN 120: $x - \frac{1}{x} = \frac{9}{11} \Rightarrow$ cubes ka difference
 $\frac{729}{1331} + \frac{27}{11} = \frac{3996}{1331}$ ✓ book ans (c)
- ⑤ EN 232: $60/75 = 4/x$
 $x = \frac{4 \times 75}{60} = 5$ ✓ book ans (d)
- ⑤ EN 111: 7 min 12 s = din ka kitna hissa
 $\frac{432}{86400} = \frac{1}{200}$ ✓ book ans (c)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (199) EN 23, 44–97, 100, 103, 105, 107–111, 114–120, 127, 137, 157–165, 167, 168, 171–181, 184, 185, 187, 190, 198, 200, 204, 217, 218, 220, 221, 223–225, 229–232 · HI 14–72, 80, 81, 86, 101, 102, 104–115, 117, 119–125, 132, 133 | Ch 1 (Sankhya Paddhati): EN 69

DF-2 — COMPARE + OPERATE + “HISSA SE POORA” MACHINE

PEHCHAN – Question mein ye dikhe

largest / smallest fraction, ascending / descending, what should be added / subtracted, remaining part, $1/(1 \cdot 2) + 1/(2 \cdot 3) + \dots$, numerator / denominator conditions; सबसे बड़ी भिन्न, क्या जोड़ा जाए, अंश, हर

MASTER FORMULA – sirf ye yaad karo

$$\frac{a}{b} \text{ vs } \frac{c}{d} : a \cdot d \text{ vs } b \cdot c \quad \frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd} \quad \text{Poora} = \frac{\text{Hissa ki value}}{\text{Hissa ki fraction}}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Badi / chhoti fraction, order	Cross multiply — bada product wali fraction badi $\cdot$ bahut saari $\Rightarrow$ sabko decimal ($\div$) $\cdot$ do ke beech: $\frac{a+c}{b+d}$
② Jod / ghata, “kya jodein”, bacha hissa	A mein kya jodein ki $B \Rightarrow B - A \cdot$ bacha hissa $= 1 - (\text{sab hisson ka jod}) \cdot \frac{3}{4} = 7/8 \text{ kg} \Rightarrow$ poora $= \frac{7}{8} \div \frac{3}{4}$
③ Cancel wali series	$\frac{1}{a(a+d)} + \dots + \frac{1}{l(l+d)} = \frac{1}{d} \left(\frac{1}{a} - \frac{1}{l+d} \right) \cdot \frac{1}{1 \cdot 2} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$
④ Ansh / har badle to fraction kya	Options daal ke check (5 sec) $\cdot$ same k dono se ghata: $\frac{a-k}{b-k} = \frac{c}{d} \Rightarrow k = \frac{bc-ad}{c-d} \cdot \text{upar} + x\%$, neche $-y\% \Rightarrow \text{naya} = \text{purana} \times \frac{100+x}{100-y}$

TERMS – Har letter ka matlab

a, c = upar (ansh) $\cdot$ b, d = neeche (har) $\cdot$ l = series ka last number $\cdot$ d = gap.

SOLVED – Asli book questions (row ke saath, verified)

① **EN 9:** 15/16, 24/25, 34/35, 19/20 mein greatest neeche — upar = 1 sab mein $\Rightarrow$ sabse bade numbers: 34/35 ✓ **book ans (c)**

① **HI 2:** 5/7, 9/14, 16/21, 29/42 में सबसे बड़ी LCM 42: 30, 27, 32, 29 $\Rightarrow$ 16/21 ✓ **book ans (d)**

① **EN 8:** 2/3, 3/4, 4/5, 5/6: greatest — least $\frac{5}{6} - \frac{2}{3} = \frac{1}{6}$ ✓ **book ans (c)**

② **EN 139:** 10/11 mein kya jodein ki 11/10 bane $\frac{11}{10} - \frac{10}{11} = \frac{121-100}{110} = \frac{21}{110}$ ✓ **book ans (a)**

② **EN 216:** Tapan 1/4, Trisha 2/3, Ravi = baaki $1 - \frac{1}{4} - \frac{2}{3} = \frac{1}{12}$ ✓ **book ans (b)**

③ **EN 102:** $1/(1 \cdot 4) + 1/(4 \cdot 7) + \dots + 1/(47 \cdot 50)$ $\frac{1}{3} \left(1 - \frac{1}{50} \right) = \frac{49}{150}$ ✓ **book ans (d)**

④ **EN 130:** 15/19 ke dono se kya ghatayein ki 3/4 $k = \frac{19 \cdot 3 - 15 \cdot 4}{3 - 4} = 3$ ✓ **book ans (d)**

④ **EN 131:** 42/45 se kya ghatayein ki 5/6 $k = \frac{45 \cdot 5 - 42 \cdot 6}{5 - 6} = 27$ ✓ **book ans (a)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (178) EN 1–22, 24–43, 98, 99, 101, 102, 104, 106, 112, 113, 121–126, 128–136, 138–156, 166, 169, 170, 182, 183, 186, 188, 189, 191–197, 199, 201–203, 205–211, 213–216, 219, 222, 226–228 $\cdot$ **HI** 1–13, 73–79, 82–85, 87–100, 103, 116, 118, 126–129, 131 | *Ch 1 (Sankhya Paddhati):* EN 173, 174, 181, 183, 341 $\cdot$ **HI** 123, 124 | *Ch 6 (Pratishatata):* EN 162 | *Ch 9 (Anupat aur Samanupat):* EN 169 | *Ch 11 (Samay aur Karya):* EN 117 | *Ch 16 (Aayu Sambandhi Prashn):* EN 217, 225, 236

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
DF-1	CONVERT MACHINE (decimal $\leftrightarrow$ fraction $\cdot$ terminating $\cdot$ shift)	112	86
DF-2	COMPARE + OPERATE + “HISSA SE POORA” MACHINE	119	46
$\rightarrow$ IS-2	ROOT + SURD MACHINE ($\sqrt{\text{nikaalo}} \cdot \text{simplify} \cdot \text{ration (Ch 3)}$)	1	1
Total (book ke sawaal)		232/232 ✓	133/133 ✓
$\rightarrow$ IS-2: EN 212 HI 130			

Chapter 3

Indices and Surds (Ghatank – Karani) घातांक तथा करणी

2 formulas (pehle 7) · English book 139 sawaal · Hindi book 103 sawaal

IS-1 — POWER MACHINE (same base $\Rightarrow$ powers barabar)

PEHCHAN – Question mein ye dikhe

find x when $6^{x+4} = 6^7, 8^{2/3}, 3^{-5}, \sqrt{(3^{38} + 3^{39})}$, nested roots $\sqrt{(214 + \sqrt{(107 + \sqrt{196})})}$, $\sqrt{(34^2 - 16^2)}$, $\sqrt{x + x} = 90$; घात, घातांक, वर्गमूल

MASTER FORMULA – sirf ye yaad karo

$$a^m a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n} \quad (a^m)^n = a^{mn} \quad a^{-n} = \frac{1}{a^n} \quad a^0 = 1 \quad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Equation mein power / value nikaalo	Dono taraf same base (2, 3, 5) banao $\Rightarrow$ powers barabar rakho · jod ho to common nikaalo: $a^m + a^{m+1} = a^m(1 + a)$ · $625 = 5^4$, $729 = 3^6$, $1024 = 2^{10}$, $0.125 = 2^{-3}$
② Root ke andar root / $\sqrt{a^2 - b^2}$	Sabse andar wala pehle · $\sqrt{a^2 - b^2} = \sqrt{(a-b)(a+b)}$ · $\sqrt{x}\sqrt{y} = \sqrt{xy}$ · $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$
③ $\sqrt{\text{wali equation}}$ / word problem	Options daalo · “har member utna de jitne member” $\Rightarrow$ members = $\sqrt{\text{total}} \cdot \frac{\sqrt{x}}{c} = d \Rightarrow x = (cd)^2$

TERMS – Har letter ka matlab

a = base · m, n = power (ghatank).

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 115:** $2^x \cdot 4^{12} \cdot 8^3 = 16^{11}$
 $x + 24 + 9 = 44 \Rightarrow x = 11$ ✓ book ans (b)
- ① **EN 112:** $6^{x+4} = 6^7$
 $x + 4 = 7 \Rightarrow 3$ ✓ book ans (d)
- ② **EN 77:** $\sqrt{(214 + \sqrt{(107 + \sqrt{196})})}$
 $\sqrt{214 + \sqrt{121}} = \sqrt{225} = 15$ ✓ book ans (b)
- ② **EN 1:** $\sqrt{(34^2 - 16^2)}$
 $\sqrt{18 \cdot 50} = 30$ ✓ book ans (d)
- ③ **EN 139:** har member utne ₹ de jitne members; total ₹62.41
 $\sqrt{6241} = 79$ ✓ book ans (d)
- ③ **EN 130:** $(1/3 - 1/4)$ of number = $\sqrt{\text{number}}$
 $\frac{x}{12} = \sqrt{x} \Rightarrow x = 144$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (88) EN 1, 8, 11, 28, 52, 55, 60, 66, 75, 77, 80, 82–85, 90–92, 96, 99, 103, 106, 112–122, 124–130, 132, 134, 137–139 · HI 5, 8, 13–15, 17, 19, 21, 24, 41, 55, 62–64, 66, 67, 69–71, 75–77, 79–83, 89, 92, 94, 95, 97–100 | Ch 22 (Beejganit): EN 254, 255, 258, 291 · HI 19, 41 | Ch 28 (Vividh): EN 12, 57

IS-2 — ROOT + SURD MACHINE ($\sqrt{\text{nikaalo}} \cdot \text{simplify} \cdot \text{rationalise}$)

PEHCHAN – Question mein ye dikhe

square root of 519841, $\sqrt{0.0324}$, $\sqrt{(576/625)}$, mixed surd $\sqrt{1350}$, $7\sqrt{48} + 7\sqrt{147}$, rationalise the denominator, $\sqrt{(8 + 2\sqrt{15})}$; वर्गमूल, करणी, परिमेयीकरण

MASTER FORMULA – sirf ye yaad karo

$$\sqrt{a^2 b} = a\sqrt{b} \quad (a + \sqrt{b})(a - \sqrt{b}) = a^2 - b \quad \sqrt{A \pm 2\sqrt{B}} = \sqrt{m} \pm \sqrt{n} \quad (m + n = A, mn = B)$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Bade number ka $\sqrt{\quad}$	Last digit: $1 \Rightarrow 1/9 \cdot 4 \Rightarrow 2/8 \cdot 9 \Rightarrow 3/7 \cdot 6 \Rightarrow 4/6 \cdot 5 \Rightarrow 5$ · last 2 digit hatao, bache se chhota square $k^2 \Rightarrow \text{root} = k$ · bacha $\geq k(k+1) \Rightarrow$ bada candidate
② Decimal / fraction ka $\sqrt{\quad}$	Point se 2-2 digit ke jode (cube: 3-3) · root mein decimal places aadhe · $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \cdot \sqrt{2} = 1.414$, $\sqrt{3} = 1.732$, $\sqrt{5} = 2.236$
③ Surd simplify / jodo	Har surd ko common surd mein: $\sqrt{48} = 4\sqrt{3}$, $\sqrt{147} = 7\sqrt{3} \Rightarrow p\sqrt{b} + q\sqrt{b} = (p+q)\sqrt{b}$
④ Rationalise / $\sqrt{A \pm 2\sqrt{B}}$	$\frac{1}{a+\sqrt{b}} = \frac{a-\sqrt{b}}{a^2-b} \cdot \sqrt{A \pm 2\sqrt{B}}$: jod A, guna B wale m, n dhoondo

TERMS – Har letter ka matlab

Surd = jiska exact root na nikle ($\sqrt{2}$, $\sqrt{3}$) · conjugate = beech ka sign ulta.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 13:** $\sqrt{519841}$
last 1 $\Rightarrow 1/9$; $5198: 72^2 = 5184 \Rightarrow 72$; $5198 < 72 \cdot 73 = 5256 \Rightarrow$ chhota $\Rightarrow$ **721 ✓ book ans (a)**
- ① **EN 34:** $\sqrt{243049}$
last 9 $\Rightarrow 3/7$; $2430: 49^2 = 2401$; $2430 < 49 \cdot 50 = 2450 \Rightarrow$ **493 ✓ book ans (c)**
- ② **EN 44:** $\sqrt{0.0324}$
 $\sqrt{324} = 18$, 4 places $\Rightarrow$ 2 places: **0.18 ✓ book ans (a)**
- ② **EN 43:** $\sqrt{0.00069169}$
 $\sqrt{69169} = 263$, 8 places $\Rightarrow$ 4: **0.0263 ✓ book ans (c)**
- ③ **EN 3:** $\sqrt{1350}$ mixed surd
 $1350 = 15^2 \times 6 \Rightarrow 15\sqrt{6}$ **✓ book ans (d)**
- ③ **EN 74:** $7\sqrt{48} + 7\sqrt{147}$
 $28\sqrt{3} + 49\sqrt{3} = 77\sqrt{3}$ **✓ book ans (d)**
- ④ **EN 78:** $\sqrt{(8+2\sqrt{15}) \times (8-2\sqrt{15})}$
 $\sqrt{64-60} = 2$ **✓ book ans (b)**
- ④ **EN 65:** $\sqrt{x} + \sqrt{y} = \sqrt{(14+8\sqrt{3})}$; $x^2 + y^2$?
 $14 + 2\sqrt{48}: m+n=14, mn=48 \Rightarrow 8,6 \Rightarrow 64+36=100$ **✓ book ans (c)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (161) EN 2-7, 9, 10, 12-27, 29-51, 53, 54, 56, 57, 59, 61-65, 67-74, 76, 78, 79, 81, 86-89, 93-95, 97, 98, 100-102, 104, 105, 107-111, 131, 133, 135, 136 · **HI** 1-4, 6, 7, 9-12, 16, 18, 22, 23, 25-40, 42-54, 56-61, 65, 68, 72, 73, 78, 84-88, 90, 91, 93, 96, 101-103 | Ch 2 (Dashamlav Bhinn): EN 212 · **HI** 130 | Ch 22 (Beejganit): **HI** 71

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
IS-1	POWER MACHINE (same base $\Rightarrow$ powers barabar)		45
IS-2	ROOT + SURD MACHINE ($\sqrt{\quad}$ nikaalo · simplify · rationalise)		92
$\rightarrow$ NS-3	DO-NUMBER + SERIES MACHINE (Ch 1)		1
$\rightarrow$ SIM-1	VBODMAS + “?” + APPROX MACHINE (Ch 4)		0
$\rightarrow$ SIM-2	IDENTITY + SHORTCUT MACHINE (bada calculation kabh (Ch 4)		1
Total (book ke sawaal)		139/139 ✓	103/103 ✓
$\rightarrow$ NS-3: EN 123 · $\rightarrow$ SIM-1: HI 20, 74 · $\rightarrow$ SIM-2: EN 58			

Chapter 4

Simplification (Saralikaran) सरलीकरण

2 formulas (pehle 8) · English book 248 sawaal · Hindi book 243 sawaal

SIM-1 — VBODMAS + “?” + APPROX MACHINE

PEHCHAN – Question mein ye dikhe

simplify, find the value of ?, closest approximate value, brackets () { } [], “of”, mixed fractions, continued fraction, 97×97 , 95×103 ; सरल कीजिए, ? के स्थान पर, लगभग मान

MASTER FORMULA – sirf ye yaad karo

$$V \rightarrow \text{Of} \rightarrow D \rightarrow M \rightarrow A \rightarrow S \quad a \square ? = b \Rightarrow ? = b \text{ (ulta } \square) a$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Lamba expression / brackets	Bracket andar se bahar () $\rightarrow$ { } $\rightarrow$ [] · “of” = $\times$ ($\div$ se pehle) $\cdot \div \times$ barabar $\Rightarrow$ left se right · sign swap wale: options ke sign badal ke check
② “?” nikaalo / decimal jod	? akela chhodo, baaki ulte operation se doosri taraf: $a + ? = b \Rightarrow b - a$, $a \times ? = b \Rightarrow b \div a$, $\sqrt{?} = k \Rightarrow k^2$ · decimal point ek line mein
③ Approximate value	Har number ko aasaan number bana do ($19.99 \rightarrow 20$, $324 \div 18$) $\Rightarrow$ VBODMAS $\Rightarrow$ sabse paas wala option
④ Fractions ki chain / continued fraction	Mixed $\rightarrow$ improper, $\div = \times$ ulta · continued fraction neeche se upar $\cdot 1 + 2 + \dots + n + \dots + 2 + 1 = n^2$
⑤ Base ke paas wale numbers ka guna	$(B \pm d)^2 = B^2 \pm 2Bd + d^2$ · $(B + x)(B + y) = B^2 + B(x + y) + xy$ · $N \times 99 \dots 9 = N \times 10^k - N \cdot ac + bc = (a + b)c$

TERMS – Har letter ka matlab

V = vinculum / bracket · O = of · D = $\div$ · M = $\times$ · A = + · S = - · B = base (100, 1000).

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 46:** $3 + [3 \times \{3 - (3 + 3) \div 6\}]$
 $(3 + 3) \div 6 = 1 \Rightarrow 3 \times 2 = 6 \Rightarrow 9$ ✓ **book ans (b)**
- ① **EN 43:** $72 \div 4 \times \{8 \times 4 - (14 - 19)\}$
 $\{32 + 5\} = 37$; $72 \div 4 = 18$; $18 \times 37 = 666$ ✓ **book ans (a)**
- ② **EN 37:** $9876 + 34.567 - ? = 9908.221$
 $? = 9910.567 - 9908.221 = 2.346$ ✓ **book ans (d)**
- ② **EN 140:** $19 \times 23 = 437 \Rightarrow 190 \times 0.023$
 1 zero aage + 3 decimal $\Rightarrow$ 2 decimal: 4.37 ✓ **book ans (d)**
- ③ **HI 115:** $19.99 - 10.01 + 24.99 \times 10.01$
 $20 - 10 + 25 \times 10 = 260$ ✓ **book ans (a)**
- ③ **HI 118:** $29.99 + 15.01 + 44.99 \times 15.01$
 $30 + 15 + 675 = 720$ ✓ **book ans (c)**
- ④ **EN 164:** $1/(2 + 1/(3 + 1/(1 + 1/4)))$
 $1 + \frac{1}{4} = \frac{5}{4} \rightarrow \frac{4}{5} \rightarrow \frac{19}{5} \rightarrow \frac{5}{19} \rightarrow \frac{43}{19} \rightarrow \frac{19}{43}$ ✓ **book ans (a)**
- ④ **EN 198:** $(1 + 2/3) \div [(1 + 1/3) \div (2/3 + 1)]$
 $\frac{5}{3} \div \frac{4}{5} = \frac{25}{12}$ ✓ **book ans (d)**
- ⑤ **EN 23:** 97×97
 $(100 - 3)^2 = 10000 - 600 + 9 = 9409$ ✓ **book ans (b)**
- ⑤ **EN 38:** 1093×1093
 $(1100 - 7)^2 = 1210000 - 15400 + 49 = 1194649$ ✓ **book ans (a)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (345) EN 1-19, 21-58, 60-81, 94-96, 101, 102, 106, 116, 118, 119, 121, 124, 126-140, 143-155, 160-189, 191-201, 210, 236, 237, 239 · HI 4, 5, 7-10, 12, 15-17, 19-28, 30-52, 54-76, 78-91, 93-106, 108-113, 115-122, 124, 125, 127-130, 132-139, 141, 143-159, 161-165, 167, 169-180, 182-189, 191-196, 199, 217, 235, 236 | Ch 1 (Sankhya Paddhati): EN 337, 342, 345 | Ch 3 (Ghatank - Karani): HI 20, 74 | Ch 28 (Vividh): EN 20

SIM-2 — IDENTITY + SHORTCUT MACHINE (bada calculation kabhi nahi)

PEHCHAN – Question mein ye dikhe

same numbers repeat ($a \times a - b \times b$, $a^3 + b^3 / (a^2 - ab + b^2)$), $x : y = 2 : 3$ then $(5x + 3y)/(5x - 3y)$, $5^{n+2} = 3125$, $\sqrt{(56 + \sqrt{(56 + \dots)})}$; सरल कीजिए, अनुपात

MASTER FORMULA – sirf ye yaad karo

$$a^2 - b^2 = (a + b)(a - b) \quad (a \pm b)^2 = a^2 \pm 2ab + b^2 \quad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Same numbers baar-baar	$\frac{a^2-b^2}{a-b} = a + b \cdot (a+b)^2 - (a-b)^2 = 4ab \cdot (a+b)^2 + (a-b)^2 = 2(a^2 + b^2) \cdot$ $\frac{a^3+b^3}{a^2-ab+b^2} = a + b \cdot a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$
② $x : y = a : b$ diya, expression ka maan	Seedha $x = a$, $y = b$ daal do $\cdot px = qy \Rightarrow x : y = q : p$
③ Power equation / infinite root	Same base $\Rightarrow$ powers barabar (IS-1) $\cdot \sqrt{N + \sqrt{N + \dots}} = k$ jahan $k(k-1) = N \cdot$ $\sqrt{N - \sqrt{N - \dots}} = k$ jahan $k(k+1) = N \cdot \sqrt{N \sqrt{N} \dots} = N$

TERMS – Har letter ka matlab

a, b = expression mein baar-baar aane wale numbers.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 84: $(3.7)^3 - 3(3.7)^2(0.7) + 3(3.7)(0.7)^2 - (0.7)^3$
 $(3.7 - 0.7)^3 = 27$ ✓ book ans (b)
- ① EN 98: $73 \times 73 + 42 \times 42 - 2 \times 73 \times 42$
 $(73 - 42)^2 = 961$ ✓ book ans (a)
- ② EN 234: $x : y = 2 : 3 \Rightarrow (5x + 3y) : (5x - 3y)$
 $(10 + 9) : (10 - 9) = 19 : 1$ ✓ book ans (d)
- ② EN 235: $a : b = 3 : 2 \Rightarrow (7a + 9b) : (5a + 7b)$
 $39 : 29$ ✓ book ans (c)
- ③ EN 215: $\sqrt{(56 + \sqrt{(56 + \dots)})}$
 $56 = 8 \times 7 \Rightarrow 8$ ✓ book ans (d)
- ③ EN 221: $\sqrt{(30 - \sqrt{(30 - \dots)})}$
 $30 = 5 \times 6 \Rightarrow 5$ ✓ book ans (a)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (157) EN 20, 59, 82-93, 97-100, 103-105, 107-115, 117, 120, 122, 123, 125, 141, 142, 156-159, 190, 202-209, 211-235, 238, 240-248 · HI 1-3, 6, 11, 13, 14, 18, 29, 53, 77, 92, 107, 114, 123, 126, 131, 140, 142, 160, 166, 168, 181, 190, 197, 198, 200-216, 218-234, 237-243 | Ch 3 (Ghatank - Karani): EN 58 | Ch 22 (Beejganit): EN 264 · HI 43, 45 | Ch 28 (Vividh): EN 21

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
SIM-1	VBODMAS + “?” + APPROX MACHINE	163	176
SIM-2	IDENTITY + SHORTCUT MACHINE (bada calculation kabhi nahi)	85	67
Total (book ke sawaal)		248/248 ✓	243/243 ✓

Chapter 5

LCM & HCF (L.S. – M.S.) लघुत्तम समापवर्त्य एवं महत्तम समापवर्तक

2 formulas (pehle 8) · English book 231 sawaal · Hindi book 126 sawaal

LH-1 — HCF–LCM NIKAALO + LINK MACHINE

PEHCHAN – Question mein ye dikhe

find HCF / LCM, HCF and LCM given find other number, ratio 3 : 4 and HCF / LCM, LCM of fractions / decimals, pairs with given HCF; ल.स., म.स., गुणनफल, अनुपात

MASTER FORMULA – sirf ye yaad karo

HCF = common primes ki **MIN** power LCM = saare primes ki **MAX** power $HCF \times LCM = a \times b$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① HCF / LCM nikaalo	Prime powers $\Rightarrow$ min / max · lagataar ya co-prime $\Rightarrow$ HCF = 1, LCM = guna · $HCF(a^m - 1, a^n - 1) = a^{HCF(m,n)} - 1$
② HCF, LCM, ek number $\Rightarrow$ doosra	$b = \frac{HCF \times LCM}{a}$ (sirf 2 numbers) · LCM = k $\times$ HCF, LCM + HCF = S $\Rightarrow$ HCF = $\frac{S}{k+1} \cdot \frac{1}{a} +$ $\frac{1}{b} = \frac{a+b}{HCF \times LCM}$
③ Ratio a : b diya	Numbers = h a, h b $\Rightarrow$ LCM = h · a · b · h = $\frac{LCM}{LCM(a,b)}$ · pairs (HCF h, LCM L) = 2^{k-1} , k = L/h ke alag primes
④ Fraction / decimal ka HCF–LCM	HCF = $\frac{HCF(\text{upar})}{LCM(\text{neeche})}$ · LCM = $\frac{LCM(\text{upar})}{HCF(\text{neeche})}$ · decimal: $\times 100$ karke nikaalo, phir $\div 100$

TERMS – Har letter ka matlab

h = HCF · a, b = ratio ke co-prime number.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 1: LCM of 98, 28, 112
 $2 \cdot 7^2, 2^2 \cdot 7, 2^4 \cdot 7 \Rightarrow 2^4 \cdot 7^2 = 784$ ✓ book ans (a)
- ① EN 54: x = $2^3 3^2 5 \cdot 7^3$, y = $2^2 3^3 5^2 7^2$, z = $2^4 3 \cdot 5^3 7$ ka HCF
 $2^2 \cdot 3 \cdot 5 \cdot 7 = 420$ ✓ book ans (c)
- ② EN 121: HCF 1080, LCM 30240, ek 4320
 $\frac{1080 \cdot 30240}{4320} = 7560$ ✓ book ans (c)
- ② EN 119: sum 54, LCM 84, HCF 6 $\Rightarrow 1/a + 1/b$
 $\frac{54}{504} = \frac{3}{28}$ ✓ book ans (c)
- ③ EN 122: ratio 5:7, HCF 8 $\Rightarrow$ LCM
 $8 \cdot 5 \cdot 7 = 280$ ✓ book ans (d)
- ③ EN 145: LCM 84, ratio 2:3 $\Rightarrow$ sum
 $h = 84/6 = 14 \Rightarrow 28 + 42 = 70$ ✓ book ans (b)
- ④ EN 106: LCM of 2/3, 4/9, 7/12, 3/5
 $\frac{LCM(2,4,7,3)}{HCF(3,9,12,5)} = \frac{84}{1}$ ✓ book ans (c)
- ④ EN 104: HCF of 2/3, 4/5, 3/2
 $\frac{HCF(2,4,3)}{LCM(3,5,2)} = \frac{1}{30}$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (200) EN 1–4, 6–23, 52, 54–57, 59–61, 67, 68, 70–81, 85, 88, 90, 104–171, 173, 175, 176, 178, 199, 201, 202, 215, 224, 225, 231 · HI 1–5, 7–11, 26, 27, 29–45, 61–104, 126

LH-2 — SAATH KAB / SABSE BADA NAAP MACHINE (application + remainder)

PEHCHAN – Question mein ye dikhe

least number which leaves remainder, greatest number which divides leaving remainders, bells toll together, largest measuring tile / can, minimum number of rooms / rows; घंटियाँ एक साथ, सबसे छोटी संख्या, बड़ी से बड़ी संख्या

MASTER FORMULA – sirf ye yaad karo

Saath / sabse chhota number $\Rightarrow$ LCM Sabse bada naap / sabse bada divisor $\Rightarrow$ HCF

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Sabse chhota N jo sab se bhaag de, shesh R	$N = k \cdot \text{LCM} + R$ ($k = 1$ sabse chhota) $\cdot$ har jagah gap same g ($d - r = g$) $\Rightarrow \text{LCM} - g$ $\cdot$ upar se m se poorate kate $\Rightarrow k = 1, 2, 3 \dots$ try
② Sabse bada number jo bhaag de, shesh r_1, r_2	$\text{HCF}(a - r_1, b - r_2, c - r_3) \cdot$ same (anjaan) shesh $\Rightarrow \text{HCF}(a - b , b - c)$
③ Sabse bada naap / kam se kam hisse / tiles	Naap = $\text{HCF} \cdot$ hisse = $\frac{q_1 + q_2 + q_3}{\text{HCF}}$ $\cdot$ square tiles = $\frac{L \times B}{\text{HCF}(L, B)^2}$
④ Ghantiyan / lights phir saath kab	Start + $\text{LCM} \cdot T$ samay mein $\left\lfloor \frac{T}{\text{LCM}} \right\rfloor$ (+1 agar shuru wala bhi gina)

TERMS – Har letter ka matlab

R = har jagah bachne wala shesh $\cdot$ q = quantities $\cdot$ T = kul samay.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 27:** 9, 8, 10, 12 se rem 3
LCM = 360 $\Rightarrow$ 363 ✓ **book ans (c)**
- ① **EN 39:** 10, 9, 8 se rem 9, 8, 7
gap 1 $\Rightarrow 360 - 1 = 359$ ✓ **book ans (d)**
- ② **EN 91:** 2000, 2200 ko rem 22, 38
 $\text{HCF}(1978, 2162) = 46$ ✓ **book ans (a)**
- ② **EN 97:** 2270, 3739, 6677 same remainder
 $\text{HCF}(1469, 2938) = 1469$ ✓ **book ans (d)**
- ③ **EN 63:** 72, 90, 144 L ke liye bada can
 $\text{HCF} = 18$ L ✓ **book ans (b)**
- ③ **EN 83:** 188, 282, 423 students: min sections
 $\text{HCF} 47 \Rightarrow 4 + 6 + 9 = 19$ ✓ **book ans (c)**
- ④ **EN 183:** 3, 6, 12, 15, 18 s; 9:58:45 par saath $\Rightarrow$ agla?
LCM 180 s = 3 min $\Rightarrow$ 10:01:45 ✓ **book ans (b)**
- ④ **EN 193:** 16, 24, 36, 42 min
LCM = 1008 min ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (158) EN 5, 24–51, 53, 58, 62–66, 69, 82–84, 86, 87, 89, 91–103, 177, 179–198, 200, 203–214, 216–221, 223, 226–230 $\cdot$ HI 6, 12–25, 28, 46–60, 106–125 | Ch 1 (Sankhya Paddhati): EN 154 $\cdot$ HI 153, 155 | Ch 28 (Vividh): EN 76 $\cdot$ HI 59

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
LH-1	HCF–LCM NIKAALO + LINK MACHINE	126	74
LH-2	SAATH KAB / SABSE BADA NAAP MACHINE (application + remainder)	102	51
$\rightarrow$ NS-1	BHAAG MACHINE (divisible $\cdot$ remainder $\cdot$ unit digit) (Ch 1)	1	0
$\rightarrow$ NS-2	GINTI + FACTOR MACHINE (kitne numbers $\cdot$ kitne fact) (Ch 1)	1	0
$\rightarrow$ NS-3	DO-NUMBER + SERIES MACHINE (Ch 1)	1	1
Total (book ke sawaal)		231/231 ✓	126/126 ✓
$\rightarrow$ NS-1: EN 222 $\cdot$ $\rightarrow$ NS-2: EN 174 $\cdot$ $\rightarrow$ NS-3: EN 172 HI 105			

Chapter 6

Percentage (Pratishatata) प्रतिशतता

2 formulas (pehle 9) · English book 247 sawaal · Hindi book 117 sawaal

PC-1 — PERCENT = HISSA MACHINE (of · marks · income · votes)

PEHCHAN – Question mein ye dikhe

x% of y, what percent, 27.5% of a number is 11, exam pass / fail marks, income–expenditure–savings, election votes, boys / girls %; प्रतिशत, अंक, आय, बचत, मत

MASTER FORMULA – sirf ye yaad karo

$$x\% \text{ of } Y = \frac{x \times Y}{100} \quad Y = \frac{\text{Value} \times 100}{x} \quad A \text{ is } \left(\frac{A}{B} \times 100 \right) \% \text{ of } B$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo ⇒ answer
① x% of, kitna %, poora number	Master formula · a% of A = b% of B ⇒ A : B = b : a · $\frac{1}{8} = 12.5\%$, $\frac{1}{6} = 16\frac{2}{3}\%$, $\frac{1}{9} = 11\frac{1}{9}\%$
② Exam: pass / fail by	Do students: Max = $\frac{(\text{fail by}) + (\text{extra})}{p_2 - p_1} \times 100$ · Ek: Max = $\frac{(\text{mila} + \text{fail by}) \times 100}{\text{pass}\%}$
③ Income – kharcha – bachat	Income = $\frac{\text{Bachat} \times 100}{\text{bachat}\%}$ · chain: bacha = $I(1 - \frac{a}{100})(1 - \frac{b}{100})$ ⇒ ulta ÷ · change wale: Income = 100 maan lo
④ Election / votes	Margin = $(W\% - L\%) \times \text{Valid} \cdot \text{winner} = \text{Total} \times \text{vote}\% \times \text{valid}\% \times W\%$
⑤ Do group ka milke %	$\frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ · ratio chahiye ⇒ alligation $\frac{n_1}{n_2} = \frac{p_2 - p}{p - p_1}$

TERMS – Har letter ka matlab

Y = poora (whole) · x = percent · W, L = jeetne / haarne wale ka % · p = percent.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 85:** 27.5% of number = 11
 $\frac{11 \times 100}{27.5} = 40$ ✓ **book ans (c)**
- ① **EN 87:** k% of 280 = 50% of 350
 $k = \frac{175 \times 100}{280} = 62.5$ ✓ **book ans (c)**
- ② **EN 1:** 45% fail by 18; 65% ⇒ 6 zyada
 $\frac{18 + 6}{20} \times 100 = 120$ ✓ **book ans (c)**
- ② **EN 35:** 40% chahiye; 70 mile, 20 se fail
 $\frac{90 \times 100}{40} = 225$ ✓ **book ans (d)**
- ③ **EN 62:** 80% kharch, ₹5000 bachat
 $\frac{5000 \times 100}{20} = 25000$ ✓ **book ans (a)**
- ③ **EN 63:** 15% rent, baaki ka 20%, phir 10% ⇒ ₹42,840 bache
 $\frac{42840}{0.85 \times 0.8 \times 0.9} = 70000$ ✓ **book ans (d)**
- ④ **EN 239:** 37% wala 5,20,000 se haara
 $(63 - 37)\% = 26\% \Rightarrow \frac{520000}{0.26} = 20,00,000$ ✓ **book ans (b)**
- ④ **EN 238:** 90% voted, 80% valid, winner 60% = 64800
 $\frac{64800}{0.9 \times 0.8 \times 0.6} = 150000$ ✓ **book ans (a)**
- ⑤ **EN 33:** boys:girls 3:2; 20%, 25% scholarship ⇒ NOT
 $\frac{60 + 50}{5} = 22\% \Rightarrow 78\%$ ✓ **book ans (a)**
- ⑤ **EN 10:** 10,000; male +5%, female +10% ⇒ 10,800; females?
 total +8% ⇒ F : M = 3 : 2 ⇒ 6000 ✓ **book ans (b)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (228) EN 1, 10, 13, 17, 20, 23, 25–63, 65, 66, 68, 69, 71–82, 84–88, 90–93, 95–100, 102–110, 112–117, 119–124, 126–130, 132–134, 165, 167–170, 172–174, 176, 178, 182–195, 200–203, 208, 209, 214, 217, 219–221, 238–247 · **HI** 15, 18, 21–36, 38, 39, 41–50, 52–67, 70, 71, 73, 74, 76, 78–82, 96, 97, 99, 100, 102, 105–109, 116, 117 | Ch 1 (Sankhya Paddhati): **EN** 194, 196 | Ch 8 (Batta / Chhoot): **EN** 49 · **HI** 74 | Ch 12 (Mishran / Alligation): **EN** 23 · **HI** 7 | Ch 28 (Vividh): **EN** 43, 53, 64 · **HI** 50

PC-2 — MULTIPLIER MACHINE (badha · ghata · do baar · population)**PEHCHAN – Question mein ye dikhe**

increased / decreased by $x\%$, becomes ... after increase, successive changes, population grows $r\%$ per year, price up so consumption down, A is 20% less than B; वृद्धि, कमी, जनसंख्या, खपत

MASTER FORMULA – sirf ye yaad karo

$$\text{Naya} = \text{Purana} \times \frac{100 \pm x}{100} \quad \text{Net}\% = a + b + \frac{a \times b}{100}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Badha / ghata / ulta / % change	$\text{Naya} = \text{purana} \times M \cdot \text{Purana} = \text{naya} \div M \cdot \% \text{ change} = \frac{\text{Naya}-\text{Purana}}{\text{Purana}} \times 100$
② Do / teen change ek saath	$\text{Net}\% = a + b + \frac{ab}{100}$ (ghata $\Rightarrow -$) · square / circle area = $2a + \frac{a^2}{100}$ · +x phir -x $\Rightarrow -\frac{x^2}{100}\%$
③ Population / growth n saal	$P \left(1 + \frac{r}{100}\right)^n$ · pehle ki $\Rightarrow \div$ · alag rate $\Rightarrow P \left(1 + \frac{a}{100}\right) \left(1 - \frac{b}{100}\right) \dots$
④ Uta % / consumption	$x\%$ zyada $\Rightarrow$ ulta $\frac{x}{100+x} \times 100\%$ kam · $x\%$ kam $\Rightarrow$ ulta $\frac{x}{100-x} \times 100\%$ zyada · price + $r\%$, kharcha same $\Rightarrow$ consumption $\frac{r}{100+r} \times 100\%$ kam

TERMS – Har letter ka matlab

M = multiplier = $(100 \pm x)/100$ · a, b = do changes (ghate to minus) · r = har saal ka rate · n = saal.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 131:** 45% badhne par 725
 $725 \times \frac{100}{145} = 500$ ✓ **book ans (a)**
- ① **EN 83:** 30% increment ke baad ₹910
 $910 \times \frac{100}{130} = 700$ ✓ **book ans (c)**
- ② **EN 137:** +10% phir +8%
 $10 + 8 + 0.8 = 18.8\%$ ✓ **book ans (c)**
- ② **EN 140:** +40% phir -40%
 $-\frac{40^2}{100} = -16\%$ ✓ **book ans (d)**
- ③ **EN 4:** 2 saal pehle 10,000; 10% growth $\Rightarrow$ ab
 $10000 \times 1.1^2 = 12100$ ✓ **book ans (a)**
- ③ **EN 3:** 15625; +8%, +12%, -22%
 $15625 \times 1.08 \times 1.12 \times 0.78 = 14742$ ✓ **book ans (c)**
- ④ **EN 197:** A, B se 20% kam $\Rightarrow$ B, A se kitna zyada
 $\frac{20}{80} \times 100 = 25\%$ ✓ **book ans (b)**
- ④ **EN 210:** tea price +20%, budget same $\Rightarrow$ consumption
 $\frac{20}{120} \times 100 = 16\frac{2}{3}\%$ ✓ **book ans (b)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (149) EN 2–9, 11, 12, 14–16, 18, 19, 21, 22, 24, 64, 67, 70, 83, 89, 94, 101, 111, 118, 125, 131, 135–161, 163, 164, 166, 175, 179, 181, 196–199, 204–207, 210–213, 215, 216, 222–237 · **HI** 1–14, 16, 17, 19, 20, 37, 40, 51, 68, 72, 75, 77, 83–95, 101, 104, 110–115 | *Ch 1 (Sankhya Paddhati):* EN 193, 334 | *Ch 7 (Labh aur Hani):* **HI** 56, 57 | *Ch 8 (Batta / Chhoot):* EN 24, 27, 35 | *Ch 10 (Saajhedaari):* EN 15 · **HI** 4 | *Ch 15 (Chakraviddhi Byaj):* EN 204 | *Ch 28 (Vividh):* EN 39

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
PC-1	PERCENT = HISSA MACHINE (of · marks · income · votes)	150	68
PC-2	MULTIPLIER MACHINE (badha · ghata · do baar · population)	92	46
$\rightarrow$ DF-2	COMPARE + OPERATE + “HISSA SE POORA” MACHINE (Ch 2)	1	0
$\rightarrow$ PL-1	CP $\rightarrow$ SP MULTIPLIER MACHINE (profit · loss · MP · “ (Ch 7)	2	1
$\rightarrow$ PL-2	GINTI · RATE · NAKLI BAAT MACHINE (Ch 7)	1	2
$\rightarrow$ SI-1	P · R · T MACHINE (simple interest) (Ch 14)	1	0
Total (book ke sawaal)		247/247 ✓	117/117 ✓
$\rightarrow$ DF-2: EN 162 · $\rightarrow$ PL-1: EN 177, 218 HI 103 · $\rightarrow$ PL-2: EN 171 HI 69, 98 · $\rightarrow$ SI-1: EN 180			

Chapter 7

Profit & Loss (Labh aur Hani) लाभ और हानि

2 formulas (pehle 10) · English book 168 sawaal · Hindi book 114 sawaal

PL-1 — CP → SP MULTIPLIER MACHINE (profit · loss · MP · “100 maan lo”)

PEHCHAN – Question mein ye dikhe

cost price / selling price, gain / loss %, sold at ₹... with x% profit, find CP, had he sold for ₹... more, marked price + discount + profit, same SP two articles; क्रय मूल्य, विक्रय मूल्य, लाभ, हानि, अंकित मूल्य

MASTER FORMULA – sirf ye yaad karo

$$SP = CP \times \frac{100 \pm P}{100} \quad P\% = \frac{SP - CP}{CP} \times 100 \quad \frac{MP}{CP} = \frac{100 + P}{100 - D}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo ⇒ answer
① CP, SP diye ⇒ P/L %	$P\% = \frac{SP - CP}{CP} \times 100$ · repair / bhaada / tax ⇒ CP mein jodo · profit = SP ka $\frac{a}{b} \Rightarrow P\% = \frac{a}{b-a} \times 100$
② Ek se doosra (CP ↔ SP), SP badla, chain A → B → C	$CP = SP \times \frac{100}{100 \pm P} \cdot SP_2 = SP_1 \times \frac{100 \pm P_2}{100 \pm P_1}$ · chain: $CP(1 \pm \frac{a}{100})(1 \pm \frac{b}{100})$
③ ₹ farak ↔ % farak	$CP = \frac{(\text{rupaye ka farak}) \times 100}{(\% \text{ ka farak})}$ (profit +, loss -: 12% gain vs 4% loss ⇒ 16) · S ₁ pe profit = S ₂ pe loss ⇒ $CP = \frac{S_1 + S_2}{2}$
④ Do cheezein, same SP, x% profit + x% loss	Hamesha loss = $\frac{x^2}{100} \%$ · a% profit, b% loss ⇒ $CP_1 : CP_2 = (100 - b) : (100 + a)$
⑤ MP, discount, profit ek saath	$\frac{MP}{CP} = \frac{100 + P}{100 - D}$ · MP = CP + x% ⇒ M = 100 + x ⇒ $P\% = \frac{M(100 - D)}{100} - 100$
⑥ Shart wale (SP double to profit triple ...)	CP = 100 maan lo ⇒ SP = 100 ± P ⇒ jo badla wo badlo ⇒ naya P% · profit SP ka r% ⇒ CP par $\frac{r}{100 - r} \times 100\%$

TERMS – Har letter ka matlab

CP = khareed · SP = bikri · P = profit % (loss ho to -) · MP = likha hua daam · D = discount %.

SOLVED – Asli book questions (row ke saath, verified)

① EN 1: CP ₹4500, SP ₹3500 ⇒ loss %

$$\frac{1000}{4500} \times 100 = 22\frac{2}{9}\% \checkmark \text{ book ans (c)}$$

① EN 2: ₹222 mein becha, ₹48 loss

$$CP = 270 \Rightarrow \frac{48}{270} \times 100 = 17\frac{7}{9}\% \checkmark \text{ book ans (a)}$$

② EN 35: ₹207 pe 28% loss ⇒ CP

$$207 \times \frac{100}{72} = 287.50 \checkmark \text{ book ans (d)}$$

② EN 69: ₹4125 pe 10% gain ⇒ 18% ke liye

$$4125 \times \frac{118}{110} = 4425 \checkmark \text{ book ans (b)}$$

③ EN 34: 12% gain aur 4% loss ka farak ₹28

$$\frac{28 \times 100}{16} = 175 \checkmark \text{ book ans (a)}$$

③ EN 44: ₹33 zyada ⇒ 12% se 14%

$$\frac{33 \times 100}{2} = 1650 \checkmark \text{ book ans (c)}$$

④ EN 119: ₹990 each; 10% gain, 10% loss

$$\frac{10^2}{100} = 1\% \text{ loss } \checkmark \text{ book ans (c)}$$

④ EN 120: ₹8,25,000 each; ±9%

$$\frac{81}{100} = 0.81\% \text{ loss } \checkmark \text{ book ans (b)}$$

⑤ EN 158: 20% discount ke baad bhi 8% profit; CP ₹500 ⇒ MP

$$500 \times \frac{108}{80} = 675 \checkmark \text{ book ans (d)}$$

⑥ EN 4: 74% profit; CP +50%, SP same

$$\frac{174 - 150}{150} = 16\% \checkmark \text{ book ans (b)}$$

IS FORMULA SE HONE WALE BOOK KE SAWAAL (176) EN 1, 2, 4-8, 11, 13-15, 17-20, 22-32, 34-37, 39, 40, 42-45, 47, 50, 52, 53, 55, 57, 60-81, 83-95, 97, 98, 100, 102-107, 109, 111, 112, 114-117, 119, 120, 122, 124, 125, 131, 132, 141, 158-162, 164, 167 · **HI** 1, 3, 4, 6, 20-22, 25, 26, 29, 30, 34, 36-38, 47, 50-52, 55, 58-62, 65-70, 72, 74-81, 85-87, 89-101, 104, 107, 108, 110, 111, 113 | *Ch 6 (Pratishatata)*: EN 177, 218 · **HI** 103 | *Ch 8 (Batta / Chhoot)*: EN 44 · **HI** 71 | *Ch 28 (Vividh)*: EN 65

PL-2 — GINTI · RATE · NAKLI BAAT MACHINE

PEHCHAN – Question mein ye dikhe

SP of 20 articles = CP of 25, bought 6 for ₹5 and sold 5 for ₹6, uses false weight 900 g for 1 kg, part sold at x% profit and rest at y% loss, overall gain; वस्तुओं का क्रय मूल्य, दर, गलत बाट

MASTER FORMULA – sirf ye yaad karo

$$P\% = \left(\frac{\text{Jitna milna chahiye}}{\text{Jitna diya}} - 1 \right) \times 100 \quad \Rightarrow \quad \text{SP of } x = \text{CP of } y : P\% = \frac{y - x}{x} \times 100$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① x cheezon ka SP = y cheezon ka CP	$P\% = \frac{y-x}{x} \times 100$ (jitni bechi woh neeche) · n bechkar k ka SP profit $\Rightarrow \frac{k}{n-k} \times 100$, k ka CP profit $\Rightarrow \frac{k}{n} \times 100$
② a cheezein ₹X mein khareedi, b ₹Y mein bechi	$P\% = \left(\frac{a \times Y}{b \times X} - 1 \right) \times 100$ (cross guna) · ₹1 mein n bechkar P% $\Rightarrow$ khareedi thi $n \times \frac{100+P}{100}$
③ Nakli baat (false weight)	$P\% = \frac{\text{Asli} - \text{Nakli}}{\text{Nakli}} \times 100$ · saath mein markup p% $\Rightarrow \left(\frac{100+p}{100} \times \frac{\text{Asli}}{\text{Nakli}} - 1 \right) \times 100$
④ Hissa x% pe, baaki y% pe $\Rightarrow$ overall	Overall = $\frac{w_1 p_1 + w_2 p_2}{w_1 + w_2}$ (loss = -) · baaki ka % = $\frac{T \cdot W - w_1 p_1}{w_2}$

TERMS – Har letter ka matlab

x, y = cheezon ki ginti · Asli = jitna dena tha (1000 g) · Nakli = jitna diya (900 g) · w = CP ka hissa · T = chahiye overall %.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 3:** 32 ka SP = 38 ka CP
 $\frac{6}{32} \times 100 = 18.75\%$ ✓ **book ans (c)**
- ① **EN 12:** 9 ka SP = 15 ka CP
 $\frac{6}{9} \times 100 = 66.6\%$ gain ✓ **book ans (b)**
- ② **EN 33:** 8 for ₹5 khareed, 5 for ₹8 bech
 $\frac{8 \times 8}{5 \times 5} = 2.56 \Rightarrow 156\%$ ✓ **book ans (c)**
- ② **EN 137:** 6 for ₹10 khareed, 4 for ₹6 bech
 $\frac{6 \times 6}{4 \times 10} = 0.9 \Rightarrow 10\%$ loss ✓ **book ans (b)**
- ③ **EN 9:** 1 kg ki jagah 800 g
 $\frac{200}{800} \times 100 = 25\%$ ✓ **book ans (a)**
- ③ **EN 130:** 25% gain $\Rightarrow$ 1 kg ki jagah kitna
 $\frac{1000 \times 100}{125} = 800$ g ✓ **book ans (a)**
- ④ **EN 101:** ₹15,000; 45% pe 40% gain, 25% pe 10% loss, baaki CP pe
 $0.45 \times 40 - 0.25 \times 10 = 15.5\% \Rightarrow 2325$ ✓ **book ans (b)**
- ④ **EN 59:** 2/3 pe 6% profit, baaki 3% loss, profit ₹540
 $4 - 1 = 3\% \Rightarrow \frac{540}{0.03} = 18000$ ✓ **book ans (b)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (109) EN 3, 9, 10, 12, 16, 21, 33, 38, 46, 48, 49, 51, 54, 56, 58, 59, 82, 96, 99, 101, 110, 113, 121, 123, 126-130, 133-140, 142-153, 155-157, 163, 165, 166, 168 · **HI** 2, 5, 7-19, 23, 24, 27, 28, 31-33, 35, 39-46, 48, 49, 53, 54, 63, 71, 73, 82-84, 88, 102, 103, 105, 106, 109, 112, 114 | *Ch 6 (Pratishatata)*: EN 171 · **HI** 69, 98 | *Ch 12 (Mishran / Alligation)*: EN 47

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
PL-1	CP → SP MULTIPLIER MACHINE (profit · loss · MP · “100 maan lo”	108	62
PL-2	GINTI · RATE · NAKLI BAAT MACHINE	56	49
→ NS-3	DO-NUMBER + SERIES MACHINE (Ch 1)	4	0
→ PC-2	MULTIPLIER MACHINE (badha · ghata · do baar · popu (Ch 6)	0	2
→ RP-2	CROSS-GUNA MACHINE (proportion · ratio badlo) (Ch 9)	0	1
Total (book ke sawaal)		168/168 ✓	114/114 ✓
→ NS-3: EN 41, 108, 118, 154 · → PC-2: HI 56, 57 · → RP-2: HI 64			

Chapter 8

Discount (Batta / Chhoot) बट्टा / छूट

1 formula (pehle 4) · English book 76 sawaal · Hindi book 79 sawaal

DS-1 — DISCOUNT MULTIPLIER MACHINE

PEHCHAN – Question mein ye dikhe

marked price / list price, discount %, successive discounts 20% and 10%, single equivalent discount, mark up and still gain, buy 3 get 1 free, GST / tax after discount; अंकित मूल्य, छूट, क्रमागत छूट

MASTER FORMULA – sirf ye yaad karo

$$SP = MP \times \frac{100 - d}{100} \quad \text{Do discount} = a + b - \frac{a \times b}{100} \quad \frac{MP}{CP} = \frac{100 + P}{100 - D}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo ⇒ answer
① MP, SP, discount (tax bhi)	$d\% = \frac{MP - SP}{MP} \times 100 \cdot MP = SP \times \frac{100}{100 - d} \cdot \text{bill} = MP(1 - \frac{d}{100})(1 + \frac{t}{100})$
② Successive discounts	$a + b - \frac{ab}{100} \cdot \text{teen} \Rightarrow [1 - (1 - \frac{a}{100})(1 - \frac{b}{100})(1 - \frac{c}{100})] \times 100 \cdot \text{compare} \Rightarrow (100 - a)(100 - b) \text{ jiska chhota woh sasta}$
③ MP, discount, profit chain	$\frac{MP}{CP} = \frac{100 + P}{100 - D} \cdot CP = 100 \text{ maan lo} \Rightarrow D\% = \frac{M - (100 + P)}{M} \times 100$
④ Buy x get y free	$\text{Discount} = \frac{y}{x + y} \times 100\%$

TERMS – Har letter ka matlab

MP = likha daam · SP = diya daam · d / D = discount % · t = tax % · P = profit %.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 11: MP ₹2850, 21% discount
 $2850 \times 0.79 = 2251.50$ ✓ book ans (d)
- ① EN 13: 25% discount ke baad ₹28,473 diye
 $\frac{28473}{0.75} = 37964$ ✓ book ans (d)
- ② EN 2: 10%, 15%, 20% ka single
 $1 - 0.9 \times 0.85 \times 0.8 = 0.388 \Rightarrow 38.8\%$ ✓ book ans (d)
- ② EN 9: 30%, 20%, 10% ka single
 $1 - 0.7 \times 0.8 \times 0.9 = 49.6\%$ ✓ book ans (c)
- ③ EN 15: CP ₹360, 10% discount, 25% gain ⇒ MP
 $360 \times \frac{125}{90} = 500$ ✓ book ans (c)
- ③ EN 17: 5% discount, 10% profit ⇒ MP kitna upar
 $\frac{110}{95} = 1.1579 \Rightarrow 15\frac{15}{19}\%$ ✓ book ans (d)
- ④ HI 52: 5 khareedo 3 free
 $\frac{3}{8} \times 100 = 37.5\%$ ✓ book ans (a)
- ④ HI 54: 3 saree par 2 free
 $\frac{2}{5} \times 100 = 40\%$ ✓ book ans (c)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (149) EN 1–23, 25, 26, 28–34, 36–43, 45–48, 50–76 · HI 1–70, 72, 73, 75–79 | Ch 28 (Vividh): EN 79

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
DS-1	DISCOUNT MULTIPLIER MACHINE	71	77
→ PC-1	PERCENT = HISSA MACHINE (of · marks · income · vot (Ch 6)	1	1
→ PC-2	MULTIPLIER MACHINE (badha · ghata · do baar · popu (Ch 6)	3	0
→ PL-1	CP → SP MULTIPLIER MACHINE (profit · loss · MP · “ (Ch 7)	1	1
Total (book ke sawaal)		76/76 ✓	79/79 ✓
→ PC-1 : EN 49 HI 74 · → PC-2 : EN 24, 27, 35 · → PL-1 : EN 44 HI 71			

Chapter 9

Ratio & Proportion (Anupat aur Samanupat) अनुपात और समानुपात

2 formulas (pehle 8) · English book 178 sawaal · Hindi book 162 sawaal

RP-1 — “1 HISSA” MACHINE (share · jodna · coins · bachat)

PEHCHAN – Question mein ye dikhe

divide ₹... in the ratio, difference of shares, A : B and B : C find A : B : C, x% of A = y% of B, coins of 1 ₹, 50 p, 25 p in ratio, income ratio and expenditure ratio; अनुपात में बाँटो, सिक्के, आय, व्यय

MASTER FORMULA – sirf ye yaad karo

$$1 \text{ hissa} = \frac{\text{Di hui value}}{\text{uske parts}} \Rightarrow \text{Hissa} = \text{Total} \times \frac{a}{a+b+c}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Baantna / ek hissa / farak	$\text{Hissa} = T \times \frac{a}{a+b+c} \cdot 1 \text{ hissa} = \frac{\text{farak}}{a-b} \cdot \text{jod S, farak D} \Rightarrow (S+D) : (S-D)$
② A : B, B : C $\Rightarrow$ A : B : C	$A : B : C = ac : bc : bd$ (beech wale ko barabar karo) $\cdot A : C = ac : bd$
③ x% of A = y% of B, $\frac{1}{2}A = \frac{1}{3}B$	$x\%A = y\%B \Rightarrow A : B = y : x$ (ulta) $\cdot \frac{A}{p} = \frac{B}{q} \Rightarrow A : B = p : q \cdot pA = qB = rC \Rightarrow \frac{1}{p} : \frac{1}{q} : \frac{1}{r}$
④ Coins / notes	$k = \frac{\text{Kul rupaye}}{av_1+bv_2+cv_3} \Rightarrow \text{coins} = ak, bk, ck$ (50 p = 0.5)
⑤ Income a : b, kharcha c : d, bachat diya	$k = \frac{S(d-c)}{ad-bc} \Rightarrow \text{income} = ak, bk$ (barabar bachat S $\Rightarrow k = \frac{S(d-c)}{ad-bc}$)

TERMS – Har letter ka matlab

T = total · a, b, c = ratio ke parts · k = 1 hissa · v = coin ki value · S = bachat.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 2: ₹1470 ko 3 : 4, Mohan (4)
 $1470 \times \frac{4}{7} = 840$ ✓ book ans (d)
- ① EN 5: 4 : 3, y ka ₹2400 $\Rightarrow$ total
 $\frac{2400}{3} \times 7 = 5600$ ✓ book ans (c)
- ② EN 28: a : b = 32 : 35, b : c = 21 : 32
 $a : c = 32 \cdot 21 : 35 \cdot 32 = 3 : 5$ ✓ book ans (c)
- ② EN 26: R : G = 15 : 32, P : R = 18 : 25 $\Rightarrow$ G : P
 $R = 75 \Rightarrow G \text{ 160, P 54} \Rightarrow 80 : 27$ ✓ book ans (a)
- ③ EN 7: boys ka 15% = girls ka $\frac{1}{3}$
 $B : G = \frac{1}{3} : 0.15 = 20 : 9$ ✓ book ans (a)
- ③ EN 20: pehle ka 25% = doosre ke 50% ka 3 guna
 $0.25A = 1.5B \Rightarrow 6 : 1$ ✓ book ans (b)
- ④ EN 147: ₹2, ₹5, ₹10 = 5 : 7 : 8, total ₹1250 $\Rightarrow$ ₹5 coins
 $k = \frac{1250}{125} = 10 \Rightarrow 70$ ✓ book ans (c)
- ④ EN 152: 25p, 10p, 5p = 1 : 2 : 3, ₹30 $\Rightarrow$ 5p
 $k = \frac{30}{0.6} = 50 \Rightarrow 150$ ✓ book ans (c)
- ⑤ EN 12: income 7 : 5, exp 9 : 7, bachat 1700, 1100
 $k = \frac{11900-9900}{49-45} = 500 \Rightarrow 3500, 2500$ ✓ book ans (d)
- ⑤ EN 163: income 5 : 7, savings 2 : 3, kharcha 35,400 & 48,600
 $k = \frac{106200-97200}{15-14} = 9000 \Rightarrow 18000$ ✓ book ans (a)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (185) EN 1, 2, 4, 5, 7–16, 20, 22, 24–69, 71–75, 78, 83, 86, 147, 148, 150–164, 166, 167, 172, 174–178 · HI 1–3, 5–9, 11–14, 16–54, 56–60, 141–157, 160, 162 | Ch 10 (Saajhedaari): EN 2–5, 7, 9–12, 23–25 | Ch 11 (Samay aur Karya): EN 189 | Ch 28 (Vividh): EN 26, 51

RP-2 — CROSS-GUNA MACHINE (proportion • ratio badlo)

PEHCHAN – Question mein ye dikhe

fourth / third / mean proportional, $a : b :: c : d$, what number added to each to make ratio, ratio becomes after adding, $(x + y)/(x - y) = \dots$, if $x/y = \dots$ then find; समानुपात, मध्यानुपाती, अनुपात बदल

MASTER FORMULA – sirf ye yaad karo

$$\frac{a}{b} = \frac{c}{d} \iff a \times d = b \times c$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Kuch joda / ghataya $\Rightarrow$ naya ratio	$\frac{ak+x}{bk+y} = \frac{c}{d} \Rightarrow k = \frac{cy-dx}{ad-bc}$ • dono mein same $x \Rightarrow k = \frac{x(c-d)}{ad-bc}$ • $a : b \rightarrow c : d$ ke liye jodo $x = \frac{bc-ad}{d-c}$
② Proportional	4th $= \frac{bc}{a} \cdot 3rd = \frac{b^2}{a}$ • mean $= \sqrt{ab}$ • duplicate $a^2 : b^2$, sub-duplicate $\sqrt{a} : \sqrt{b}$
③ Ratio algebra	$\frac{x+y}{x-y} = \frac{p}{q} \Rightarrow \frac{x}{y} = \frac{p+q}{p-q}$ • $x : y = a : b \Rightarrow$ seedha $x = a$, $y = b$ daalo

TERMS – Har letter ka matlab

$a : b :: c : d$ — a, d = bahar wale (extremes), b, c = beech wale (means) • $k = 1$ hissa.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 79:** $2 : 3$, dono mein 4 $\Rightarrow 7 : 10$; farak
 $k = \frac{4(7-10)}{20-21} = 12 \Rightarrow 12$ ✓ **book ans (c)**
- ① **EN 82:** $7 : 11$ mein kya jodein ki $4 : 5$
 $x = \frac{44-35}{1} = 9$ ✓ **book ans (b)**
- ② **EN 118:** 10 aur 50 ka third proportional
 $\frac{50^2}{10} = 250$ ✓ **book ans (c)**
- ② **EN 126:** 15, 10, 12 ka fourth proportional
 $\frac{10 \times 12}{15} = 8$ ✓ **book ans (d)**
- ③ **EN 19:** $(m + n) : (m - n) = 7 : 3 \Rightarrow (m^3 + n^3) : (m^3 - n^3)$
 $m : n = 10 : 4 = 5 : 2 \Rightarrow 133 : 117$ ✓ **book ans (a)**
- ③ **EN 21:** $(3x + 2y) : (3x - 2y) = 5 : 3 \Rightarrow x : y$
 $3x : 2y = 8 : 2 \Rightarrow x : y = 8 : 3$ ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (174) EN 3, 6, 17–19, 21, 23, 70, 76, 77, 79–82, 84, 85, 87–146, 165, 168, 170, 171, 173 • HI 10, 15, 55, 61–140, 158, 159, 161 | Ch 7 (Labb aur Hani): HI 64 | Ch 10 (Saajhedaari): EN 13 | Ch 11 (Samay aur Karya): EN 205 | Ch 26 (Sankhyiki aur Prayikta): EN 212 • HI 92 | Ch 28 (Vividh): EN 70 • HI 55

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
RP-1	“1 HISSA” MACHINE (share • jodna • coins • bachat)	95	75
RP-2	CROSS-GUNA MACHINE (proportion • ratio badlo)	81	86
$\rightarrow$ NS-3	DO-NUMBER + SERIES MACHINE (Ch 1)	1	1
$\rightarrow$ DF-2	COMPARE + OPERATE + “HISSA SE POORA” MACHINE (Ch 2)	1	0
Total (book ke sawaal)		178/178 ✓	162/162 ✓
$\rightarrow$ DF-2: EN 169 • $\rightarrow$ NS-3: EN 149 HI 4			

Chapter 10

Partnership (Saajhedaari) साझेदारी

1 formula (pehle 2) · English book 54 sawaal · Hindi book 25 sawaal

PT-1 — PROFIT $\propto$ PAISA $\times$ SAMAY MACHINE

PEHCHAN – Question mein ye dikhe

A and B invest / started a business, profit share, joined after m months, withdrew, share of A, capital of B; साझेदारी, लाभ में हिस्सा, निवेश

MASTER FORMULA – sirf ye yaad karo

$$P_A : P_B : P_C = C_A t_A : C_B t_B : C_C t_C \quad \text{Hissa} = \text{Profit} \times \frac{C_i t_i}{\sum C t}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Profit baanto	Har partner: paisa $\times$ mahine (beech mein badla $\Rightarrow$ tukde jodo) $\cdot$ join after m mahine $\Rightarrow$ $t = 12 - m \cdot$ pehle salary / x% alag, baaki ratio se
② Anjaan paisa / samay	$C_B = \frac{C_A t_A \cdot P_B}{P_A \cdot t_B} \cdot t_B = \frac{C_A t_A \cdot P_B}{P_A \cdot C_B} \Rightarrow$ join after $12 - t_B$

TERMS – Har letter ka matlab

C = capital (paisa) $\cdot$ t = kitne mahine laga $\cdot$ P = profit ka hissa.

SOLVED – Asli book questions (row ke saath, verified)

① EN 26: ₹4000, ₹8000, ₹6000; Sharma 6 mahine baad gaya; 8 mahine ka ₹34,000
 $24 : 64 : 48 \Rightarrow 34000 \times \frac{64}{136} = 16000$ ✓ book ans (d)

① EN 30: ₹20,000 & ₹30,000; Hareesh ne 3 mahine baad ₹5000 nikale
 $240000 : (90000 + 225000) = 16 : 21$ ✓ book ans (a)

① EN 38: ₹12,000, ₹6000; C ₹15,000 8 mahine baad; 2 saal ka ₹33,600
 $288 : 144 : 240 \Rightarrow 33600 \times \frac{240}{672} = 12000$ ✓ book ans (b)

② EN 16: P ₹3500 $\times$ 12; Q 5 mahine baad; Q ka profit 50% zyada
 $C_Q = \frac{42000 \times 1.5}{7} = 9000$ ✓ book ans (c)

② EN 14: A ₹28,000 $\times$ 12; B 5 mahine baad, C 7 baad; 4 : 2 : 3
 $B = \frac{336000 \times 2}{4 \times 7} = 24000$, $C = 50400$ ✓ book ans (a)

② HI 16: A ₹7250 $\times$ 12; B 3 mahine baad; 8 : 9
 $B = \frac{87000 \times 9}{8 \times 9} = 10875$ ✓ book ans (a)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (59) EN 1, 6, 8, 14, 16–19, 26–54 · HI 1–3, 6–17, 19–25

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
PT-1	PROFIT $\propto$ PAISA $\times$ SAMAY MACHINE	37	22
$\rightarrow$ NS-3	DO-NUMBER + SERIES MACHINE (Ch 1)	3	2
$\rightarrow$ PC-2	MULTIPLIER MACHINE (badha $\cdot$ ghata $\cdot$ do baar $\cdot$ popu (Ch 6)	1	1
$\rightarrow$ RP-1	“1 HISSA” MACHINE (share $\cdot$ jodna $\cdot$ coins $\cdot$ bachat) (Ch 9)	12	0
$\rightarrow$ RP-2	CROSS-GUNA MACHINE (proportion $\cdot$ ratio badlo) (Ch 9)	1	0
Total (book ke sawaal)		54/54 ✓	25/25 ✓
$\rightarrow$ NS-3: EN 20–22 HI 5, 18 $\cdot$ $\rightarrow$ PC-2: EN 15 HI 4 $\cdot$ $\rightarrow$ RP-1: EN 2–5, 7, 9–12, 23–25 $\cdot$ $\rightarrow$ RP-2: EN 13			

Chapter 11

Work & Time (Samay aur Karya) समय और कार्य

2 formulas (pehle 7) · English book 206 sawaal · Hindi book 106 sawaal

WT-1 — LCM UNIT MACHINE (Kaam = Rate × Din)

PEHCHAN – Question mein ye dikhe

A can do a work in a days, B in b days, together, alone, A + B, B + C, C + A, twice as efficient, left after, joined after, alternate days, wages; काम, दिन, अकेले, साथ मिलकर, मजदूरी

MASTER FORMULA – sirf ye yaad karo

$$\text{Kul kaam} = \text{LCM}(\text{din}) \quad \text{Rate} = \frac{\text{Kaam}}{\text{Din}} \quad \text{Din} = \frac{\text{Kaam}}{\text{Rates ka jod}} \quad (\text{do log: } \frac{ab}{a+b})$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo ⇒ answer
① Saath / akela	Saath = $\frac{ab}{a+b}$ · akela = $\frac{aT}{a-T}$ · teen: $\frac{1}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$ · bacha kaam = $1 - t(\frac{1}{a} + \frac{1}{b})$
② A + B, B + C, C + A diye	Teeno saath = $\frac{2}{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}$ · A akela = $\frac{1}{\frac{1}{T} - (\frac{1}{B+C})}$ · beech wala $\frac{1}{B} = \frac{1}{A+B} + \frac{1}{B+C} - \frac{1}{T}$
③ k guna achha / x% zyada efficient	Din $\propto \frac{1}{\text{efficiency}}$ · A = k · B, saath T ⇒ $B = T(k+1)$, $A = \frac{T(k+1)}{k}$ · x% zyada ⇒ din × $\frac{100}{100+x}$
④ Beech mein chhoda / juda	$\sum \frac{\text{uske din}}{\text{uska akela time}} = 1$ · A k din pehle chhode ⇒ kul = $\frac{(a+k)b}{a+b}$
⑤ Baari-baari (alternate)	1 cycle ka kaam = rates ka jod ⇒ poori cycles, phir bacha kaam agle wale se (aakhri cycle beech mein khatam ho sakti hai)
⑥ Mazdoori (wages)	Hissa $\propto$ kiya hua kaam · saath poora kiya ⇒ A : B = b : a (din ka ulta) · C ka = kul ₹ × $[1 - T(\frac{1}{a} + \frac{1}{b})]$

TERMS – Har letter ka matlab

a, b, c = akele ke din · T = saath ke din · rate = 1 din ka kaam (units).

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 5: A 15, B 20 ⇒ saath
 $\frac{15 \times 20}{15+20} = 8\frac{4}{7}$ din ✓ book ans (c)
- ① EN 10: saath 6, Ranga 18 ⇒ Raju
 $\frac{18 \times 6}{18-6} = 9$ ✓ book ans (a)
- ② EN 44: A+B 10, B+C 15, A+C 18 ⇒ teeno
 $\frac{2}{\frac{1}{10} + \frac{1}{15} + \frac{1}{18}} = 9$ ✓ book ans (c)
- ② EN 46: 10, 20, 30 ⇒ teeno
 $\frac{2 \times 60}{11} = \frac{120}{11}$ ✓ book ans (a)
- ③ EN 16: Himanshu 2 × Ankit, saath 16 ⇒ Ankit
 $16 \times 3 = 48$ ✓ book ans (b)
- ③ EN 19: Ishan 2 × Kamal, saath 29 ⇒ Kamal
 $29 \times 3 = 87$ ✓ book ans (c)
- ④ EN 124: A 8, B 12; A 3 din pehle chhodta
 $\frac{(8+3) \times 12}{20} = 6\frac{3}{5}$ ✓ book ans (a)
- ④ EN 132: Janaki 45, Mansi 72; Janaki 33 din pehle
 $\frac{78 \times 72}{117} = 48$ ✓ book ans (b)
- ⑤ EN 7: A 10, B 12, alternate, A se shuru
 LCM 60: $6 + 5 = 11/2$ din ⇒ 5 cycle = 55 ⇒ $10\frac{5}{6}$ ✓ book ans (d)
- ⑥ EN 166: ₹600; A 4, B 6; C ke saath 2 din ⇒ C
 $600 [1 - 2 \times \frac{5}{12}] = 100$ ✓ book ans (a)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (252) EN 1–69, 99–116, 118–150, 152, 154–188, 190–193, 195–200, 203, 204 · **HI** 1–40, 45–48, 50–72, 74, 75, 86–88, 90, 96, 97, 99, 102–106 | *Ch 28 (Vividh)*: EN 29, 71 · **HI** 56

WT-2 — M·D·H MACHINE (log · din · ghante)

PEHCHAN – Question mein ye dikhe

12 men can do in 18 days working 8 hours, how many men / days, men and women / boys, food (provisions) for n days; पुरुष, महिलाएँ, घंटे प्रतिदिन, भोजन

MASTER FORMULA – sirf ye yaad karo

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Log, din, ghante, kaam	Jo nahi diya use x maano, baaki daalo · “x men = y women” $\Rightarrow$ ek ko doosre mein badlo · khana: $\log \times \text{din} = \text{constant}$ (beete din pehle ghatao)

TERMS – Har letter ka matlab

$M = \log \cdot D = \text{din} \cdot H = \text{ghante roz} \cdot W = \text{kaam}$ (same kaam ho to W hatao).

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 70:** 12 men 24.5 din 1 wall $\Rightarrow$ 5 walls 49 din
 $\frac{12 \times 24.5 \times 5}{49} = 30$ ✓ **book ans (b)**
- ① **EN 71:** 6M + 8W 14 din; 8M + 12W 10 din $\Rightarrow$ 1 man
 $84m + 112w = 80m + 120w \Rightarrow m = 2w \Rightarrow 140$ ✓ **book ans (a)**
- ① **EN 78:** 15 men or 25 women, 22 din $\Rightarrow$ 9M + 18W
 $9M = 15W \Rightarrow 33W \Rightarrow \frac{25 \times 22}{33} = 16\frac{2}{3}$ ✓ **book ans (a)**
- ① **EN 86:** 8 men $\times$ 9 h $\times$ 20 din $\Rightarrow$ 7 men $\times$ 10 h
 $\frac{8 \times 9 \times 20}{70} = \frac{144}{7}$ ✓ **book ans (d)**
- ① **EN 90:** 36 men 24 din; 8 din baad 4 gaye
 $\frac{36 \times 16}{32} = 18$ ✓ **book ans (a)**
- ① **EN 80:** M men 50 din; 5 zyada $\Rightarrow$ 10 din kam
 $50M = 40(M + 5) \Rightarrow 20$ ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (60) EN 70–98, 151, 153, 194, 201, 202, 206 · **HI** 41–44, 49, 73, 76–85, 89, 91–95, 98, 100, 101

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
WT-1	LCM UNIT MACHINE (Kaam = Rate $\times$ Din)	168	81
WT-2	M·D·H MACHINE (log · din · ghante)	35	25
$\rightarrow$ DF-2	COMPARE + OPERATE + “HISSA SE POORA” MACHINE (Ch 2)	1	0
$\rightarrow$ RP-1	“1 HISSA” MACHINE (share · jodna · coins · bachat) (Ch 9)	1	0
$\rightarrow$ RP-2	CROSS-GUNA MACHINE (proportion · ratio badlo) (Ch 9)	1	0
Total (book ke sawaal)		206/206 ✓	106/106 ✓
$\rightarrow$ DF-2: EN 117 · $\rightarrow$ RP-1: EN 189 · $\rightarrow$ RP-2: EN 205			

Chapter 12

Alligation / Mixture (Mishran / Alligation) मिश्रण / एलीगेशन

2 formulas (pehle 4) · English book 58 sawaal · Hindi book 16 sawaal

MX-1 — ALLIGATION CROSS MACHINE (milaane ka ratio)

PEHCHAN – Question mein ye dikhe

in what ratio must be mixed, rice at ₹... and ₹... to get ₹..., mixture of two kinds, mean price, average of mixture, alloy; किस अनुपात में मिलाएँ, मिश्रण

MASTER FORMULA – sirf ye yaad karo

$$\frac{\text{Sasta}}{\text{Mehenga}} = \frac{d - m}{m - c} \quad m = \frac{q_1 c + q_2 d}{q_1 + q_2}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo ⇒ answer
① Kis ratio mein milaayein	Cross rule · SP + profit diya ⇒ pehle $m = SP \times \frac{100}{100+P}$ · ratio wale mixture ⇒ ek component ka fraction lo
② Mixtures jodo ⇒ naya rate / ratio	$m = \frac{\sum q_i p_i}{\sum q_i}$ · component = $\Sigma(\text{quantity} \times \text{uska fraction})$

TERMS – Har letter ka matlab

c = saste ka rate / % · d = mehenge ka · m = mixture ka (mean) · q = quantity.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 10:** 80% juice kitna milayein 36 L ke 25% mein ⇒ 60%
 $25 : 80 \Rightarrow (80 - 60) : (60 - 25) = 20 : 35 \Rightarrow 36 \times \frac{35}{20} = 63$ ✓ **book ans (a)**
- ① **EN 28:** ₹45 & ₹52 sugar; ₹55.20 pe 15% profit
 $m = \frac{55.2}{1.15} = 48 \Rightarrow (52 - 48) : (48 - 45) = 4 : 3$ ✓ **book ans (a)**
- ① **EN 38:** ₹50 & ₹60 rice; ₹70 pe 20% profit
 $m = 58\frac{1}{3} \Rightarrow 1\frac{2}{3} : 8\frac{1}{3} = 1 : 5$ ✓ **book ans (c)**
- ② **EN 24:** 2 L (3 : 2) + 3 L (2 : 3)
 milk $1.2 + 1.2 = 2.4$, water $2.6 \Rightarrow 12 : 13$ ✓ **book ans (b)**
- ② **EN 27:** gold : platinum 5 : 2 aur 5 : 7, barabar
 $\frac{5}{7} + \frac{5}{12} : \frac{2}{7} + \frac{7}{12} = 95 : 73$ ✓ **book ans (c)**
- ② **EN 34:** 4 vessels 80, 75, 60, 50% milk ⇒ water : milk
 milk $2.65 \Rightarrow 1.35 : 2.65 = 27 : 53$ ✓ **book ans (b)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (49) EN 2, 3, 5, 8, 10, 14, 21, 24–34, 37–46, 48–50, 52, 54–58 · **HI** 1, 3–5, 8–12, 15, 16 | Ch 28 (Vividh): EN 4

MX-2 — “JO NAHI BADLA” + REPLACEMENT MACHINE

PEHCHAN – Question mein ye dikhe

how much water to be added / evaporated, milk : water becomes, x litres taken out and replaced by water n times; पानी मिलाया, निकाला, बदला

MASTER FORMULA – sirf ye yaad karo

$$C_1 V_1 = C_2 V_2 \quad \text{Bacha shuddh} = \text{Shuru} \times \left(1 - \frac{x}{V}\right)^n$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Paani joda / udaya, ek cheez joda	Jo cheez nahi badli woh fix: $V_2 = \frac{C_1 V_1}{C_2} \Rightarrow \text{add} = V_2 - V_1$ · ratio a : b $\rightarrow$ c : d sirf B jodkar $\Rightarrow$ naya B = $A \times \frac{d}{c}$
② Baar-baar nikaalo, paani daalo	$\text{Shuru} \times \left(1 - \frac{x}{V}\right)^n$ · paani = V – bacha

TERMS – Har letter ka matlab

C = concentration (%) · V = volume · x = har baar nikaala · n = kitni baar.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 1:** 90 ml 38% $\Rightarrow$ 17.1% ke liye paani
 $V_2 = \frac{90 \times 38}{17.1} = 200 \Rightarrow 110 \text{ ml} \checkmark$ **book ans (c)**
- ① **EN 4:** 80 L, 3 : 1 $\Rightarrow$ 2 : 3, paani
milk 60 fix $\Rightarrow$ water $60 \times \frac{3}{2} = 90 \Rightarrow 70 \checkmark$ **book ans (c)**
- ① **EN 6:** 10 L se 2 L paani uda, ab 6% namak
 $\frac{8 \times 6}{10} = 4.8\% \checkmark$ **book ans (a)**
- ① **EN 11:** 8 L mein 4% paani $\Rightarrow$ 2% ke liye pure milk
paani 0.32 fix $\Rightarrow \frac{0.32}{0.02} = 16 \Rightarrow 8 \text{ L} \checkmark$ **book ans (c)**
- ② **EN 35:** 50 L milk, 10 L $\times$ 3 baar $\Rightarrow$ water : milk
milk = $50 \left(\frac{4}{5}\right)^3 = 25.6 \Rightarrow 24.4 : 25.6 = 61 : 64 \checkmark$ **book ans (c)**
- ② **EN 53:** 80 L, 8 L $\times$ 3 baar
 $80 \times 0.9^3 = 58.32 \checkmark$ **book ans (a)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (23) EN 1, 4, 6, 7, 9, 11–13, 15–20, 22, 35, 36, 51, 53 · **HI** 2, 6, 13, 14

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
MX-1	ALLIGATION CROSS MACHINE (milaane ka ratio)	37	11
MX-2	“JO NAHI BADLA” + REPLACEMENT MACHINE	19	4
$\rightarrow$ PC-1	PERCENT = HISSA MACHINE (of · marks · income · vot (Ch 6)	1	1
$\rightarrow$ PL-2	GINTI · RATE · NAKLI BAAT MACHINE (Ch 7)	1	0
Total (book ke sawaal)		58/58 ✓	16/16 ✓
$\rightarrow$ PC-1: EN 23 HI 7 · $\rightarrow$ PL-2: EN 47			

Chapter 13

Pipe & Cistern (Nal aur Tanki) नल और टंकी

1 formula (pehle 6) · English book 155 sawaal · Hindi book 95 sawaal

PP-1 — NET RATE MACHINE (WT-1 hi hai: bharo +, khaali −)

PEHCHAN – Question mein ye dikhe

pipe / tap fills a tank in ... hours, outlet / leak empties, both opened together, closed after, opened alternately, capacity in litres; नल, टंकी, रिसाव, भरना, खाली करना

MASTER FORMULA – sirf ye yaad karo

$$\frac{1}{T} = \sum \frac{1}{\text{bharne wale}} - \sum \frac{1}{\text{khaali karne wale}} \quad (\text{bharo } a, \text{ khaali } b : T = \frac{ab}{b-a})$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Kai pipes saath	LCM units: bharo +, khaali $- \Rightarrow$ net $\Rightarrow$ LCM $\div$ net $\cdot$ do: $\frac{ab}{a+b}$ $\cdot$ teen: $\frac{XYZ}{YZ+XZ-XY}$
② Leak	Leak khaali karega = $\frac{a \times b}{b-a}$ (a = bina leak, b = leak ke saath)
③ Saath T; A ko a zyada, B ko b zyada	$T = \sqrt{a \times b}$
④ Beech mein band / khola	$\sum \frac{\text{pipe kitni der khula}}{\text{uska akela time}} = 1$ (khaali wala $-$)
⑤ Baari-baari ghante	2 ghante ka net = $\frac{1}{a} - \frac{1}{b} \Rightarrow$ cycles $\cdot$ trap: aakhri ghante mein bharne wala akela khatam kar deta hai
⑥ Capacity (litre)	$C = \frac{r}{\left \frac{1}{T} - \frac{1}{a}\right } \cdot$ fractions ka farak = litres ka farak $\Rightarrow C = \frac{L}{f_1 - f_2}$

TERMS – Har letter ka matlab

a, b = pipe ka akela time $\cdot$ T = saath ka time $\cdot$ r = litre / ghanta $\cdot$ C = capacity.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 1: A 21 h, B 18 h $\Rightarrow$ saath
 $\frac{21 \times 18}{39} = 9\frac{9}{13}$ h ✓ book ans (a)
- ① EN 23: fill 20 min, empty 60 min
 $\frac{20 \times 60}{40} = 30$ ✓ book ans (d)
- ② EN 29: 9 h, leak se 10 h
 $\frac{9 \times 10}{1} = 90$ ✓ book ans (d)
- ② EN 38: 1.5 h, leak se 2.25 h
 $\frac{1.5 \times 2.25}{0.75} = 4.5$ h ✓ book ans (b)
- ③ EN 2: saath 20 h; A, B se 9 h tez $\Rightarrow$ B
 $ab = 400, b - a = 9 \Rightarrow 16, 25 \Rightarrow B = 45$ ✓ book ans (a)
- ③ EN 7: saath 45/4 min; bada 12 min kam $\Rightarrow$ bada
 $a = \frac{-12 + \sqrt{144 + 506.25}}{2} = 6.75 \Rightarrow 18$ ✓ book ans (c)
- ④ EN 85: A 16, B 32; kul 16 min $\Rightarrow$ A kab band
 $\frac{t}{16} + \frac{16}{32} = 1 \Rightarrow t = 8$ ✓ book ans (a)
- ④ EN 87: A 36, B 48; 30 min mein bhari $\Rightarrow$ B kab band
 $\frac{30}{36} + \frac{t}{48} = 1 \Rightarrow 8$ ✓ book ans (c)
- ⑤ EN 118: inlet 15 h, outlet 21 h, baari-baari
 LCM 105: +7, $-5 \Rightarrow 49$ cycles = 98 $\Rightarrow +1 = 99$ h ✓ book ans (d)
- ⑥ EN 128: P 20 h mein khaali; Q 20 L/h; dono 12 h
 $C \left(\frac{1}{12} - \frac{1}{20} \right) = 20 \Rightarrow 600$ ✓ book ans (d)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (250) EN 1–155 · HI 1–95

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
PP-1	NET RATE MACHINE (WT-1 hi hai: bharo +, khaali –)	155	95
Total (book ke sawaal)		155/155 ✓	95/95 ✓

Chapter 14

Simple Interest (Sadharan Byaj) साधारण ब्याज

1 formula (pehle 6) · English book 200 sawaal · Hindi book 76 sawaal

SI-1 — P·R·T MACHINE (simple interest)

PEHCHAN – Question mein ye dikhe

simple interest, amount after T years, at what rate, in how many years, doubles / triples itself, amount ₹A₁ in T₁ years and ₹A₂ in T₂ years, rate increased by 2%, lent in two parts, instalment; साधारण ब्याज, मिश्रधन, दर, दुगुना

MASTER FORMULA – sirf ye yaad karo

$$SI = \frac{P \times R \times T}{100} \quad A = P + SI \quad (\text{jo poocha use akela karo: } P = \frac{100 SI}{RT}, R = \frac{100 SI}{PT})$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo ⇒ answer
① P, R, T, SI, A mein se ek	Seedha master formula · mahine ⇒ $\div 12 \cdot P = \frac{A \times 100}{100 + RT}$
② T ₁ pe A ₁ , T ₂ pe A ₂	1 saal ka SI = $\frac{A_2 - A_1}{T_2 - T_1} \Rightarrow P = A_1 - T_1 \times (1 \text{ saal SI})$
③ Rate / time badla, SI ka farak	$P = \frac{\Delta SI \times 100}{\Delta R \times T}$ (dono badle ⇒ $R_1 T_1 - R_2 T_2$)
④ n guna / SI = P ka hissa	$R \times T = (n - 1) \times 100 \cdot SI = f \times P \Rightarrow RT = 100f \cdot R = T \Rightarrow R = \sqrt{100f}$
⑤ Do hisson mein diya	SI barabar ⇒ $P_1 : P_2 = \frac{1}{R_1 T_1} : \frac{1}{R_2 T_2}$ · kul SI diya ⇒ alligation (rates par)
⑥ Kisht / present worth	Kisht $x = \frac{100 D}{100n + \frac{Rn(n-1)}{2}} \cdot PW = \frac{A \times 100}{100 + RT}$

TERMS – Har letter ka matlab

P = mool · R = dar % saalana · T = saal · A = mishradhan · n = kitne guna / kishte · D = chukana.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 2: ₹48,750, 16%, 73 din
 $\frac{48750 \times 16 \times 73}{36500} = 1560$ ✓ book ans (a)
- ① EN 9: ₹2000, 8.25%, 7 Feb → 20 Apr 2022
 21 + 31 + 20 = 72 din ⇒ $\frac{2000 \times 8.25 \times 72}{36500} \approx 33$ ✓ book ans (d)
- ② EN 39: 8 saal ₹24,500, 13 saal ₹32,000 ⇒ P
 $\frac{7500}{5} = 1500 \Rightarrow 24500 - 12000 = 12500$ ✓ book ans (a)
- ② EN 111: 2 saal ₹1250, 5 saal ₹2000 ⇒ R
 250/saal, P = 750 ⇒ 33.33% ✓ book ans (b)
- ③ EN 1: 1 saal, A = ₹1484; rate +2% ⇒ ₹26.50 zyada ⇒ SI
 $P = \frac{26.5 \times 100}{2} = 1325 \Rightarrow 159$ ✓ book ans (d)
- ③ EN 42: 3 saal, rate +2% ⇒ ₹84 zyada
 $\frac{84 \times 100}{6} = 1400$ ✓ book ans (c)
- ④ EN 109: 8 saal mein double
 $\frac{100}{8} = 12.5\%$ ✓ book ans (d)
- ④ EN 115: 8 saal mein triple
 $\frac{200}{8} = 25\%$ ✓ book ans (c)
- ⑤ EN 26: ₹6600; $10\% \times 3 = 9\% \times 4$ (SI barabar)
 $P_1 : P_2 = 36 : 30 = 6 : 5 \Rightarrow 3600$ ✓ book ans (a)
- ⑥ EN 193: ₹9600, 5 saal, 10% ⇒ kisht
 $\frac{960000}{500 + 100} = 1600$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (280) EN 1–9, 11–137, 139–200 · HI 1–76 | Ch 6 (Pratishatata): EN 180 | Ch 15 (Chakravardhi Byaj): EN 54, 202, 207 · HI 65, 69

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
SI-1	P·R·T MACHINE (simple interest)	198	76
→ CI-2	CI – SI FARAK MACHINE (Ch 15)	2	0
Total (book ke sawaal)		200/200 ✓	76/76 ✓
→ CI-2: EN 10, 138			

Chapter 15

Compound Interest (Chakravridhi Byaj) चक्रवृद्धि ब्याज

2 formulas (pehle 6) · English book 206 sawaal · Hindi book 70 sawaal

CI-1 — CI AMOUNT MACHINE ($A = P \times \text{multiplier}^n$)

PEHCHAN – Question mein ye dikhe

compound interest, compounded annually / half-yearly / quarterly, amount after n years, at what rate, doubles in, instalments, different rates in successive years; चक्रवृद्धि ब्याज, अर्धवार्षिक, तिमाही

MASTER FORMULA – sirf ye yaad karo

$$A = P \left(1 + \frac{R}{100} \right)^n \quad CI = A - P \quad \begin{array}{l|l|l} \text{half-yearly} & R/2 & 2n \\ \text{quarterly} & R/4 & 4n \end{array}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① A / CI / P nikaalo	Master formula · alag rates $\Rightarrow P(1 + \frac{r_1}{100})(1 + \frac{r_2}{100}) \cdot 2$ saal $CI\% = 2r + \frac{r^2}{100}$ (10% $\Rightarrow$ 21%, 3 saal 33.1%)
② Rate nikaalo	$1 + \frac{R}{100} = \left(\frac{A}{P} \right)^{1/n}$ · lagataar saal: $R = \frac{A_2 - A_1}{A_1} \times 100 \cdot 1.21 = 1.1^2, 1.331 = 1.1^3, 1.44 = 1.2^2$
③ Time / k guna	k guna T saal mein $\Rightarrow k^m$ guna mT saal mein · A/P ko multiplier ki power pehchano
④ Kisht / present worth	$P = \frac{x}{1+i} + \frac{x}{(1+i)^2} + \dots \cdot 2$ kisht: $x = \frac{P(1+i)^2}{2+i}$ · PW = $\frac{A}{(1+i)^n}$
⑤ SI vs CI / yearly vs half-yearly ka fayda	$P \times [(\text{naya multiplier})^n - (\text{purana})] \cdot \text{farak diya} \Rightarrow P = \text{farak} \div \text{bracket}$

TERMS – Har letter ka matlab

$P = \text{mool} \cdot R = \text{dar \%} \cdot n = \text{saal} \cdot i = R/100 \cdot x = \text{kisht}$.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 1: 10% pe 2 saal mein ₹10,164 $\Rightarrow P$
 $\frac{10164}{1.21} = 8400$ ✓ book ans (b)
- ① EN 15: ₹2000, 40%; teesre saal ka CI
 $2000 \times 0.4 \times 1.4^2 = 1568$ ✓ book ans (c)
- ② EN 163: lagataar saal ka interest ₹225, ₹236.25
 $\frac{11.25}{225} = 5\%$ ✓ book ans (d)
- ② EN 168: 2 saal ₹9680, 3 saal ₹10,648
 $\frac{968}{9680} = 10\%$ ✓ book ans (a)
- ③ EN 183: 6 saal mein 2 guna $\Rightarrow$ 8 guna
 $8 = 2^3 \Rightarrow 18$ ✓ book ans (a)
- ③ EN 184: 10 saal mein double $\Rightarrow$ 16 guna
 $16 = 2^4 \Rightarrow 40$ ✓ book ans (c)
- ④ EN 7: ₹28,700, 2 saal, 5% $\Rightarrow$ kisht
 $\frac{28700 \times 1.1025}{2.05} = 15435$ ✓ book ans (d)
- ④ EN 16: ₹16,400, 2 kisht, 5%
 $\frac{16400 \times 1.1025}{2.05} = 8820$ ✓ book ans (d)
- ⑤ EN 3: 40%, 2 saal; half-yearly se ₹4544 zyada $\Rightarrow P$
 $P(1.2^4 - 1.4^2) = 0.1136P = 4544 \Rightarrow 40000$ ✓ book ans (b)
- ⑤ EN 19: 20%, 2 saal; half-yearly ₹482 zyada
 $P(1.1^4 - 1.2^2) = 0.0241P \Rightarrow 20000$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (197) EN 1–53, 55–66, 68–77, 79–98, 100, 105, 124–126, 162–201, 203, 205, 206 · HI 1–5, 8–34, 37, 39–42, 52–64, 66–68, 70

CI-2 — CI – SI FARAK MACHINE

PEHCHAN – Question mein ye dikhe

difference between CI and SI for 2 / 3 years, CI and SI for 2 years are ... find rate / sum; चक्रवृद्धि और साधारण ब्याज का अंतर

MASTER FORMULA – sirf ye yaad karo

$$\text{2 saal: } CI - SI = P \left(\frac{R}{100} \right)^2 \quad \text{3 saal: } CI - SI = P \left(\frac{R}{100} \right)^2 \left(3 + \frac{R}{100} \right)$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho

Value aise daalo $\Rightarrow$ answer

① Farak diya / poocha

$$\text{Master formula} \cdot 2 \text{ saal ka SI aur CI diye } \Rightarrow R = \frac{2(CI-SI)}{SI} \times 100 \cdot CI = SI \left(1 + \frac{R}{200} \right)$$

TERMS – Har letter ka matlab

P = mool $\cdot$ R = dar %.

SOLVED – Asli book questions (row ke saath, verified)

① **EN 132:** 2 saal 9%, farak ₹97.2 $\Rightarrow$ P

$$\frac{97.2 \times 10^4}{81} = 12000 \quad \checkmark \text{ book ans (d)}$$

① **EN 134:** ₹5000, 10%, 3 saal $\Rightarrow$ farak

$$\frac{5000 \times 100 \times 310}{10^6} = 155 \quad \checkmark \text{ book ans (d)}$$

① **EN 144:** 10%, 3 saal, farak ₹620 $\Rightarrow$ P

$$\frac{620 \times 10^6}{100 \times 310} = 20000 \quad \checkmark \text{ book ans (d)}$$

① **EN 137:** ₹24,000, 2 saal, farak ₹60 $\Rightarrow$ r

$$r^2 = \frac{60 \times 10^4}{24000} = 25 \Rightarrow 5\% \quad \checkmark \text{ book ans (c)}$$

① **EN 119:** SI ₹660, CI ₹696.30 (2 saal) $\Rightarrow$ r

$$\frac{2 \times 36.3}{660} \times 100 = 11\% \quad \checkmark \text{ book ans (b)}$$

① **EN 112:** SI ₹500, 2 saal, 10% $\Rightarrow$ CI

$$500 \times 1.05 = 525 \quad \checkmark \text{ book ans (b)}$$

IS FORMULA SE HONE WALE BOOK KE SAWAAL (75) EN 67, 99, 101–104, 106–123, 127–161 $\cdot$ **HI** 6, 7, 35, 36, 38, 43–51 | Ch 14 (Sadharan Byaj): **EN** 10, 138

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
CI-1	CI AMOUNT MACHINE ($A = P \times \text{multiplier}^n$)	143	54
CI-2	CI – SI FARAK MACHINE	59	14
$\rightarrow$ PC-2	MULTIPLIER MACHINE (badha $\cdot$ ghata $\cdot$ do baar $\cdot$ popu (Ch 6)	1	0
$\rightarrow$ SI-1	P $\cdot$ R $\cdot$ T MACHINE (simple interest) (Ch 14)	3	2
Total (book ke sawaal)		206/206 ✓	70/70 ✓
$\rightarrow$ PC-2: EN 204 $\cdot$ $\rightarrow$ SI-1: EN 54, 202, 207 HI 65, 69			

Chapter 16

Problems on Age (Aayu Sambandhi Prashn) आयु संबंधी प्रश्न

1 formula (pehle 4) · English book 241 sawaal · Hindi book 122 sawaal

AG-1 — UMAR MACHINE (ratio + “farak kabhi nahi badalta”)

PEHCHAN – Question mein ye dikhe

ratio of present ages, after / before t years ratio, father is m times son, sum of ages, average age of family, product of ages; आयु, अनुपात, वर्ष पहले, वर्ष बाद

MASTER FORMULA – sirf ye yaad karo

$$\text{Umar} = ax, bx \quad \frac{ax+t}{bx+t} = \frac{c}{d} \Rightarrow x = \frac{t(c-d)}{ad-bc} \quad (t \text{ saal pehle} \Rightarrow t \text{ negative})$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Do samay ke do ratio	Master formula se $x \Rightarrow$ umar ax, bx · “ m guna” = ratio $m : 1$
② Farak / “jab main tumhari umar ka tha”	Farak $d \Rightarrow 1 \text{ unit} = \frac{d}{a-b}$ · kisi samay $A = m \times B \Rightarrow B = \frac{d}{m-1}$ · janm ke samay maa ki umar = farak
③ Jod / average	n logon ka jod t saal baad = $S + nt$ (pehle: $S - nt$) · naya member = naya jod – purana jod
④ Equation / product / bacha hua	Options daalo (5 sec) · jod S , product $P \Rightarrow$ factor pair jiska jod S

TERMS – Har letter ka matlab

$a : b$ = abhi ka ratio · $c : d$ = t saal baad ka · $x = 1 \text{ unit}$ · d = farak · n = log.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 5:** abhi 3 guna, 10 saal baad 2 guna $\Rightarrow$ beti
 $k = \frac{10(2-1)}{3-2} = 10$ ✓ **book ans (d)**
- ① **EN 2:** 6 saal pehle 6 guna, 3 saal baad 3 guna $\Rightarrow$ beti
 $k = \frac{9 \times 2}{3} = 6 \Rightarrow 6 + 6 = 12$ ✓ **book ans (b)**
- ② **EN 14:** farak 20; 5 saal pehle 5 guna $\Rightarrow$ Arun
 $\frac{20}{4} = 5 \Rightarrow$ Kiran 10, Arun 30 ✓ **book ans (c)**
- ② **EN 1:** 12 saal bada; 17 saal pehle 4 guna $\Rightarrow$ Mohit
 $\frac{12}{3} = 4 \Rightarrow 4 + 17 = 21$ ✓ **book ans (d)**
- ③ **EN 10:** jod 30; 5 saal baad $3 : 2 \Rightarrow A$
 $40 \times \frac{3}{5} = 24 \Rightarrow 19$ ✓ **book ans (d)**
- ③ **EN 34:** jod 45; 5 saal pehle $6 : 1 \Rightarrow$ pita
 $35 \times \frac{6}{7} = 30 \Rightarrow 35$ ✓ **book ans (c)**
- ④ **EN 20:** Varun 3 guna; 6 saal baad gunanfal 231
 $231 = 21 \times 11 \Rightarrow x + 6 = 11 \Rightarrow$ Varun 15 ✓ **book ans (a)**
- ④ **EN 101:** $\frac{2}{3}$ meri = $\frac{3}{4}$ cousin; mera (-3) = cousin $(+1)$
 $9 : 8 \Rightarrow 9k - 3 = 8k + 1 \Rightarrow k = 4 \Rightarrow 36$ ✓ **book ans (a)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (360) EN 1–216, 218–224, 226–235, 237–241 · HI 1–53, 55–123

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
AG-1	UMAR MACHINE (ratio + “farak kabhi nahi badalta”)	238	122
$\rightarrow$ DF-2	COMPARE + OPERATE + “HISSA SE POORA” MACHINE (Ch 2)	3	0
Total (book ke sawaal)		241/241 ✓	122/122 ✓
$\rightarrow$ DF-2: EN 217, 225, 236			

Chapter 17

Average (Ausat) औसत

1 formula (pehle 6) · English book 116 sawaal · Hindi book 82 sawaal

AV-1 — TOTAL MACHINE (Total = Average × n)

PEHCHAN – Question mein ye dikhe

average of, average increases / decreases when one joins / leaves / replaced, average of first n natural numbers, combined average, required runs / marks, average of first five is ..., last five is ...; औसत

MASTER FORMULA – sirf ye yaad karo

$$\text{Total} = \text{Average} \times n \quad \Rightarrow \quad \text{Naya aadmi} = \text{Naya total} - \text{Purana total}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Series ka average	$\frac{\text{pehla} + \text{aakhri}}{2} \cdot 1 \dots n \Rightarrow \frac{n+1}{2} \cdot \text{pehle } n \text{ odd} \Rightarrow n \cdot \text{even} \Rightarrow n + 1$
② Missing number / galti sudhaar	Missing = $nA - \sum(\text{baaki})$ · sahi avg = $A + \frac{\text{sahi-galat}}{n}$ · sab mein +k $\Rightarrow$ avg +k
③ Do group ka milke average	$\frac{n_1a_1 + n_2a_2}{n_1 + n_2}$ · ratio chahiye $\Rightarrow$ alligation
④ Aaya / gaya / badla	Aaya = $A + (n+1)d$ · Gaya = $A - (n-1)d$ · Badla = purana + nd (d = avg ka badlav, ghata to -)
⑤ Pehle k ka, aakhri k ka, sab ka	Common (overlap) = $k_1a_1 + k_2a_2 - NA$ · beech wala = $NA - (k_1a_1 + k_2a_2)$
⑥ Kitne chahiye (runs / marks)	Chahiye = target avg $\times$ kul ginti – abhi tak ka total

TERMS – Har letter ka matlab

A = average · n = ginti · d = average mein badlav · N = kul ginti.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 2: 11 ke pehle 12 multiples
 $\frac{11 \times 13}{2} = 71.5$ ✓ book ans (c)
- ① EN 6: pehle 10 natural ke squares ka mean
 $\frac{11 \times 21}{6} = \frac{77}{2}$ ✓ book ans (d)
- ② EN 4: 7 ka avg 53; 58 ki jagah 16 padha
 $53 + \frac{42}{7} = 59$ ✓ book ans (c)
- ② EN 25: 40 ka avg 68; 25 ko 45 padha
 $68 - \frac{20}{40} = 67.5$ ✓ book ans (d)
- ③ EN 23: 100 @25 + 50 @70
 $\frac{2500 + 3500}{150} = 40$ ✓ book ans (d)
- ③ EN 27: 20 @50%, 40 @60%, 60 @70%
 $\frac{1000 + 2400 + 4200}{120} = 63\%$ ✓ book ans (d)
- ④ EN 47: 35 students avg 15; teacher se +1
 $15 + 36 \times 1 = 51$ ✓ book ans (b)
- ④ EN 46: 14 ka avg 20; ek gaya, avg -2
 $20 + 13 \times 2 = 46$ ✓ book ans (d)
- ⑤ EN 73: 21 ka 42; pehle 11 ka 50, aakhri 11 ka 35 $\Rightarrow$ 11th
 $550 + 385 - 882 = 53$ ✓ book ans (b)
- ⑥ EN 28: Reena 16×26 ; Shreya $12 \times 24 \Rightarrow$ baaki 4 ka avg
 $\frac{416 - 288}{4} = 32$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (199) EN 1-116 · HI 1-82 | Ch 28 (Vividh): EN 28

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
AV-1	TOTAL MACHINE (Total = Average $\times$ n)	116	82
Total (book ke sawaal)		116/116 ✓	82/82 ✓

Chapter 18

Speed, Time & Distance (Chaal, Samay aur Doori) चाल, समय और दूरी

2 formulas (pehle 7) · English book 149 sawaal · Hindi book 78 sawaal

ST-1 — D = S × T MACHINE (units · late-early · ratio · do sawari)

PEHCHAN – Question mein ye dikhe

distance, speed, time, km/h to m/s, reaches late / early, if speed is 3/4 of usual, covers part by bus and rest on foot; दूरी, चाल, समय, देर से, जल्दी

MASTER FORMULA – sirf ye yaad karo

$$D = S \times T \quad \text{km/h} \xrightarrow{\times \frac{5}{18}} \text{m/s} \quad \text{Late-Early: } D = \frac{S_1 S_2}{S_2 - S_1} \times \Delta t$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo ⇒ answer
① Seedha D, S, T / units / tukde	Jo poocha use akela: $S = \frac{D}{T}$, $T = \frac{D}{S}$ · km/h × 5/18 = m/s · bacha fraction = diya km ⇒ total = km ÷ fraction
② Late / early (do speeds)	$D = \frac{S_1 S_2}{S_2 - S_1} \times \Delta t$ · Δt = late + early (ek late, ek early) ya farak (dono late) · minutes ÷ 60
③ Speed ka ratio / “usual speed ka 3/4”	Same D ⇒ $\frac{S_1}{S_2} = \frac{T_2}{T_1}$ · speed × a/b ⇒ time × b/a ⇒ farak = $T \left \frac{b}{a} - 1 \right $
④ Do sawari / do speeds ka batwara	Tez sawari ka samay = $\frac{D - S_{\text{dheema}} \times T}{S_{\text{tez}} - S_{\text{dheema}}}$ · walk + ride T_1 , walk + walk T_2 ⇒ ride + ride = $2T_1 - T_2$

TERMS – Har letter ka matlab

D = doori · S = speed · T = samay · Δt = samay ka farak (ghante).

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 3: 57.6 km in 48 min ⇒ m/s
 $\frac{57.6}{0.8} = 72 \text{ km/h} \times \frac{5}{18} = 20 \checkmark \text{ book ans (d)}$
- ① EN 4: 8 h @75; same doori 6 h mein
 $\frac{75 \times 8}{6} = 100 \checkmark \text{ book ans (b)}$
- ② EN 77: 30 km/h 5 min late, 40 km/h 3 min early
 $\frac{30 \cdot 40}{10} \times \frac{8}{60} = 16 \checkmark \text{ book ans (d)}$
- ② EN 63: 2.5 km/h 6 min late, 3.5 km/h 6 min early
 $\frac{8.75}{1} \times \frac{12}{60} = 1.75 \checkmark \text{ book ans (c)}$
- ③ EN 42: 3/4 speed pe 15 min late ⇒ usual
 $T \left(\frac{4}{3} - 1 \right) = 15 \Rightarrow 45 \checkmark \text{ book ans (c)}$
- ③ HI 14: 2/5 speed pe 30 min late
 $T \left(\frac{5}{2} - 1 \right) = 30 \Rightarrow 20 \checkmark \text{ book ans (a)}$
- ④ EN 40: 64 km, 7 h; paidal 7, cycle 12
 $\frac{64 - 49}{5} = 3 \text{ h} \checkmark \text{ book ans (b)}$
- ④ EN 48: 300 km, 5 h; 75 aur 55
 $\frac{300 - 275}{20} = 1.25 \text{ h} = 1 \text{ h } 15 \text{ min} \checkmark \text{ book ans (c)}$

IS FORMULA SE HONE WALE BOOK KE SAWAAL (153) EN 1–6, 8, 11, 13–16, 22, 24, 26–28, 30, 33–35, 38–40, 42, 45–51, 53–60, 62–67, 69, 71–74, 76–78, 80, 82, 84–91, 95, 97–99, 101, 102, 104, 108, 144–146, 148, 149 · HI 1–9, 13, 14, 16–28, 30, 34, 36–39, 42–47, 50, 52–54 | Ch 19 (Railgaadi): EN 2–15, 17–20, 138, 144, 149, 152–157 · HI 1–3, 5, 6, 79, 80 | Ch 28 (Vividh): EN 37, 46

ST-2 — AVERAGE + RELATIVE SPEED MACHINE

PEHCHAN – Question mein ye dikhe

average speed, goes at x and returns at y, two persons start towards each other / in same direction, meet after, overtakes, after meeting reach in t₁ and t₂ hours; औसत चाल, आमने-सामने, पीछा

MASTER FORMULA – sirf ye yaad karo

$$\text{Avg speed} = \frac{\text{Kul doori}}{\text{Kul samay}} \left(\text{barabar doori: } \frac{2xy}{x+y} \right) \quad \text{Relative} = S_1 + S_2 \text{ (aamne-saamne)}, S_1 - S_2 \text{ (peeche)}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Average speed	Kul doori $\div$ kul samay $\cdot$ aana-jaana $\frac{2xy}{x+y}$ $\cdot$ teen barabar hisse $\frac{3xyz}{xy+yz+zx}$ $\cdot$ barabar samay $\frac{x+y}{2}$
② Milna / pakadna	Milne ka samay $= \frac{\text{Gap}}{S_1+S_2}$ $\cdot$ pakadne ka $= \frac{\text{Gap}}{S_1-S_2}$ $\cdot$ head start gap $= \text{speed} \times \text{deri}$
③ Milne ke baad t_1, t_2	$\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$ $\cdot$ milne tak $= \sqrt{t_1 t_2}$

TERMS – Har letter ka matlab

x, y, z = alag hisson ki speed $\cdot$ Gap = beech ki doori $\cdot$ t_1, t_2 = milne ke baad ka samay.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 112:** 45 jaana, 36 aana
 $\frac{2 \cdot 45 \cdot 36}{81} = 40$ ✓ **book ans (c)**
- ① **EN 124:** triangle ke sides 300, 200, 240
 $\frac{3 \cdot 300 \cdot 200 \cdot 240}{180000} = 240$ ✓ **book ans (a)**
- ② **EN 29:** 240 km, speeds 5 : 7, B = 84
 $\frac{240}{60+84} \text{ h} = 100 \text{ min}$ ✓ **book ans (b)**
- ② **EN 36:** A 1:00 @40, B 1:40 @50
gap $\frac{80}{3} \Rightarrow \frac{80/3}{10} \text{ h} = 160 \text{ min}$ ✓ **book ans (d)**
- ③ **EN 147:** baad mein 9 h aur 16 h
 $\sqrt{16} : \sqrt{9} = 4 : 3$ ✓ **book ans (b)**
- ③ **EN 19:** Parvej 60, baad mein 12 h; Gautam 3 h
 $60\sqrt{\frac{12}{3}} = 120$ ✓ **book ans (a)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (122) EN 7, 9, 10, 12, 17–21, 23, 25, 29, 31, 32, 36, 37, 43, 44, 52, 61, 68, 70, 75, 79, 81, 83, 92, 94, 96, 100, 103, 105–107, 110–143, 147 $\cdot$ **HI** 10–12, 15, 29, 31–33, 35, 40, 41, 48, 49, 51, 55–78 | *Ch 19 (Railgaadi):* EN 16, 135–137, 139, 140, 142, 143, 145, 148 $\cdot$ **HI** 4, 7, 74, 78 | *Ch 28 (Vividh):* EN 40

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
ST-1	$D = S \times T$ MACHINE (units $\cdot$ late-early $\cdot$ ratio $\cdot$ do sawari)	77	40
ST-2	AVERAGE + RELATIVE SPEED MACHINE	69	38
$\rightarrow TR-1$	<i>TRAIN MACHINE (ST-1 hi hai: $D = \text{dono lambai}, S = r$ (Ch 19))</i>	3	0
Total (book ke sawaal)		149/149 ✓	78/78 ✓
$\rightarrow$ TR-1: EN 41, 93, 109			

Chapter 19

Train (Railgaadi) रेलगाड़ी

1 formula (pehle 6) · English book 159 sawaal · Hindi book 80 sawaal

TR-1 — TRAIN MACHINE (ST-1 hi hai: D = dono lambai, S = relative)

PEHCHAN – Question mein ye dikhe

train crosses a pole / man / platform / bridge / tunnel / another train, length of train, speed of train, crosses a man walking, two trains in opposite / same direction; रेलगाड़ी, खंभा, प्लेटफार्म, पुल

MASTER FORMULA – sirf ye yaad karo

$$\text{Samay} = \frac{L_{\text{train}} + L_{\text{cheez}}}{\text{Relative speed}} \quad (\text{pole / aadmi / ped: } L_{\text{cheez}} = 0; \text{ km/h} \times \frac{5}{18})$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Pole / khada aadmi	$t = \frac{L}{S} \cdot L = S \times t$
② Platform / pul / surang	$t = \frac{L+P}{S} \cdot P = St - L$
③ Ek train, do crossing	$S = \frac{P_2 - P_1}{t_2 - t_1}$ (pole $\Rightarrow P = 0$) $\Rightarrow$ phir $L = St_1 - P_1$
④ Do trains	$t = \frac{L_1 + L_2}{S_1 + S_2}$ (ulti disha) $\cdot t = \frac{L_1 + L_2}{S_1 - S_2}$ (ek disha)
⑤ Chalta aadmi / gaadi	$t = \frac{L}{S \mp s}$ (ek disha $-$, ulti $+$)
⑥ Bina lambai: milna / peecha	ST-2 wala: $\frac{\text{Gap}}{S_1 \pm S_2} \cdot x$ km zyada chali $\Rightarrow D = \frac{x(S_1 + S_2)}{S_1 - S_2}$

TERMS – Har letter ka matlab

L = train ki lambai · P = platform / pul · S = train speed · s = aadmi / doosri gaadi ki speed.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 21:** 72 km/h, pole 12 s $\Rightarrow$ lambai
 $20 \times 12 = 240$ m ✓ **book ans (d)**
- ① **EN 25:** 300 m, pole 5 s $\Rightarrow$ speed
 $60 \text{ m/s} \times \frac{18}{5} = 216$ ✓ **book ans (c)**
- ② **EN 1:** 725 m train, 235 m surang, 48 s
 $\frac{960}{48} = 20 \text{ m/s} = 72 \text{ km/h}$ ✓ **book ans (b)**
- ② **EN 52:** 110 m @90 km/h, 180 m pul
 $\frac{290}{25} = 11.6$ s ✓ **book ans (b)**
- ③ **EN 57:** 69 km/h; pole 8 s, platform 32 s $\Rightarrow$ platform
 $\frac{69-5}{18} \times 24 = 460$ ✓ **book ans (a)**
- ③ **EN 49:** 200 m platform 40 s, aadmi 20 s $\Rightarrow$ train
 $\frac{200-20}{20} = 200$ ✓ **book ans (c)**
- ④ **EN 104:** 80 m + 120 m, 30 & 20 m/s ulti
 $\frac{200}{50} = 4$ s ✓ **book ans (b)**
- ④ **EN 94:** 200 m + 300 m, 120 & 100 km/h ek disha
 $\frac{500}{50/9} = 90$ s = 1.5 min ✓ **book ans (d)**
- ⑤ **EN 43:** 110 m @60, aadmi 6 ulti
 $\frac{110}{66-5/18} = 6$ s ✓ **book ans (b)**
- ⑥ **EN 106:** 30 & 35 km/h, ek ne 30 km zyada
 $\frac{30-65}{5} = 390$ ✓ **book ans (c)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (194) EN 1, 21–134, 141, 146, 147, 150, 151, 158, 159 · **HI** 8–73, 75–77 | Ch 18 (Chaal, Samay aur Doori): EN 41, 93, 109

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
TR-1	TRAIN MACHINE (ST-1 hi hai: D = dono lambai, S = relative)	122	69
→ ST-1	$D = S \times T$ MACHINE (units · late-early · ratio · do (Ch 18)	27	7
→ ST-2	AVERAGE + RELATIVE SPEED MACHINE (Ch 18)	10	4
Total (book ke sawaal)		159/159 ✓	80/80 ✓
→ ST-1: EN 2–15, 17–20, 138, 144, 149, 152–157 HI 1–3, 5, 6, 79, 80 · → ST-2: EN 16, 135–137, 139, 140, 142, 143, 145, 148 HI 4, 7, 74, 78			

Chapter 20

Boat & Stream (Naav aur Dhaara) नाव और धारा

1 formula (pehle 4) · English book 47 sawaal · Hindi book 13 sawaal

BS-1 — D / U MACHINE (dhaara ke saath / khilaaf)

PEHCHAN – Question mein ye dikhe

boat, stream, current, downstream, upstream, speed in still water, rows to a place and back, takes twice as long upstream; नाव, धारा, धारा के अनुकूल / प्रतिकूल, शांत जल

MASTER FORMULA – sirf ye yaad karo

$$D = B + S \quad U = B - S \quad \Rightarrow \quad B = \frac{D+U}{2}, \quad S = \frac{D-U}{2}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Doori / samay se B, S	D = down doori $\div$ samay, U = up doori $\div$ samay $\Rightarrow$ master formula
② Up ka samay = n $\times$ down	$\frac{B}{S} = \frac{n+1}{n-1}$
③ Jaana + aana kul T	$d = \frac{T(B^2 - S^2)}{2B}$
④ Do trip, do equation	$\frac{a_1}{D} + \frac{b_1}{U} = T_1, \frac{a_2}{D} + \frac{b_2}{U} = T_2 \Rightarrow$ options ka B daal ke check / Cramer

TERMS – Har letter ka matlab

B = boat (shaant paani) · S = stream (dhaara) · D = downstream speed · U = upstream speed.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 2: B 14; 28 km downstream $1\frac{3}{4}$ h
 $D = 16 \Rightarrow S = 2$ ✓ book ans (a)
- ① EN 4: B 11; 19 km upstream 2 h
 $U = 9.5 \Rightarrow S = 1.5$ ✓ book ans (c)
- ② EN 1: B 3; up = $2 \times$ down
 $S = 3 \cdot \frac{1}{3} = 1$ ✓ book ans (d)
- ② EN 6: B $28/3$; up = $3 \times$ down
 $\frac{28}{3} \cdot \frac{2}{4} = \frac{14}{3}$ ✓ book ans (d)
- ③ EN 14: B 20; 364 km jaana-aana 40 h
 $400 - S^2 = \frac{364 \cdot 40}{40} \Rightarrow S = 6$ ✓ book ans (d)
- ③ EN 12: still 15 h, bahti nadi 16 h; B - S = 15
 $\frac{B^2}{B^2 - S^2} = \frac{16}{15} \Rightarrow B = 4S \Rightarrow S = 5$ ✓ book ans (d)
- ④ EN 40: 63 down + 30 up = 7 h; 28 down + 48 up = 6 h $\Rightarrow$ 35 down + 27 up
 $D = 14, U = 12 \Rightarrow 2.5 + 2.25 = 4$ h 45 min ✓ book ans (d)
- ④ HI 1: 30 up + 44 down = 10 h; 40 up + 55 down = 13 h
 $U = 5, D = 11 \Rightarrow S = 3$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (60) EN 1-47 · HI 1-13

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
BS-1	D / U MACHINE (dhaara ke saath / khilaaf)	47	13
Total (book ke sawaal)		47/47 ✓	13/13 ✓

Chapter 21

Mensuration (Kshetramiti) क्षेत्रमिति

3 formulas (pehle 10) · English book 354 sawaal · Hindi book 279 sawaal

MN-1 — 2-D SHAPE TABLE (area · perimeter)

PEHCHAN – Question mein ye dikhe

area / perimeter of triangle, rectangle, square, rhombus, trapezium, circle, sector, path around a field, tiles, carpet, wheel revolutions;
क्षेत्रफल, परिमाप, वृत्त, पहिया, रास्ता

MASTER FORMULA – sirf ye yaad karo

Shape	Area	Perimeter / aur
Triangle	$\frac{1}{2}bh, \sqrt{s(s-a)(s-b)(s-c)}$	$a + b + c, s = \frac{a+b+c}{2}$
Equilateral	$\frac{\sqrt{3}}{4}a^2$	$h = \frac{\sqrt{3}}{2}a$
Rectangle	$l b$	$2(l + b), d = \sqrt{l^2 + b^2}$
Square	$a^2 = \frac{d^2}{2}$	$4a, d = a\sqrt{2}$
Rhombus	$\frac{1}{2}d_1d_2$	$a^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$
Parallelogram	$b \times h$	$2(a + b)$
Trapezium	$\frac{1}{2}(a + b)h$	
Circle	πr^2	$2\pi r$
Sector (θ)	$\frac{\theta}{360}\pi r^2$	$\text{arc} = \frac{\theta}{360}2\pi r$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Triangle	Table · right: $\frac{1}{2}pb, r = \frac{p+b-h}{2}, R = \frac{h}{2}$ · triplets 3-4-5, 5-12-13, 8-15-17, 7-24-25
② 4 bhuja wale	Table · $(l + b)^2 = d^2 + 2A$ · same perimeter mein max area = square $\left(\frac{P}{4}\right)^2$
③ Path / border / ring / tiles	Bahar path $(l + 2w)(b + 2w) - lb$ · andar $lb - (l - 2w)(b - 2w)$ · cross roads $w(l + b) - w^2$ · ring $\pi(R^2 - r^2)$ · tiles = area $\div$ ek tile
④ Circle / wheel / semicircle	Table · semicircle perimeter = $\frac{36}{7}r$ · chakkar = doori $\div 2\pi r$ · square circle mein: diagonal = $2r$

TERMS – Har letter ka matlab

b = base · h = height · l = lambai · d = diagonal · r = radius · w = path ki chaudai · θ = kon.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 3:** hyp 13, ek side 5
doosri $12 \Rightarrow \frac{1}{2} \cdot 5 \cdot 12 = 30$ ✓ **book ans (c)**
- ① **HI 23:** sides 32, 60, 68
 $8-15-17 \times 4 \Rightarrow \frac{1}{2} \cdot 32 \cdot 60 = 960$ ✓ **book ans (a)**
- ② **EN 53:** rhombus diagonals 48, 64
 $\frac{48 \cdot 64}{2} = 1536$ ✓ **book ans (b)**
- ② **EN 60:** rhombus P 164, d₁ 80
a 41, half 40 $\Rightarrow 9 \Rightarrow d_2 = 18 \Rightarrow 720$ ✓ **book ans (b)**
- ③ **EN 124:** 18 m square park, 3 m road bahar
 $24^2 - 18^2 = 252$ ✓ **book ans (c)**
- ③ **EN 72:** 1386 m² park (r 21), 7 m path andar
 $\frac{22}{7}(441 - 196) = 770$ ✓ **book ans (c)**
- ④ **EN 76:** area 154 $\Rightarrow$ circumference
 $r = 7 \Rightarrow 44$ ✓ **book ans (b)**
- ④ **EN 78:** wheel d 88 cm, 8712 m
 $\frac{8712}{2.766} = 3150$ ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (351) EN 1-85, 87-113, 115, 116, 119-130, 132-137, 139-145, 147-149, 152-172, 174-227, 244, 349 · **HI** 1-118, 136, 258-260, 263, 265, 267, 274-276, 278, 279 | Ch 28 (Vividh): EN 54 · **HI** 23

MN-2 — 3-D SHAPE TABLE (volume · surface · pighlao)

PEHCHAN – Question mein ye dikhe

volume / surface area / lateral surface of cube, cuboid, cylinder, cone, sphere, hemisphere, melted and recast, how many balls / coins, wire drawn, water level rises; आयतन, पृष्ठीय क्षेत्रफल, घन, बेलन, शंकु, गोला, पिघलाकर

MASTER FORMULA – sirf ye yaad karo

Shape	Volume	Surface
Cube	a^3	$6a^2$ (4 deewar $4a^2$), $d = a\sqrt{3}$
Cuboid	lbh	$2(lb + bh + hl)$, $d = \sqrt{l^2 + b^2 + h^2}$
Cylinder	$\pi r^2 h$	$2\pi rh$, $2\pi r(r + h)$
Cone	$\frac{1}{3}\pi r^2 h$	πrl , $\pi r(r + l)$, $l = \sqrt{r^2 + h^2}$
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$
Hemisphere	$\frac{2}{3}\pi r^3$	$2\pi r^2$, $3\pi r^2$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Cube / cuboid / tank / deewar	Table · teen faces $x, y, z \Rightarrow V = \sqrt{xyz} \cdot 4$ deewar $2h(l + b)$ · paani ka level = daala volume $\div$ base area
② Cylinder / pipe	Table · hollow $\pi(R^2 - r^2)h \cdot \frac{CSA}{TSA} = \frac{h}{h+r}$
③ Cone / frustum	Table · sector se cone: $l = R$ · frustum $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$
④ Sphere / hemisphere	Table · hollow $\frac{4}{3}\pi(R^3 - r^3)$
⑤ Pighla ke naya banaya / jode	Volume same: $n = \frac{V_{\text{bada}}}{V_{\text{chhota}}} \cdot$ wire lambai $= \frac{V}{\pi r^2} \cdot$ gole se gole $n = \left(\frac{R}{r}\right)^3 \cdot$ prism = base area $\times h$

TERMS – Har letter ka matlab

a = cube ki side · l, b, h = lambai, chaudi, unchai · r = radius · l (cone) = slant height · R, r = bahar / andar.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 229:** painting ₹588 @ ₹2/cm²
TSA 294 $\Rightarrow$ a 7 $\Rightarrow$ 343 ✓ **book ans (a)**
- ① **EN 252:** ratio 4 : 3 : 5, edges 144 $\Rightarrow$ TSA
12, 9, 15 $\Rightarrow$ 2(108+135+180)=846 ✓ **book ans (b)**
- ② **EN 275:** base circumference 176, h 12 $\Rightarrow$ TSA
r 28 $\Rightarrow$ 176 $\times$ 40 = 7040 ✓ **book ans (c)**
- ② **EN 277:** r + h = 39, TSA 1716 $\Rightarrow$ V
r 7, h 32 $\Rightarrow$ 4928 ✓ **book ans (d)**
- ③ **EN 317:** l 13, base 144 $\Rightarrow$ V
r 12, h 5 $\Rightarrow$ 240 π ✓ **book ans (c)**
- ③ **EN 319:** r 5, h 12 $\Rightarrow$ TSA
l 13 $\Rightarrow$ $\pi \cdot 5 \cdot 18 = 90\pi$ ✓ **book ans (b)**
- ④ **EN 338:** SA 38.5 $\Rightarrow$ r
 $r^2 = 3.0625 \Rightarrow 1.75$ ✓ **book ans (d)**
- ④ **EN 340:** hemisphere TSA 1848 $\Rightarrow$ diameter
 $3\pi r^2 \Rightarrow r = 14 \Rightarrow 28$ ✓ **book ans (d)**
- ⑤ **EN 342:** 6, 8, 10 cm gole mila ke ek
 $R^3 = 1728 \Rightarrow 12$ ✓ **book ans (b)**
- ⑤ **EN 350:** r 6 se r 0.3 ki goliyan
 $20^3 = 8000$ ✓ **book ans (c)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (259) EN 131, 138, 146, 150, 151, 228-237, 241, 242, 245, 247-266, 268-277, 279-290, 292-297, 299-326, 328-333, 335-344, 346-348, 350-354 · **HI** 119-126, 128-135, 137, 139-147, 149-158, 160-182, 184-208, 210-214, 216, 217, 219-234, 236-256, 262, 264, 266, 268-273, 277 | Ch 28 (Vividh): EN 2, 24, 52

MN-3 — SIZE-CHANGE MACHINE (power dekho, ratio guna)

PEHCHAN – Question mein ye dikhe

radius doubled, side increased by $x\%$, ratio of volumes / areas, if each edge is halved, two cylinders radii ratio and heights ratio; अनुपात, दोगुना, प्रतिशत वृद्धि

MASTER FORMULA – sirf ye yaad karo

$$\text{Lambai} \times k \Rightarrow \text{Area} \times k^2, \quad \text{Volume} \times k^3 \quad \% \text{ change (area)} = 2x + \frac{x^2}{100}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Size badla / do shapes ka ratio	Formula mein power dekho: $V_{\text{cyl}} \propto r^2 h \Rightarrow \text{ratio} = \frac{r_1^2 h_1}{r_2^2 h_2} \cdot \text{volume ratio } a^3 : b^3 \Rightarrow \text{side } a : b \Rightarrow \text{area } a^2 : b^2$

TERMS – Har letter ka matlab

k = kitne guna $\cdot$ x = $\%$ badlav.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 86**: radius $+5\%$
 $10 + 0.25 = 10.25\%$ ✓ **book ans (a)**
- ① **EN 240**: edge $+50\% \Rightarrow$ surface
 $1.5^2 = 2.25 \Rightarrow 125\%$ ✓ **book ans (c)**
- ① **EN 278**: same volume, radius 20% kam $\Rightarrow$ height
 $\frac{1}{0.64} = 1.5625 \Rightarrow 56.25\%$ ✓ **book ans (b)**
- ① **EN 238**: cubes ka volume $64 : 1331 \Rightarrow$ surface
 $4 : 11 \Rightarrow 16 : 121$ ✓ **book ans (a)**
- ① **EN 291**: cylinders: $r \ 2 : 3, h \ 5 : 3$
 $4 \cdot 5 : 9 \cdot 3 = 20 : 27$ ✓ **book ans (b)**
- ① **EN 327**: cones: $V \ 1 : 10, h \ 2 : 5 \Rightarrow r$
 $r^2 = \frac{V}{h} = \frac{1}{2} : 2 \Rightarrow 1 : 2$ ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (30) EN 86, 114, 117, 118, 173, 238–240, 243, 246, 267, 278, 291, 298, 327, 334, 345 · **HI** 127, 138, 148, 159, 183, 209, 215, 218, 235, 257, 261 | *Ch 28 (Vividh)*: EN 67 · **HI** 52

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
MN-1	2-D SHAPE TABLE (area $\cdot$ perimeter)	219	130
MN-2	3-D SHAPE TABLE (volume $\cdot$ surface $\cdot$ pighlao)	118	138
MN-3	SIZE-CHANGE MACHINE (power dekho, ratio guna)	17	11
Total (book ke sawaal)		354/354 ✓	279/279 ✓

Chapter 22

Algebra (Beejganit) बीजगणित

4 formulas (pehle 9) · English book 294 sawaal · Hindi book 268 sawaal

AL-1 — “VALUE DAALO” IDENTITY MACHINE

PEHCHAN – Question mein ye dikhe

if $x + 1/x = \dots$, $a + b = \dots$, $ab = \dots$, $a + b + c = 0$, simplify / expand / factorise, value of expression, HCF / LCM of polynomials; यदि ... तो ... का मान

MASTER FORMULA – sirf ye yaad karo

$$x + \frac{1}{x} = k \Rightarrow x^2 + \frac{1}{x^2} = k^2 - 2, \quad x^3 + \frac{1}{x^3} = k^3 - 3k \quad a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Do variable / $x \pm \frac{1}{x}$	$x - \frac{1}{x} = k \Rightarrow x^2 + \frac{1}{x^2} = k^2 + 2$, $x^3 - \frac{1}{x^3} = k^3 + 3k$ · $a^2 + b^2 = (a+b)^2 - 2ab$ · $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$
② Teen variable (a, b, c)	$(a+b+c)^2 = \sum a^2 + 2\sum ab$ · squares ka jod = 0 $\Rightarrow$ har bracket = 0 · shart wale aasaan numbers daalo (a = 1, b = 1, c = -2)
③ Expand / simplify / factorise	x = 1 (ya 2) daalo , sawaal aur option dono ka maan milao — 10 sec · $x^2 + px + q$: jod p, guna q wale do number
④ Expressions ka HCF / LCM	Factorise $\Rightarrow$ HCF = common, chhoti power · LCM = sab, badi power

TERMS – Har letter ka matlab

k = diya hua maan · Value daalo = jo shart di hai use satisfy karne wale aasaan number rakh ke answer milao.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 64:** $x + 1/x = 3 \Rightarrow x^2 + 1/x^2$
 $9 - 2 = 7$ ✓ **book ans (a)**
- ① **EN 66:** $x + 1/x = 42 \Rightarrow x^3 + 1/x^3$
 $42^3 - 126 = 73962$ ✓ **book ans (b)**
- ② **EN 65:** $p + q + r = 13$, $pq + qr + rp = 30$
 $13(169 - 90) = 1027$ ✓ **book ans (c)**
- ② **EN 87:** $a + b + c = 9$, $a^2 + b^2 + c^2 = 29$
 $Q = 26 \Rightarrow 9(29 - 26) = 27$ ✓ **book ans (b)**
- ③ **EN 133:** $x(2x - 5) + 6(x^2 - 4) + 18$
 $8x^2 - 5x - 6$ ✓ **book ans (c)**
- ③ **EN 155:** $x^2 + 11xy + 24y^2$
 $8 \times 3 = 24$, $8 + 3 = 11 \Rightarrow (x + 8y)(x + 3y)$ ✓ **book ans (c)**
- ④ **EN 42:** LCM of $15x^3y^4$, $12x^2y^5$
 $60x^3y^5$ ✓ **book ans (a)**
- ④ **EN 45:** HCF of $6xy^2z$, $8x^2y^3z^2$, $12x^3y^3z^3$
 $2xy^2z$ ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (212) EN 36–50, 62–133, 155, 162, 165–167, 169–171, 244–251, 253, 256, 257, 261, 263, 266, 290, 292–294 · HI 13–18, 20–27, 29, 37–40, 42, 44, 46–70, 72–101, 103–116, 129, 138, 141, 143, 144, 234, 244, 268 | Ch 28 (Vividh): EN 10

AL-2 — EQUATION MACHINE (linear · quadratic · remainder)

PEHCHAN – Question mein ye dikhe

solve, roots of the equation, sum / product of roots, equal roots, no real roots, value of k, pair of linear equations has unique / no / infinite solutions, remainder when divided by $(x - a)$, factor; मूल, समीकरण, शेषफल, गुणनखंड

MASTER FORMULA – sirf ye yaad karo

$$ax^2 + bx + c = 0 : \quad \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad D = b^2 - 4ac \quad P(x) \div (x - a) \Rightarrow \text{shesh} = P(a)$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Do linear equations	Cramer: $x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}$ · unique: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ · koi nahi: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ · anant: teeno barabar
② Quadratic	$D > 0$ do real · $D = 0$ barabar · $D < 0$ real nahi · roots se equation: $x^2 - (\text{jod})x + \text{guna} = 0$ · simplify ke baad x ki sabse badi power 2 $\Rightarrow$ quadratic · $\alpha^2 + \beta^2 = S^2 - 2P$
③ Remainder / factor	$x = a$ daalo $\Rightarrow$ shesh · factor $\Leftrightarrow P(a) = 0$ · $(px + q) \Rightarrow x = -\frac{q}{p}$

TERMS – Har letter ka matlab

α, β = roots · S = jod · P = guna · D = discriminant.

SOLVED – Asli book questions (row ke saath, verified)

① EN 51: $2x + 3y + 11 = 0, 6x + ky + 33 = 0$ coincident
 $\frac{2}{6} = \frac{3}{k} \Rightarrow 9$ ✓ book ans (b)

① EN 56: $16x - 12y + 9, 12x + ky - 11$: no solution
 $\frac{16}{12} = \frac{-12}{k} \Rightarrow -9$ ✓ book ans (c)

② EN 176: $2y^2 - 6y - 7 = 0$ ka jod
 $\frac{6}{2} = 3$ ✓ book ans (c)

② EN 180: $x^2 - kx + 169$ equal roots
 $k^2 = 676 \Rightarrow \pm 26$ ✓ book ans (b)

③ EN 136: $27x^3 - 9x^2 + 3x - 8 \div (3x + 2)$
 $x = -2/3 \Rightarrow -8 - 4 - 2 - 8 = -22$ ✓ book ans (a)

③ EN 145: $4x^3 - 2x^2 + 5x - 8 \div (x - 2)$
 $32 - 8 + 10 - 8 = 26$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (272) EN 51–61, 134–154, 156–161, 163, 164, 168, 172–243, 252, 259, 260, 262, 265 · **HI** 28, 30–36, 102, 117–128, 130–137, 139, 140, 142, 145–233, 235–243, 245–267 | Ch 28 (Vividh): EN 31

AL-3 — SERIES MACHINE (AP · GP)**PEHCHAN – Question mein ye dikhe**

n-th term, sum of n terms, how many terms, arithmetic / geometric progression, AM / GM; समांतर श्रेणी, गुणोत्तर श्रेणी, पद

MASTER FORMULA – sirf ye yaad karo

$$\text{AP: } T_n = a + (n - 1)d, S_n = \frac{n}{2}(a + l) \quad \text{GP: } T_n = ar^{n-1}, S_n = a \frac{r^n - 1}{r - 1}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Term / jod / kitne term	$n = \frac{l-a}{d} + 1$ · AP ke 3 term: $a - d, a, a + d$ · AM = $\frac{a+b}{2}$, GM = $\sqrt{ab}$

TERMS – Har letter ka matlab

a = pehla term · d = farak · r = ratio · l = aakhri term · n = kitne term.

SOLVED – Asli book questions (row ke saath, verified)① **EN 1:** pehle 25 odd ka jod

$$25^2 = 625 \checkmark \text{ book ans (b)}$$

① **EN 8:** 3, 8, 13, ... ka 50th term

$$3 + 49 \times 5 = 248 \checkmark \text{ book ans (b)}$$

① **EN 2:** 1000–3000 ke beech 7 se divisible

$$428 - 142 = 286 \checkmark \text{ book ans (c)}$$

① **EN 10:** GP a 2, r 3 $\Rightarrow$ 5th

$$2 \cdot 3^4 = 162 \checkmark \text{ book ans (d)}$$

① **EN 34:** $6 + 11 + \dots + 71$

$$n = 14 \Rightarrow 7 \times 77 = 539 \checkmark \text{ book ans (a)}$$

① **EN 29:** $1 + 4 + 7 + \dots + X = 782$

$$\frac{n}{2}(3n - 1) = 782 \Rightarrow n = 23, X = 67 \checkmark \text{ book ans (d)}$$

IS FORMULA SE HONE WALE BOOK KE SAWAAL (47) EN 1–35 · HI 1–12**AL-4 — VENN MACHINE (sets)****PEHCHAN – Question mein ye dikhe**

students like tea / coffee / both / neither, passed in both, sets, union, intersection; दोनों, कोई नहीं, समुच्चय

MASTER FORMULA – sirf ye yaad karo

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Dono / koi nahi / sirf ek	Neither = total – union · sirf A = A – dono · teen: $A + B + C - (AB + BC + CA) + ABC$

TERMS – Har letter ka matlab $\cup$ = koi bhi ek (ya) · $\cap$ = dono.**SOLVED – Asli book questions (row ke saath, verified)**① **EN 283:** 14 ne A, 13 ne B, 2 ne dono

$$14 + 13 - 2 = 25 \checkmark \text{ book ans (d)}$$

① **EN 282:** 15 English, 5 dono $\Rightarrow$ only English

$$15 - 5 = 10 \checkmark \text{ book ans (b)}$$

① **EN 289:** 5000 mein 3200 tea, 2500 coffee, 1500 dono $\Rightarrow$ neither

$$5000 - 4200 = 800 \checkmark \text{ book ans (a)}$$

① **EN 288:** 5000; coffee 3500, tea 2500, dono 2000 $\Rightarrow$ neither

$$5000 - 4000 = 1000 \checkmark \text{ book ans (b)}$$

① **EN 277:** 500: cricket 350, kabaddi 125, neither 75 $\Rightarrow$ dono %

$$475 - 425 = 50 \Rightarrow 10\% \checkmark \text{ book ans (d)}$$

IS FORMULA SE HONE WALE BOOK KE SAWAAL (24) EN 267–289 | Ch 28 (Vividh): EN 75

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
AL-1	“VALUE DAALO” IDENTITY MACHINE	113	98
AL-2	EQUATION MACHINE (linear · quadratic · remainder)	118	153
AL-3	SERIES MACHINE (AP · GP)	35	12
AL-4	VENN MACHINE (sets)	23	0
→ IS-1	POWER MACHINE (same base $\Rightarrow$ powers barabar) (Ch 3)	4	2
→ IS-2	ROOT + SURD MACHINE ($\sqrt{}$ nikaalo · simplify · ration (Ch 3)	0	1
→ SIM-2	IDENTITY + SHORTCUT MACHINE (bada calculation kabh (Ch 4)	1	2
Total (book ke sawaal)		294/294 ✓	268/268 ✓
→ IS-1: EN 254, 255, 258, 291 HI 19, 41 · → IS-2: HI 71 · → SIM-2: EN 264 HI 43, 45			

Chapter 23

Trigonometry (Trikonmiti) त्रिकोणमिति

3 formulas (pehle 8) · English book 221 sawaal · Hindi book 124 sawaal

TG-1 — “0 DAALO” + VALUE TABLE MACHINE

PEHCHAN – Question mein ye dikhe

simplify trigonometric expression, prove / equals, $\sec \theta + \tan \theta = k$, if $\tan \theta = p/q$ find, minimum / maximum value of $a \sin^2 \theta + b \cos^2 \theta$, eliminate θ ; त्रिकोणमिति, का मान

MASTER FORMULA – sirf ye yaad karo

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

$\sin^2 \theta + \cos^2 \theta = 1$, $\sec^2 \theta - \tan^2 \theta = 1$, $\csc^2 \theta - \cot^2 \theta = 1$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Expression simplify / “equals”	$\theta = 45^\circ$ daalo (ya 30°), sawaal aur options dono ka maan milao · do option same aayein to doosra angle
② $\sec + \tan = k$ (cosec + cot)	$\sec - \tan = \frac{1}{k} \Rightarrow \sec = \frac{k+1}{2}$, $\sin \theta = \frac{k^2-1}{k^2+1} \cdot \tan + \cot = k \Rightarrow \tan^2 + \cot^2 = k^2 - 2$
③ Ek ratio diya $\Rightarrow$ baaki	Triangle banao: $\tan = p/b \Rightarrow h = \sqrt{p^2 + b^2} \Rightarrow \sin = p/h$, $\cos = b/h \cdot \frac{a \sin + b \cos}{c \sin + d \cos} \Rightarrow \cos$ se bhaag
④ $a \sin^2 + b \cos^2$ (min / max)	Min = chhota(a, b), max = bada(a, b) · $a \sin \theta + b \cos \theta$: max $\sqrt{a^2 + b^2}$
⑤ θ hatao / equation	Square karke jodo / ghatao taaki $\sin^2 + \cos^2 = 1$ bane · equation: options daalo

TERMS – Har letter ka matlab

p = perpendicular (saamne) · b = base · h = hypotenuse · cosec = $1/\sin$ · sec = $1/\cos$ · cot = $1/\tan$.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 14: $1/(1 + \sin \theta) + 1/(1 - \sin \theta)$
 $\frac{2}{\cos^2 \theta} = 2 \sec^2 \theta$ ✓ book ans (c)
- ① EN 105: $(1 - \cos^2 \theta)(\cot^2 \theta + 1) - 1$
 $\sin^2 \theta \csc^2 \theta - 1 = 0$ ✓ book ans (a)
- ② EN 41: $\sec + \tan = 1.25 \Rightarrow \sec - \tan$
 $\frac{1}{1.25} = 0.8$ ✓ book ans (b)
- ② EN 37: $\sec + \tan = 3.2 \Rightarrow \sec$
 $\frac{3.2+0.3125}{2} = 1.75625$ ✓ book ans (c)
- ③ EN 66: $\tan \theta = 4/3 \Rightarrow \sin + \cos$
 $\frac{4}{5} + \frac{3}{5} = \frac{7}{5}$ ✓ book ans (b)
- ③ EN 72: $5 \tan \theta = 4 \Rightarrow (3 \sin - 2 \cos)/(2 \sin + 3 \cos)$
 $\frac{12/5-2}{8/5+3} = \frac{2}{23}$ ✓ book ans (b)
- ④ EN 97: least of $2 \sin^2 + 3 \cos^2$
 2 ✓ book ans (d)
- ④ EN 98: minimum of $9 \sin^2 + 10 \cos^2$
 9 ✓ book ans (c)
- ⑤ EN 207: $x = 3 \cos A \cos B$, $y = 3 \cos A \sin B$, $z = 3 \sin A$
 $3^2 = 9$ ✓ book ans (a)
- ⑤ EN 213: $x = a \sin \theta$, $y = b \tan \theta \Rightarrow a^2/x^2 - b^2/y^2$
 $\operatorname{cosec}^2 - \cot^2 = 1$ ✓ book ans (d)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (227) EN 1–10, 13–15, 18, 22, 23, 26–49, 51–54, 56–63, 65–72, 74, 76, 78–87, 89–106, 108–111, 113–115, 117, 120, 130, 131, 133, 140, 141, 144–146, 149–151, 166, 168, 169, 206–221 · **HI** 1–15, 17–42, 44–51, 53–65, 68–83, 85–94, 99, 101, 106–109, 112, 114, 115 | *Ch 28 (Vividh)*: **EN** 23

TG-2 — ANGLE BADLO MACHINE (90° jodi $\cdot 180^\circ \cdot A \pm B$)

PEHCHAN – Question mein ye dikhe

$\sin 37^\circ / \cos 53^\circ, \tan 1^\circ \tan 2^\circ \dots \tan 89^\circ$, if $\sec x = \operatorname{cosec} y$ then $x + y$, $\cos(180^\circ - \theta)$, $\sin 75^\circ$, $\tan 15^\circ$, $\cos 2\theta$; पूरक कोण

MASTER FORMULA – sirf ye yaad karo

$$A + B = 90^\circ \Rightarrow \sin A = \cos B, \tan A \tan B = 1 \quad \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① 90° jodi / 180° , 360°	Jodi banao: $\tan 1^\circ \cdot \tan 89^\circ = 1 \cdot \sec x = \operatorname{cosec} y \Rightarrow x + y = 90^\circ \cdot 180^\circ - \theta$: sin same, cos / tan ka sign ulta $\cdot 360^\circ + \theta$: sab same
② $A \pm B$ / double angle	$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B \cdot \tan 15^\circ = 2 - \sqrt{3}$, $\tan 75^\circ = 2 + \sqrt{3} \cdot \sin 2\theta = 2 \sin \theta \cos \theta$, $\cos 2\theta = 1 - 2 \sin^2 \theta$

TERMS – Har letter ka matlab

A, B = do kon.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 155**: $\tan 1^\circ \cdot \tan 2^\circ \dots \tan 89^\circ$
jodiyani = 1 ✓ **book ans (c)**
- ① **EN 158**: $\cos 1^\circ \cdot \cos 2^\circ \dots \cos 179^\circ$
 $\cos 90^\circ = 0 \Rightarrow 0$ ✓ **book ans (a)**
- ① **EN 153**: $\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 80^\circ + \sin^2 85^\circ$
 $1 + 1 = 2$ ✓ **book ans (c)**
- ② **EN 77**: $\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$
 $(2 - \sqrt{3})^2 + (2 + \sqrt{3})^2 = 14$ ✓ **book ans (d)**
- ② **EN 50**: $(1 + \tan A)(1 + \tan B) = 2 \Rightarrow \tan(A + B)$
 $A + B = 45^\circ \Rightarrow 1$ ✓ **book ans (a)**
- ② **EN 64**: $\tan \alpha = \frac{1}{2}, \tan \beta = \frac{1}{3} \Rightarrow \alpha + \beta$
 $\frac{5/6}{1-1/6} = 1 \Rightarrow 45^\circ$ ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (75) EN 11, 12, 16, 17, 19–21, 24, 25, 50, 55, 64, 73, 75, 77, 88, 107, 112, 116, 118, 119, 121–129, 132, 134–139, 142, 143, 147, 148, 152–165, 167, 170 · **HI** 16, 43, 52, 66, 67, 84, 95–98, 100, 102–105, 110, 111, 113

TG-3 — UNCHAI-DOORI MACHINE (heights & distances)

PEHCHAN – Question mein ye dikhe

angle of elevation / depression, height of tower, shadow, ladder, observer moves towards tower; उन्नयन कोण, अवनमन कोण, मीनार, छाया

MASTER FORMULA – sirf ye yaad karo

$$\tan \theta = \frac{\text{Unchai}}{\text{Doori}} \quad 30^\circ : h = \frac{d}{\sqrt{3}} \quad 45^\circ : h = d \quad 60^\circ : h = d\sqrt{3}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Tower / chhaya / ladder	Master formula $\cdot$ do jagah se complementary kon (a, b doori) $\Rightarrow h = \sqrt{ab} \cdot 30^\circ \rightarrow 60^\circ$ chala d $\Rightarrow h = \frac{\sqrt{3}}{2} d \cdot$ ladder: Pythagoras

TERMS – Har letter ka matlab

$h = \text{unchai} \cdot d = \text{zameen par doori} \cdot \theta = \text{kon.}$

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 174:** 18 m aur 32 m se complementary angles
 $\sqrt{18 \cdot 32} = 24$ ✓ **book ans (d)**
- ① **EN 179:** 1.5 m aadmi, 24.5 m door, 26 m tower
 $\tan = 24.5/24.5 = 1 \Rightarrow 45^\circ$ ✓ **book ans (d)**
- ① **EN 178:** $200\sqrt{3}$ m tower, 200 m door se
 $\tan = \sqrt{3} \Rightarrow 60^\circ$ ✓ **book ans (b)**
- ① **EN 198:** pillars 5 m, 10 m, 12 m door $\Rightarrow$ tops ki doori
 $\sqrt{144 + 25} = 13$ ✓ **book ans (c)**
- ① **EN 181:** ladder 45° , foot $4\sqrt{2}$ m door
 $4\sqrt{2} \cdot \sqrt{2} = 8$ ✓ **book ans (b)**
- ① **EN 191:** tower $32\sqrt{3}$, depression 60°
 $\frac{32\sqrt{3}}{\sqrt{3}} = 32$ ✓ **book ans (a)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (44) EN 171–205 · HI 116–124

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
TG-1	“ θ DAALO” + VALUE TABLE MACHINE	129	97
TG-2	ANGLE BADLO MACHINE (90° jodi $\cdot 180^\circ \cdot A \pm B$)	57	18
TG-3	UNCHAI–DOORI MACHINE (heights & distances)	35	9
Total (book ke sawaal)		221/221 ✓	124/124 ✓

Chapter 24

Co-ordinate Geometry (Nirdeshank Jyamiti) निर्देशांक ज्यामिति

2 formulas (pehle 5) · English book 49 sawaal · Hindi book 17 sawaal

CG-1 — POINT MACHINE (doori · section · area)

PEHCHAN – Question mein ye dikhe

distance between points, midpoint, divides in the ratio, centroid, reflection, area of triangle, collinear; दूरी, मध्य बिंदु, अनुपात में विभाजित, क्षेत्रफल

MASTER FORMULA – sirf ye yaad karo

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right) \quad \text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Section / midpoint / centroid / reflection	Midpoint = average · centroid = teeno ka average · x-axis mein reflection (x, -y), y-axis (-x, y)
② Doorri / triangle kaaisa	Teeno side nikaalo: do barabar $\Rightarrow$ isosceles · $a^2 + b^2 = c^2 \Rightarrow$ right
③ Area / collinear	Master area formula · area = 0 $\Rightarrow$ ek line par · axes aur line ka triangle $\frac{1}{2}ab$

TERMS – Har letter ka matlab

(x₁, y₁), (x₂, y₂) = points · m : n = ratio.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 7: (-4, 4), (4, 0) ko 3 : 1
 $\left(\frac{12-4}{4}, \frac{0+4}{4} \right) = (2, 1)$ ✓ book ans (b)
- ① EN 8: (-1, 9), (11, 1) ko 3 : 1
 $\left(\frac{33-1}{4}, \frac{3+9}{4} \right) = (8, 3)$ ✓ book ans (d)
- ② EN 39: square ke consecutive vertices (2, -2), (5, 2)
 $\sqrt{9+16} = 5$ ✓ book ans (d)
- ② EN 40: (x, 7), (1, 15) ki doori 10
 $(x-1)^2 = 36 \Rightarrow 7, -5$ ✓ book ans (d)
- ③ EN 16: (0, 2), (2, 3), (3, 1)
 $\frac{1}{2}|0-2-3| = 2.5$ ✓ book ans (c)
- ③ EN 23: (1, 2), (-4, -3), (4, 1)
 $\frac{1}{2}|-4+4+20| = 10$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (46) EN 1, 3-18, 20-29, 35, 38-40 · HI 1-12, 14-16

CG-2 — LINE + CIRCLE MACHINE (slope)

PEHCHAN – Question mein ye dikhe

slope, equation of line, parallel / perpendicular, intercept, angle with x-axis, centre and radius of circle; ढाल, रेखा का समीकरण, वृत्त

MASTER FORMULA – sirf ye yaad karo

$$m = \frac{y_2 - y_1}{x_2 - x_1} = -\frac{a}{b} \quad y = mx + c \quad \parallel: m_1 = m_2, \quad \perp: m_1 m_2 = -1$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Line	$ax + by + c = 0 \Rightarrow m = -a/b$, x-intercept $= -c/a$, y-intercept $= -c/b$ · $\tan \theta = m$ · point line par $\Rightarrow$ daal ke satisfy
② Circle	$(x - h)^2 + (y - k)^2 = r^2$ · $x^2 + y^2 + 2gx + 2fy + c = 0 \Rightarrow$ centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$

TERMS – Har letter ka matlab

$m = \text{slope}$ · $c = \text{y-intercept}$ · $(h, k) = \text{centre}$.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 30:** $3x - 2y - 11 = 0$ aur $x + y = 7 \Rightarrow 3\alpha + 5\beta$
 $(5, 2) \Rightarrow 25$ ✓ **book ans (c)**
- ① **EN 31:** $kx - 3y = 6$ through $(3, 2)$
 $3k - 6 = 6 \Rightarrow 4$ ✓ **book ans (a)**
- ① **EN 34:** $(-2, 5)$ aur $(1, 3)$ se line
 $m = -2/3 \Rightarrow 2x + 3y - 11 = 0$ ✓ **book ans (d)**
- ② **EN 32:** centre $(-2, 3)$, radius 4
 $x^2 + y^2 + 4x - 6y - 3 = 0$ ✓ **book ans (b)**
- ② **EN 43:** $(2, 3)$ se $x^2 + y^2 = 4$ par tangent
 $\sqrt{4 + 9 - 4} = 3$ ✓ **book ans (b)**
- ② **EN 44:** $(2, 5)$ se tangent $\sqrt{37}$, circle $x^2 + y^2 - 5x + 4y + k$
 $39 + k = 37 \Rightarrow -2$ ✓ **book ans (a)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (20) EN 2, 19, 30–34, 36, 37, 41–49 · **HI** 13, 17

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
CG-1	POINT MACHINE (doori · section · area)		31
CG-2	LINE + CIRCLE MACHINE (slope)		18
Total (book ke sawaal)		49/49 ✓	17/17 ✓

Chapter 25

Geometry (Jyamiti) ज्यामिति

3 formulas (pehle 8) · English book 276 sawaal · Hindi book 207 sawaal

GE-1 — KON (ANGLE) MACHINE (line · triangle · polygon · quadrilateral)

PEHCHAN – Question mein ye dikhe

complementary / supplementary angle, parallel lines and transversal, angles of a triangle, exterior angle, bisectors meet at O find $\angle BOC$, regular polygon interior / exterior angle, number of sides / diagonals, parallelogram / cyclic quadrilateral angles; कोण, समांतर रेखाएँ, बहुभुज

MASTER FORMULA – sirf ye yaad karo

Seedhi rekha = 180° Triangle = 180° Bahari kon = door ke do andar ke kon ka jod n -gon ka jod = $(n - 2) \times 180^\circ$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Lines & angles	Complementary 90° · supplementary 180° · vertically opposite / alternate (Z) / corresponding (F) barabar · co-interior jod 180°
② Triangle ke kon	$\angle BOC$: incentre $90^\circ + \frac{A}{2}$ · excentre $90^\circ - \frac{A}{2}$ · circumcentre $2A$ · orthocentre $180^\circ - A$
③ Polygon	Exterior = $\frac{360^\circ}{n}$ · interior = $180^\circ - \frac{360^\circ}{n}$ · diagonals = $\frac{n(n-3)}{2}$ · $n = \frac{360}{180 - \text{interior}}$
④ Chaar bhuja (quadrilateral)	Jod 360° · cyclic: saamne ke kon ka jod 180° · parallelogram: saamne barabar, paas ke jod 180° · rhombus: $a^2 = (\frac{d_1}{2})^2 + (\frac{d_2}{2})^2$

TERMS – Har letter ka matlab

n = bhujaon ki ginti · A = triangle ka kon.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 236: supplementary, bada 36° zyada
 $\frac{180-36}{2} = 72^\circ$ ✓ book ans (a)
- ① EN 237: complementary: bada = $3 \times$ chhota – 6
 $4x - 6 = 90 \Rightarrow x = 24 \Rightarrow 66^\circ$ ✓ book ans (c)
- ② EN 1: $\angle B, \angle C$ ke bisectors P par; $\angle BPC = 128^\circ$
 $90 + \frac{A}{2} = 128 \Rightarrow A = 76^\circ$ ✓ book ans (b)
- ② EN 5: $\angle A = 75^\circ$, bisectors D par $\Rightarrow \angle BDC$
 $90 + 37.5 = 127.5^\circ$ ✓ book ans (d)
- ③ EN 181: regular octagon ka interior
 $180 - 45 = 135^\circ$ ✓ book ans (c)
- ③ EN 182: interior $168^\circ \Rightarrow n$
 $\frac{360}{12} = 30$ ✓ book ans (c)
- ④ EN 67: cyclic, saamne ke kon 3 : 7
 $180 \cdot \frac{7}{10} = 126^\circ$ ✓ book ans (d)
- ④ EN 70: teen kon 110, 55, 125
 $360 - 290 = 70^\circ$ ✓ book ans (c)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (336) EN 1, 3–6, 17–19, 21–24, 26, 27, 29, 30, 33, 34, 37–39, 44, 47, 54, 59, 65, 67–121, 138, 181–260, 262, 264–269, 271, 272, 274–276 · HI 1–38, 42–44, 46, 47, 49, 51, 53, 54, 61, 62, 65, 66, 69–71, 73, 77, 78, 80, 81, 83, 86–89, 91, 97, 101–160, 173–207 | Ch 28 (Vividh): HI 29

GE-2 — LAMBAI MACHINE (Pythagoras + similar triangle)

PEHCHAN – Question mein ye dikhe

similar triangles, $DE \parallel BC$, angle bisector divides side, mid-point theorem, area ratio, inradius / circumradius, centroid, median, Pythagoras; समरूप त्रिभुज, माधिका, अंतःत्रिज्या

MASTER FORMULA – sirf ye yaad karo

$$c^2 = a^2 + b^2 \quad \text{Similar: side ratio } k \Rightarrow \text{area ratio } k^2$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Similar / BPT / bisector / midpoint	BPT: $\frac{AD}{DB} = \frac{AE}{EC}$ · bisector: $\frac{BD}{DC} = \frac{AB}{AC}$ · midpoints $\Rightarrow DE = \frac{1}{2} BC$ · congruent (SSS, SAS, ASA, RHS): letters same kram mein milao
② r, R, centroid, triplets	Right: $r = \frac{a+b-c}{2}$, $R = \frac{c}{2}$ · equilateral: $r = \frac{a}{2\sqrt{3}}$, $R = \frac{a}{\sqrt{3}}$ · koi bhi: $r = \frac{\text{Area}}{s}$ · centroid 2 : 1

TERMS – Har letter ka matlab

c = hypotenuse · k = sides ka ratio · s = semi-perimeter.

SOLVED – Asli book questions (row ke saath, verified)

① **EN 2:** PQ 12, PR 15, QR 18; bisector S $\Rightarrow$ SR
 $18 \cdot \frac{15}{27} = 10$ ✓ **book ans (c)**

① **EN 8:** AB 4, AC 5, BC 6; bisector AD $\Rightarrow$ BD
 $6 \cdot \frac{4}{9} = 2.67$ ✓ **book ans (d)**

① **EN 11:** D, E midpoints, BC = 6
DE = 3 ✓ **book ans (b)**

② **EN 12:** 3-4-5 ka inradius : circumradius
1 : 2.5 = 2 : 5 ✓ **book ans (b)**

② **EN 32:** right triangle 4 & 3 $\Rightarrow$ inradius
 $\frac{3+4-5}{2} = 1$ ✓ **book ans (a)**

② **EN 64:** 9, 40, 41 ka inradius
 $\frac{9+40-41}{2} = 4$ ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (76) EN 2, 7–12, 14–16, 20, 25, 28, 31, 32, 35, 36, 40–43, 45, 46, 48–53, 55–58, 60–64, 66, 155, 263, 273 · **HI** 39–41, 45, 48, 50, 52, 55–60, 63, 64, 67, 68, 72, 74–76, 79, 82, 84, 85, 90, 92–96, 98–100

GE-3 — CIRCLE MACHINE (chord · tangent · kon)**PEHCHAN – Question mein ye dikhe**

chord at distance from centre, angle subtended at centre / circumference, angle in semicircle, tangent from external point, two tangents, common tangent, intersecting chords; जीवा, स्पर्श रेखा, केंद्र

MASTER FORMULA – sirf ye yaad karo

$$r^2 = d^2 + \left(\frac{\text{chord}}{2}\right)^2 \quad \text{Tangent}^2 = d^2 - r^2 \quad \angle \text{centre} = 2 \times \angle \text{paridhi}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Chord / kon	Semicircle mein kon 90° · same segment ke kon barabar · chords P par katen: $PA \cdot PB = PC \cdot PD$
② Tangent	Bahar ke point se do tangent barabar · $\angle APB + \angle AOB = 180^\circ$ · tangent-chord ka kon = doosre (alternate) segment ka kon · direct common $\sqrt{d^2 - (r_1 - r_2)^2}$, transverse $\sqrt{d^2 - (r_1 + r_2)^2}$

TERMS – Har letter ka matlab

r = radius · d = centre se doori · chord = jeeva.

SOLVED – Asli book questions (row ke saath, verified)① **EN 129:** chord 24, centre se 5

$$\sqrt{144 + 25} = 13 \checkmark \text{ book ans (a)}$$

① **EN 130:** $\angle ACB = 46.5^\circ \Rightarrow \angle AOB$

$$93^\circ \checkmark \text{ book ans (b)}$$

① **EN 131:** PQ diameter, $\angle RPO = 42^\circ \Rightarrow \angle RQP$

$$90 - 42 = 48^\circ \checkmark \text{ book ans (a)}$$

② **EN 156:** tangent 21, centre se 29 $\Rightarrow r$

$$\sqrt{841 - 441} = 20 \checkmark \text{ book ans (a)}$$

② **EN 159:** AP 15, r 8 $\Rightarrow OP$

$$17 \checkmark \text{ book ans (c)}$$

② **EN 161:** r 4, point 5 door $\Rightarrow$ tangent

$$3 \checkmark \text{ book ans (d)}$$

IS FORMULA SE HONE WALE BOOK KE SAWAAL (72) EN 13, 122–137, 139–154, 156–180, 261, 270 · HI 161–172**COVERAGE CHECK – Har sawaal ka formula**

Formula	Naam	EN Qs	HI Qs
GE-1	KON (ANGLE) MACHINE (line · triangle · polygon · quadrilateral)	174	161
GE-2	LAMBAI MACHINE (Pythagoras + similar triangle)	42	34
GE-3	CIRCLE MACHINE (chord · tangent · kon)	60	12
Total (book ke sawaal)		276/276 ✓	207/207 ✓

Chapter 26

Statistics & Probability (Sankhyiki aur Prayikta) सांख्यिकी और प्रायिकता

3 formulas (pehle 6) · English book 265 sawaal · Hindi book 121 sawaal

SP-1 — MEAN–MEDIAN–MODE MACHINE

PEHCHAN – Question mein ye dikhe

mean / average of data, frequency table, median, mode, grouped data, missing frequency, relation between mean, median and mode; माध्य, माध्यिका, बहुलक

MASTER FORMULA – sirf ye yaad karo

$$\bar{x} = \frac{\sum f x}{\sum f} \quad \text{Median} = \text{sort karke beech wala} \quad \text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Mean	$\frac{\sum x}{n}$ · table $\Rightarrow \frac{\sum f x}{\sum f}$ (grouped: class ka mid-value) · missing k: $\sum f x = \bar{x} \sum f$
② Median	n odd $\Rightarrow \left(\frac{n+1}{2}\right)^{th}$ · even $\Rightarrow$ beech ke do ka average · grouped $l + \frac{\frac{n}{2} - cf}{f} \times h$
③ Mode	Sabse zyada baar · grouped $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
④ Teeno ka rishta	Mode = 3 Median - 2 Mean · Median = $\frac{2 \text{ Mean} + \text{Mode}}{3}$

TERMS – Har letter ka matlab

f = frequency · l = class ki lower limit · h = class size · cf = pichli cumulative frequency.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 1: 10 values ka mean, har ek +10
 $14 + 10 = 24$ ✓ book ans (d)
- ① EN 3: 33, x, 47, 83, 109 ka mean 67 $\Rightarrow$ 50, 64, 100, 126, x ka mean
 $x = 63 \Rightarrow 403/5 = 80.6$ ✓ book ans (c)
- ② EN 63: 15, 2, 7, 8, 11, 5, 14
 2, 5, 7, 8, 11, 14, 15 ✓ book ans (a)
- ② EN 61: 1, 1.1, 2.3, 0.1, 0.9, 5, 3.9, 2.5, 4.2, 4.6
 $\frac{2.3+2.5}{2} = 2.4$ ✓ book ans (b)
- ③ EN 111: 5, 4, 4, 6, 7, 6, 9, 7, 6, 5
 6 teen baar ✓ book ans (d)
- ③ EN 112: 15, 26, 15, 29, 19, 18, 19, 15, 24, 23, 15, 19
 15 chaar baar ✓ book ans (d)
- ④ EN 175: mean 42, mode 60 $\Rightarrow$ median
 $\frac{84+60}{3} = 48$ ✓ book ans (b)
- ④ EN 176: mode 24.6, mean 20.1 $\Rightarrow$ median
 $\frac{40.2+24.6}{3} = 21.6$ ✓ book ans (c)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (285) EN 1–17, 19–65, 67–70, 72–133, 171–183, 187, 188, 191–193, 195, 196, 198–206, 213–215, 218–224, 226, 231, 233, 234, 236, 239, 241 · HI 1–52, 54–87, 93–107, 109–113, 116–118

SP-2 — SPREAD MACHINE (range · variance · SD)

PEHCHAN – Question mein ye dikhe

range, variance, standard deviation, coefficient of variation, class mark, class size; परास, प्रसरण, मानक विचलन

MASTER FORMULA – sirf ye yaad karo

$$\text{Range} = \max - \min \quad \sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 \quad \sigma = \sqrt{\sigma^2} \quad CV = \frac{\sigma}{\bar{x}} \times 100$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho**Sawaal mein ye ho****Value aise daalo $\Rightarrow$ answer**① **Range / SD / class**Sab mein $+k \Rightarrow$ SD same $\cdot \times k \Rightarrow$ SD $\times k \cdot$ class mark $= \frac{\text{lower} + \text{upper}}{2} \cdot 1 \dots n$ ka $\sigma^2 = \frac{n^2 - 1}{12}$ **TERMS – Har letter ka matlab** σ = standard deviation $\cdot \bar{x}$ = mean.**SOLVED – Asli book questions (row ke saath, verified)**① **EN 134:** mean 25, SD 5 $\Rightarrow$ CV20% ✓ **book ans (c)**① **EN 137:** {11, 7, 10, 13, 9} ka SDmean 10, $\sigma^2 = \frac{20}{5} = 4 \Rightarrow 2$ ✓ **book ans (b)**① **EN 139:** variance 36, har value double $\Rightarrow$ SD $6 \times 2 = 12$ ✓ **book ans (a)**① **EN 152:** 3, 6, 5, 2, 4 ka variance $\frac{1+4+1+4+0}{5} = 2$ ✓ **book ans (c)**① **EN 157:** 9, 5, 9, 3, 4, 7, 8, 4, 8, 9, 5, 9 ka range $9 - 3 = 6$ ✓ **book ans (c)**① **EN 159:** SD 4.5, har value $+K \Rightarrow$ variance $4.5^2 = 20.25$ ✓ **book ans (b)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (72) EN 18, 66, 71, 134–170, 184–186, 189, 190, 194, 197, 207–211, 216, 217, 225, 227–230, 232, 235, 237, 238, 240 $\cdot$ HI 53, 88, 108, 114, 115, 119–121

SP-3 — PROBABILITY MACHINE**PEHCHAN – Question mein ye dikhe**

probability, coin, dice, cards, balls drawn, at least one; प्रायिकता, सिक्का, पासा, ताल

MASTER FORMULA – sirf ye yaad karo

$$P = \frac{\text{Favourable}}{\text{Total}} \quad P(\text{nahi}) = 1 - P \quad {}^nC_r = \frac{n!}{r!(n-r)!}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho**Sawaal mein ye ho****Value aise daalo $\Rightarrow$ answer**① **Coin / dice / cards / balls**n coins $2^n \cdot$ 2 dice 36 $\cdot$ cards 52 (red 26, suit 13, face 12, ace 4) $\cdot$ balls: $\frac{\text{fav } C}{\text{total } C} \cdot$ at least one $= 1 -$ koi nahi**TERMS – Har letter ka matlab**Favourable = jo chahiye wale tarike $\cdot$ Total = saare tarike.

SOLVED – Asli book questions (row ke saath, verified)① **EN 254:** 3 coins, kam se kam ek head

$$1 - \frac{1}{8} = \frac{7}{8} \quad \checkmark \text{ book ans (d)}$$

① **EN 248:** alphabet mein vowel

$$\frac{5}{26} \quad \checkmark \text{ book ans (a)}$$

① **EN 246:** 2 black, 6 green, 4 yellow; 2 balls same (green)

$$\frac{{}^6C_2}{{}^{12}C_2} = \frac{15}{66} = \frac{5}{22} \quad \checkmark \text{ book ans (d)}$$

① **EN 253:** 2 cards: ek red, ek black

$$\frac{26 \cdot 26}{13 \cdot 26} = \frac{26}{51} \quad \checkmark \text{ book ans (a)}$$

① **EN 260:** dice do baar, sum 10

$$\frac{3}{36} = \frac{1}{12} \quad \checkmark \text{ book ans (c)}$$

① **EN 261:** face card

$$\frac{12}{52} = \frac{3}{13} \quad \checkmark \text{ book ans (c)}$$

IS FORMULA SE HONE WALE BOOK KE SAWAAL (27) EN 242–265 · HI 89–91**COVERAGE CHECK – Har sawaal ka formula**

Formula	Naam	EN Qs	HI Qs
SP-1	MEAN–MEDIAN–MODE MACHINE	176	109
SP-2	SPREAD MACHINE (range · variance · SD)	64	8
SP-3	PROBABILITY MACHINE	24	3
→ RP-2	CROSS-GUNA MACHINE (proportion · ratio badlo) (Ch 9)	1	1
Total (book ke sawaal)		265/265 ✓	121/121 ✓
→ RP-2: EN 212 HI 92			

Chapter 27

Data Interpretation (Aankdon ka Vishleshan) आंकड़ों का विश्लेषण

1 formula (pehle 4) · English book 217 sawaal · Hindi book 174 sawaal

DI-1 — PADHO-UTHAO-DAALO MACHINE (pie · % · ratio · average)

PEHCHAN – Question mein ye dikhe

pie chart, bar graph, table, line graph — percentage increase, ratio, difference, total, average, maximum / minimum; पाई चार्ट, आलेख, तालिका

MASTER FORMULA – sirf ye yaad karo

$$1\% = 3.6^\circ \quad \% \text{ change} = \frac{\text{Naya} - \text{Purana}}{\text{Purana}} \times 100 \quad \text{Ratio} = \frac{\sum \text{pehla}}{\sum \text{doosra}} \quad \text{Avg} = \frac{\text{Jod}}{\text{Ginti}}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Pie chart	Hissa = $\frac{\text{kon}}{360} \times \text{Total} = \frac{\%}{100} \times \text{Total}$ · kon = % $\times 3.6$ · do hisson ka ratio = kon ka ratio
② % / % badlav	% change master formula (base = purana) · A is $\frac{A}{B} \times 100\%$ of B
③ Ratio / farak / total	Sirf zaroori numbers uthao · common lakh / hazaar cancel · har row ka farak likh ke max / min chuno
④ Average / max-min / seedha padho	Jod $\div$ ginti · “average se zyada kitne” $\Rightarrow$ gino · max / min $\Rightarrow$ chart se seedha padho

TERMS – Har letter ka matlab

Purana = base value · kon = sector ka angle.

SOLVED – Asli book questions (row ke saath, verified)

- ① **EN 3:** Delhi 30% $\Rightarrow$ central angle
 $30 \times 3.6 = 108^\circ$ ✓ **book ans (c)**
- ① **EN 55:** Rice 40°, total 9000 tonnes
 $\frac{40}{360} \times 9000 = 1000$ ✓ **book ans (b)**
- ② **EN 93:** CO₂: 2016 = 110, 2017 = 150
 $\frac{40}{110} \times 100 = 36.36\%$ ✓ **book ans (b)**
- ② **EN 97:** year 1 total 36, year 3 total 42
 $\frac{6}{36} \times 100 = 16.67\%$ ✓ **book ans (b)**
- ③ **EN 82:** Physics 300, 425, 475, 515; Chemistry 450, 550, 590, 700 $\Rightarrow$ farak sabse kam
 150, 125, 115, 185 $\Rightarrow$ 2005 ✓ **book ans (a)**
- ③ **EN 100:** quarter price: 36/40, 80/80, 65/40, 43/40
 4, 0, 25, 3 $\Rightarrow$ quarter 3 ✓ **book ans (d)**
- ④ **EN 85:** bulbs 225, 100, 150, 200, 75, 90 $\Rightarrow$ average ke paas
 avg 140 $\Rightarrow$ Wednesday (150) ✓ **book ans (a)**
- ④ **EN 102:** goals 2, 1, 0, 4, 3, 2 $\Rightarrow$ average
 $12 \div 6 = 2$ ✓ **book ans (d)**

IS FORMULA SE HONE WALE BOOK KE SAWAAL (399) EN 1-217 · HI 1-174 | Ch 28 (Vividh): EN 22, 56, 59, 60, 68 · HI 48, 49, 53

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
DI-1	PADHO-UTHAO-DAALO MACHINE (pie · % · ratio · average)	217	174
Total (book ke sawaal)		217/217 ✓	174/174 ✓

Chapter 28

Miscellaneous (Vividh) विविध

1 formula (pehle 4) · English book 82 sawaal · Hindi book 59 sawaal

MS-1 — POORA–HISSA + EK-KA-KITNA + ULTA CHALO MACHINE

PEHCHAN – Question mein ye dikhe

fraction of the remaining, map scale 1 : n, unitary method, doubles every day, monkey climbs and slips, digit rule (odd digit +1 ...), key presses; शेष का भाग, पैमाना, ऐकिक नियम

MASTER FORMULA – sirf ye yaad karo

$$\text{Poora} = \frac{\text{Bacha hua}}{(1-a)(1-b)(1-c)\dots} \quad 1 \text{ ka} = \frac{\text{Kul}}{\text{Ginti}} \quad \text{Puzzle} \Rightarrow \text{aakhri se ulta chalo / options daalo}$$

VALUE DAALO – sawaal jaisi row chuno, numbers rakho

Sawaal mein ye ho	Value aise daalo $\Rightarrow$ answer
① Hisse diye, bacha diya	“Baaki ka” $\Rightarrow$ guna chain $(1-a)(1-b)$ · “total ka” $\Rightarrow$ jod-ghata · bacha fraction = diya number $\Rightarrow$ poora = number $\div$ fraction
② Units / scale / ek ka kitna	Scale 1 : n $\Rightarrow$ asli = map $\times$ n · pehle ek ka, phir guna: $\frac{a_1}{b_1} = \frac{a_2}{b_2} \cdot 1 \text{ km} = 10^5 \text{ cm}$
③ Puzzle patterns	Doubling: poora t din $\Rightarrow$ aadha $(t-1)$ · climb-slip: $\frac{H-u}{u-d}$ cycles + aakhri chadhai · A+B, B+C, C+A $\Rightarrow$ teeno = jod $\div$ 2
④ Digit rule puzzles	Har ank ke neeche naya ank likho, phir sirf jo poocha wahi padho (sabse bada, 2nd from left ...)

TERMS – Har letter ka matlab

a, b, c = baari-baari diye hisse · H = unchai · u = upar · d = phisalna.

SOLVED – Asli book questions (row ke saath, verified)

- ① EN 7: $\frac{1}{4}$ bhara, 10 cup daale $\Rightarrow \frac{3}{4}$
 $\frac{\frac{10}{3}-\frac{1}{4}}{\frac{3}{4}-\frac{1}{4}} = 20$ ✓ book ans (b)
- ① EN 36: drum $\frac{3}{4} \rightarrow 9$ litre nikaale $\rightarrow \frac{1}{2}$
 $\frac{9}{\frac{1}{4}} = 36$ ✓ book ans (b)
- ② EN 25: 1 : 50000, map par 12 cm
 600000 cm = 6 km ✓ book ans (b)
- ② EN 30: 1 : 150000, map par 6 cm
 9 km ✓ book ans (c)
- ③ EN 16: har minute double, 30 min mein poori $\Rightarrow$ aadhi
 29 min ✓ book ans (a)
- ③ EN 80: 12 m ped, +2 –1
 $\frac{2-10}{1} + 1 = 21$ ✓ book ans (b)
- ④ HI 1: 4723361: sam $\div$ 2, vishham + 2 $\Rightarrow$ bada + chhota
 2 9 1 5 5 3 3 $\Rightarrow 9 + 1 = 10$ ✓ book ans (b)
- ④ HI 8: 42514563: vishham + 3, sam – 1 $\Rightarrow$ bada – chhota
 3 1 8 4 3 8 5 6 $\Rightarrow 8 - 1 = 7$ ✓ book ans (b)

IS FORMULA SE HONE WALE BOOK KE SAWAAL (95) EN 1, 3, 5–9, 11, 13–19, 25, 27, 30, 32–36, 38, 41, 44, 45, 47–50, 55, 58, 61–63, 66, 69, 72–74, 77, 78, 80–82 · HI 1–22, 24–28, 30–47, 51, 54, 57, 58

COVERAGE CHECK – Har sawaal ka formula

Formula	Naam	EN Qs	HI Qs
MS-1	POORA-HISSA + EK-KA-KITNA + ULTA CHALO MACHINE	46	49
→ NS-3	DO-NUMBER + SERIES MACHINE (Ch 1)	1	0
→ IS-1	POWER MACHINE (same base ⇒ powers barabar) (Ch 3)	2	0
→ SIM-1	VBODMAS + “?” + APPROX MACHINE (Ch 4)	1	0
→ SIM-2	IDENTITY + SHORTCUT MACHINE (bada calculation kabh (Ch 4)	1	0
→ LH-2	SAATH KAB / SABSE BADA NAAP MACHINE (application + (Ch 5)	1	1
→ PC-1	PERCENT = HISSA MACHINE (of · marks · income · vot (Ch 6)	3	1
→ PC-2	MULTIPLIER MACHINE (badha · ghata · do baar · popu (Ch 6)	1	0
→ PL-1	CP → SP MULTIPLIER MACHINE (profit · loss · MP · “ (Ch 7)	1	0
→ DS-1	DISCOUNT MULTIPLIER MACHINE (Ch 8)	1	0
→ RP-1	“1 HISSA” MACHINE (share · jodna · coins · bachat) (Ch 9)	2	0
→ RP-2	CROSS-GUNA MACHINE (proportion · ratio badlo) (Ch 9)	1	1
→ WT-1	LCM UNIT MACHINE (Kaam = Rate × Din) (Ch 11)	2	1
→ MX-1	ALLIGATION CROSS MACHINE (milaane ka ratio) (Ch 12)	1	0
→ AV-1	TOTAL MACHINE (Total = Average × n) (Ch 17)	1	0
→ ST-1	D = S × T MACHINE (units · late-early · ratio · do (Ch 18)	2	0
→ ST-2	AVERAGE + RELATIVE SPEED MACHINE (Ch 18)	1	0
→ MN-1	2-D SHAPE TABLE (area · perimeter) (Ch 21)	1	1
→ MN-2	3-D SHAPE TABLE (volume · surface · pighlao) (Ch 21)	3	0
→ MN-3	SIZE-CHANGE MACHINE (power dekho, ratio guna) (Ch 21)	1	1
→ AL-1	“VALUE DAALO” IDENTITY MACHINE (Ch 22)	1	0
→ AL-2	EQUATION MACHINE (linear · quadratic · remainder) (Ch 22)	1	0
→ AL-4	VENN MACHINE (sets) (Ch 22)	1	0
→ TG-1	“0 DAALO” + VALUE TABLE MACHINE (Ch 23)	1	0
→ GE-1	KON (ANGLE) MACHINE (line · triangle · polygon · q (Ch 25)	0	1
→ DI-1	PADHO-UTHAO-DAALO MACHINE (pie · % · ratio · avera (Ch 27)	5	3
Total (book ke sawaal)		82/82 ✓	59/59 ✓
→ AL-1: EN 10 · → AL-2: EN 31 · → AL-4: EN 75 · → AV-1: EN 28 · → DI-1: EN 22, 56, 59, 60, 68 HI 48, 49, 53 · → DS-1: EN 79 · → GE-1: HI 29 · → IS-1: EN 12, 57 · → LH-2: EN 76 HI 59 · → MN-1: EN 54 HI 23 · → MN-2: EN 2, 24, 52 · → MN-3: EN 67 HI 52 · → MX-1: EN 4 · → NS-3: EN 42 · → PC-1: EN 43, 53, 64 HI 50 · → PC-2: EN 39 · → PL-1: EN 65 · → RP-1: EN 26, 51 · → RP-2: EN 70 HI 55 · → SIM-1: EN 20 · → SIM-2: EN 21 · → ST-1: EN 37, 46 · → ST-2: EN 40 · → TG-1: EN 23 · → WT-1: EN 29, 71 HI 56			

Appendix A – Unseen Test (random sawaal, sirf naye cards se)

Har formula ke 3 sawaal computer ne random chune (seed 20261002), sirf unme se jo pehle kabhi hal nahi kiye gaye the. Har ek ko sirf us formula ki VALUE DAALO row se hal kiya aur book ki answer key se milaya: **159/159 ✓**.

Formula • row	Sawaal	Values daale $\Rightarrow$ answer	Book
NS-1 ④	EN 297 Ch 1	$1 \Rightarrow 1^{102} \times 1^{52} \Rightarrow$ unit digit 1	✓ (b)
NS-1 ①	HI 7 Ch 1	$4 \Rightarrow$ last 2 digits $\div 4$	✓ (b)
NS-1 ④	EN 288 Ch 1	$1 \times 1 \Rightarrow 1$	✓ (c)
NS-2 ③	HI 63 Ch 1	$1771 = 7 \times 11 \times 23 \Rightarrow 23 + 59 = 82 \Rightarrow 23 \times 59 = 1357$	✓ (a)
NS-2 ③	HI 116 Ch 1	beech wala = $HCF(357, 609) = 21 \Rightarrow 17, 21, 29 \Rightarrow 67$	✓ (b)
NS-2 ②	HI 146 Ch 1	odd $\Rightarrow 2$ hatao: $(5+1)(4+1)(6+1) = 210$	✓ (c)
NS-3 ②	EN 70 Ch 1	options daalo: $25 \times 23 = 575$ ✓	✓ (c)
NS-3 ①	EN 167 Ch 1	$8x + 4 = 100 \Rightarrow x = 12$	✓ (a)
NS-3 ①	HI 92 Ch 1	$x + \frac{x}{3} = 12 \Rightarrow x = 12 \div \frac{4}{3} = 9$	✓ (b)
DF-1 ②	EN 86 Ch 2	$12/30 = 2/5 \Rightarrow$ neeche sirf 5 $\Rightarrow$ terminating	✓ (b)
DF-1 ①	EN 71 Ch 2	$4 + \frac{567-5}{990} = 4\frac{281}{495}$	✓ (a)
DF-1 ①	HI 20 Ch 2	$\frac{7900-7}{9990} \Rightarrow 0.7900$	✓ (c)
DF-2 ②	HI 98 Ch 2	$\frac{5}{4} - \frac{3}{5} = \frac{13}{20}$	✓ (b)
DF-2 ②	HI 99 Ch 2	$\frac{1}{5} - \frac{1}{12} = \frac{7}{60}$	✓ (c)
DF-2 ①	EN 20 Ch 2	decimal: 0.2, 0.25, 0.125, 0.025 $\Rightarrow 25/1000$	✓ (b)
IS-1 ②	EN 106 Ch 3	$\sqrt{\frac{45 \times 20}{12 \times 3}} = \sqrt{25} = 5$	✓ (d)
IS-1 ③	HI 8 Ch 3	$x = \frac{\sqrt{455625}}{27} = 25 \Rightarrow \sqrt{x} = 5$	✓ (c)
IS-1 ③	EN 82 Ch 3	$x^2 + 4x^2 = 625 \Rightarrow x = \sqrt{125}$	✓ (c)
IS-2 ②	EN 97 Ch 3	$\frac{0.0013}{13} = 0.0001$	✓ (c)
IS-2 ①	HI 34 Ch 3	last digit 4 $\Rightarrow 2/8$; $1831 \Rightarrow 42$ ($42 \times 43 = 1806 \leq 1831$) $\Rightarrow 428$	✓ (c)
IS-2 ③	HI 86 Ch 3	$8\sqrt{5} = 17.84 \Rightarrow 11\sqrt{5} = 24.53$	✓ (b)
SIM-1 ④	EN 153 Ch 4	$\frac{1}{3} \times \frac{7}{3} \times \frac{24}{5} \div \frac{16}{15} = \frac{7}{2}$	✓ (d)
SIM-1 ①	HI 41 Ch 4	$56 \div 7 - 21 = -13$	✓ (c)
SIM-1 ④	EN 173 Ch 4	of pehle: $\frac{11}{5} - \left(\frac{2}{5} - \frac{1}{5}\right) + \frac{3}{2} = \frac{7}{2}$	✓ (a)
SIM-2 ③	EN 105 Ch 4	$\frac{(2\sqrt{2}+\sqrt{7})^2}{8-7} = 15 + 4\sqrt{14} \Rightarrow y = 4$	✓ (d)
SIM-2 ③	EN 240 Ch 4	$\sum n^3 = 36^2 \Rightarrow (6^4)^{-5/2} = 6^{-10}$	✓ (c)
SIM-2 ③	EN 206 Ch 4	$27 + 41 + 24 - 43 = 49$	✓ (c)

LH-1 ①	EN 140 Ch 5	$4752 = 2^4 \cdot 3^3 \cdot 11 \Rightarrow \text{teesra} = 2 \cdot 3^3 = 54$	✓ (a)
LH-1 ①	EN 55 Ch 5	do prime $\Rightarrow$ HCF = 1	✓ (a)
LH-1 ②	EN 127 Ch 5	$N = \frac{9 \times 180}{36} = 45$	✓ (c)
LH-2 ④	EN 186 Ch 5	$\text{LCM}(3, 7) = 21 \text{ s} \Rightarrow 60 \div 21 \Rightarrow 2$	✓ (a)
LH-2 ①	EN 214 Ch 5	$\text{LCM} = 4725 \Rightarrow 4725 - 3$	✓ (d)
LH-2 ①	HI 19 Ch 5	$\text{LCM } 1001 \Rightarrow 9009 + 4$	✓ (d)
PC-1 ①	EN 71 Ch 6	$\frac{40000}{200000} \times 100 = 20\%$	✓ (a)
PC-1 ④	EN 240 Ch 6	Total = $6800 \div 34\% = 20000 \Rightarrow \text{margin } (48 - 34)\% = 2800$, invalid $18\% = 3600$	✓ (b)
PC-1 ①	HI 7 Ch 12	$20\% \text{ of } 9000 \text{ g} = 1800 \text{ g}$	✓ (a)
PC-2 ②	HI 113 Ch 6	$2(10) + \frac{10^2}{100} = 21\%$	✓ (d)
PC-2 ③	EN 24 Ch 8	$\frac{162000}{(0.9)^2} = 200000$	✓ (a)
PC-2 ②	EN 233 Ch 6	$2(50) + \frac{50^2}{100} = 125\%$	✓ (c)
PL-1 ③	EN 92 Ch 7	$CP = \frac{56 \times 100}{22.5 + 12.5} = 160 \Rightarrow SP = 160 \times 1.25 = 200$	✓ (d)
PL-1 ②	HI 79 Ch 7	$2223 \times \frac{100}{117} = 1900$	✓ (d)
PL-1 ③	HI 58 Ch 7	$CP \times \frac{28 - 11.6}{100} = 18 + 50 \times 1.116 \Rightarrow CP = 450$	✓ (c)
PL-2 ②	HI 31 Ch 7	CP ₹8 / kg, SP ₹10 $\Rightarrow 25\%$	✓ (a)
PL-2 ④	EN 165 Ch 7	chahiye $SP = 5 \times 1.4 = 7 \text{ lakh} \Rightarrow (7 - 1) \div 3 = 2 \text{ lakh}$	✓ (c)
PL-2 ①	HI 28 Ch 7	$\frac{42 - 36}{36} \times 100 = 16\frac{2}{3}\% \text{ laabh}$	✓ (a)
DS-1 ①	HI 77 Ch 8	$160 \times 0.88 = 140.80$	✓ (a)
DS-1 ③	HI 47 Ch 8	$M = 118, D = 10 \Rightarrow 118 \times 0.9 - 100 = 6.2\%$	✓ (d)
DS-1 ①	EN 52 Ch 8	$160 \times 0.88 = 140.80$	✓ (a)
RP-1 ④	EN 160 Ch 9	$k = \frac{1920}{10 + 60 + 250} = 6 \Rightarrow 3k = 18$	✓ (c)
RP-1 ③	EN 51 Ch 28	$G : B = 3 : 2 \Rightarrow \frac{2}{5} \times 3 \div 5 = \frac{6}{25}$	✓ (d)
RP-1 ②	EN 69 Ch 9	Zinc barabar (21): 51 : 21 : 28	✓ (b)
RP-2 ②	HI 78 Ch 9	$x = \frac{3 \times 8}{12} = 2$	✓ (b)
RP-2 ②	HI 128 Ch 9	$4\text{th} = \frac{24 \times 45}{12} = 90$	✓ (a)
RP-2 ①	EN 103 Ch 9	$k = \frac{3(-5) - 4(5)}{5 \cdot 4 - 7 \cdot 3} = 35 \Rightarrow 12k = 420$	✓ (b)
PT-1 ①	EN 44 Ch 10	$10000 \times 12 : 15000 \times 8 = 1 : 1$	✓ (c)
PT-1 ①	EN 35 Ch 10	$8000 \times 36 : 10000 \times 36 : 6000 \times 30 \Rightarrow Z = 9660 \times \frac{180}{828} = 2100$	✓ (b)
PT-1 ①	HI 11 Ch 10	$3 : 2 : 5 \Rightarrow 1680, 1120, 2800$	✓ (c)
WT-1 ④	EN 112 Ch 11	6 din mein $6 \left(\frac{1}{30} + \frac{1}{20} \right) = \frac{1}{2} \Rightarrow B: \frac{1}{2} \times 20 = 10$	✓ (c)

WT-1 ③	EN 23 Ch 11	$k = 2, T = 17 \Rightarrow B = 17 \times 3 = 51$	✓ (b)
WT-1 ②	HI 21 Ch 11	$\frac{1}{B} = \frac{1}{97.2} + \frac{1}{162} - \frac{1}{81} \Rightarrow 243$	✓ (a)
WT-2 ①	HI 49 Ch 11	$7 \times 12 - 7 \times 5 = 49 \text{ man-days} \div 4 = 12\frac{1}{4}$	✓ (d)
WT-2 ①	EN 81 Ch 11	$36m + 54b = 45m + 30b \Rightarrow m : b = 8 : 3 \Rightarrow \frac{18(16+9)}{32+6} = 11\frac{16}{19}$	✓ (a)
WT-2 ①	EN 201 Ch 11	$\log \times \text{din const: } 105 \times 17 = 119 \times x \Rightarrow x = 15$	✓ (a)
MX-1 ②	EN 49 Ch 12	$1 \text{ hissa} = 140 \div 4 = 35 \Rightarrow \text{copper } 3 \times 35$	✓ (c)
MX-1 ②	EN 50 Ch 12	$1 \text{ hissa} = 90 \div 6 = 15 \Rightarrow 20 \times 15$	✓ (d)
MX-1 ②	EN 56 Ch 12	$\frac{35.26}{1.25+2.015} = 10.80$	✓ (c)
MX-2 ①	EN 7 Ch 12	acid fix: $2 \rightarrow 5 \text{ hisse} = 10 \text{ L} \Rightarrow 1 \text{ hissa } 10/3 \Rightarrow \text{acid } 10$	✓ (a)
MX-2 ①	HI 13 Ch 12	doodh 6 fix $\Rightarrow \text{paani } 6 \times \frac{3}{2} = 9 \Rightarrow \text{jodo } 9 - 4 = 5$	✓ (c)
MX-2 ①	EN 20 Ch 12	doodh $14 \div 25 = 56\%$	✓ (b)
PP-1 ④	EN 79 Ch 13	aadha 2 h $\Rightarrow \text{baaki aadha } 4 \text{ tanker se } \frac{1}{2} \text{ h} \Rightarrow 2 \text{ h } 30 \text{ min}$	✓ (d)
PP-1 ④	EN 93 Ch 13	P ne $24/32 = \frac{3}{4} \Rightarrow \text{Q ne } \frac{1}{4} \Rightarrow \frac{1}{4} \times 48 = 12 \text{ min}$	✓ (a)
PP-1 ①	HI 94 Ch 13	$\frac{1}{t} + \frac{2}{t} - \frac{1}{10} = \frac{1}{2.5} \Rightarrow t = 6$	✓ (a)
SI-1 ⑤	HI 31 Ch 14	$P_1 : P_2 = \frac{1}{30} : \frac{1}{36} = 6 : 5 \Rightarrow 3600, 3000$	✓ (a)
SI-1 ①	EN 152 Ch 14	$R = \frac{27 \times 100}{1080 \times \frac{4}{12}} = 10\%$	✓ (d)
SI-1 ⑥	HI 75 Ch 14	$SI = 4500 \Rightarrow 4500 \div 18000 = 25\%$	✓ (c)
CI-1 ①	EN 96 Ch 15	$8000(1.04^2 - 1) = 652.8$	✓ (a)
CI-1 ①	HI 19 Ch 15	$y = \frac{612}{0.0816} = 7500 \Rightarrow A = 8112$	✓ (b)
CI-1 ②	EN 177 Ch 15	$\left(\frac{7986}{6000}\right)^{1/3} = 1.1 \Rightarrow 10\%$	✓ (d)
CI-2 ①	EN 120 Ch 15	$R = 100 \div 8 = 12.5\% \Rightarrow 8000(1.125^2 - 1) = 2125$	✓ (c)
CI-2 ①	EN 110 Ch 15	half-yearly 5%, 2 baar: $P(0.05)^2 = 50 \Rightarrow P = 20000$	✓ (d)
CI-2 ①	EN 111 Ch 15	$SI = \frac{CI}{1 + \frac{200}{200}} = 4400$	✓ (d)
AG-1 ②	EN 169 Ch 16	$x = \frac{5(10-7)}{5.7-3.10} = 3 \Rightarrow \text{farak } 2x = 6$	✓ (b)
AG-1 ①	EN 12 Ch 16	options daalo: $G = 15 \Rightarrow M = 75 \Rightarrow 78 = 4(18) + 6$ ✓	✓ (b)
AG-1 ③	HI 45 Ch 16	abhi jod $28 + 2 \times 5 = 38 \Rightarrow \text{chhota } \frac{38-12}{2} = 13$	✓ (b)
AV-1 ⑤	EN 113 Ch 17	$1050 - 360 - 126 = 564 \Rightarrow \div 3 = 188$	✓ (c)
AV-1 ④	EN 32 Ch 17	$200 - 114 = 86 \Rightarrow \text{doosra max } 45 \Rightarrow 41$	✓ (a)
AV-1 ⑤	EN 78 Ch 17	$625 - 300 - 220 = 105$	✓ (c)
ST-1 ③	EN 148 Ch 18	$\frac{D/T}{(D/2)/(2T)} = 4 : 1$	✓ (d)
ST-1 ①	HI 39 Ch 18	$450 \div 150 = 3 \text{ m/s} = 10.8 \text{ km/h}$	✓ (b)

ST-1 ③	EN 2 Ch 18	car 66 $\Rightarrow$ bus 66 $\times \frac{7}{11} = 42$	✓ (a)
ST-2 ①	HI 63 Ch 18	$\frac{1}{\frac{1}{25} + \frac{1}{30} + \frac{5}{50}} = 33\frac{1}{3}$	✓ (b)
ST-2 ①	EN 130 Ch 18	$\frac{2 \cdot x \cdot 2x}{3x} = 24 \Rightarrow x = 18$	✓ (a)
ST-2 ②	HI 40 Ch 18	relative 1 km/h $\times 6$ min $= 100$ m $\Rightarrow 150 - 100 = 50$	✓ (b)
TR-1 ③	EN 141 Ch 19	$S = \frac{195-155}{18-16} = 20$ m/s $= 72$ km/h	✓ (b)
TR-1 ②	HI 10 Ch 19	10 m/s $\times 53 = 530 \Rightarrow 530 - 110 = 420$	✓ (b)
TR-1 ③	HI 29 Ch 19	$P = S(t - t_0): 200 = S \times 20 \Rightarrow S = 10 \Rightarrow L = 10 \times 20 = 200$	✓ (c)
BS-1 ③	HI 5 Ch 20	$\frac{T}{T_0} = \frac{B^2}{B^2 - S^2} = \frac{16}{15} \Rightarrow B = 4S \Rightarrow 3S = 15 \Rightarrow S = 5$	✓ (d)
BS-1 ③	EN 45 Ch 20	$\frac{d}{x-y} - \frac{d}{x+y} = t \Rightarrow d = \frac{(x^2 - y^2)t}{2y}$ (x = 5, y = 1, t = 1 daal ke option milao)	✓ (b)
BS-1 ②	EN 16 Ch 20	n = 2 $\Rightarrow \frac{B}{S} = \frac{3}{1} \Rightarrow S = 9 \div 3 = 3$	✓ (b)
MN-1 ③	HI 110 Ch 21	$2800 \times 15 \text{ cm}^2 = 4.2 \text{ m}^2 \times ₹25 = ₹105$	✓ (d)
MN-1 ②	EN 198 Ch 21	max sides 35.5, 25.5 $\Rightarrow 2(61) = 122$	✓ (b)
MN-2 ⑤	HI 202 Ch 21	volume same: $\frac{4}{3}\pi(6^3 - 4^3) = \pi 8^2 h \Rightarrow h = 3.17$	✓ (c)
MN-2 ②	HI 182 Ch 21	$2\pi r(r + h) = 22(3.5 + h) = 269.5 \Rightarrow h = 8.75 \Rightarrow 8.75 : 7 = 5 : 4$	✓ (d)
MN-2 ②	EN 287 Ch 21	r = 10.5 $\Rightarrow 2\pi r(r + h) = 66 \times 31.5 = 2079$	✓ (d)
MN-3 ①	EN 334 Ch 21	$r_X l_X = 5r_Y(5l_Y) \Rightarrow r_X = 25r_Y \Rightarrow \text{area } 625 : 1$	✓ (c)
MN-3 ①	HI 183 Ch 21	$V \propto r^2 \Rightarrow 4 : 9$	✓ (c)
AL-1 ②	HI 37 Ch 22	x = 1, y = 2, z = 3 daalo: $(0.5 + 6 - 1)^2 = 30.25 = \text{option (b)}$	✓ (b)
AL-1 ③	HI 13 Ch 22	x = 1, y = 2, z = 3 daalo: $9 - 24 = -15 = \text{option (d)}$	✓ (d)
AL-1 ③	HI 47 Ch 22	x = 2 daalo: $3 \times 1 \times 5 = 15 = 2^4 - 1$	✓ (d)
AL-2 ①	EN 54 Ch 22	unique $\Leftrightarrow \frac{2}{2} \neq \frac{3}{p} \Rightarrow p \neq 3$	✓ (c)
AL-2 ②	HI 241 Ch 22	$(a - b)^2 = S^2 - 4P = 25 + 24 = 49$	✓ (a)
AL-2 ②	HI 248 Ch 22	simplify ke baad power: (a) 3, (b) $\sqrt{\text{hata ke 2}}$, (c) 1, (d) $1 \Rightarrow$ (b)	✓ (b)
AL-3 ①	EN 9 Ch 22	$5 + 8 \times 4 = 37$	✓ (b)
AL-3 ①	EN 12 Ch 22	$2 + 9 \times 5 = 47$	✓ (c)
AL-3 ①	HI 6 Ch 22	$3 \times 2^4 = 48$	✓ (a)
AL-4 ①	EN 272 Ch 22	English: $30 + 20 + x = 60 \Rightarrow x = 10$ (Hindi: $25 + 20 + 10 = 55$ ✓)	✓ (c)
AL-4 ①	EN 286 Ch 22	sirf A = $33 - 10 - 8 + x = 21 \Rightarrow x = 6$	✓ (c)
AL-4 ①	EN 285 Ch 22	$100 - 65 - 30 = 5\% \text{ of } 60 = 3$	✓ (b)
TG-1 ①	HI 21 Ch 23	$\theta = 30^\circ$ daalo: $\frac{1}{64} + \frac{27}{64} = \frac{7}{16} = 1 - 3 \cdot \frac{1}{4} \cdot \frac{3}{4} \Rightarrow$ (a)	✓ (a)
TG-1 ①	EN 108 Ch 23	$\theta = 30^\circ$ daalo $\Rightarrow 0$	✓ (c)

TG-1 ②	EN 60 Ch 23	$\sec = \frac{k+\frac{1}{k}}{2} = \frac{(2-\sqrt{3})+(2+\sqrt{3})}{2} = 2 \Rightarrow \theta = 60^\circ \Rightarrow \frac{3}{2} + \frac{1}{2} = 2$ (book: $\sec = 2$ se $\theta = 60^\circ$ liya; strictly $\tan \theta$ negative hai) $\sec \theta + \tan \theta = 2 - \sqrt{3}$ se $\tan \theta$ negative aata hai (strictly answer -1 , option mein nahi); book ne $\sec \theta = 2 \Rightarrow \theta = 60^\circ$ maan ke (b) 2 diya	✓ (b)
TG-2 ②	EN 135 Ch 23	options daalo: $\cos 60^\circ = \frac{1}{2} = \sin 30^\circ \Rightarrow 30^\circ$	✓ (b)
TG-2 ②	EN 16 Ch 23	$A = 60^\circ, B = 30^\circ$ daalo: $-3.73 + 3.73 = 0$	✓ (b)
TG-2 ①	EN 24 Ch 23	$(48^\circ + \theta) + (42^\circ - \theta) = 90^\circ \Rightarrow \sin = \cos \Rightarrow 0$	✓ (d)
TG-3 ①	EN 189 Ch 23	$d = 960 \times 1/60 = 16 \text{ km} \Rightarrow h = \frac{\sqrt{3}}{2} \times 16 = 13.86$	✓ (b)
TG-3 ①	EN 183 Ch 23	same doori d: $x = d \tan 60^\circ, y = d \tan 30^\circ \Rightarrow 3 : 1$	✓ (a)
TG-3 ①	EN 185 Ch 23	$45^\circ \Rightarrow h = d = 28.5 + \text{observer } 1.5 = 30$	✓ (a)
CG-1 ②	EN 6 Ch 24	side 2, midpoint (2, 5), height $\sqrt{3} \Rightarrow (2 \pm \sqrt{3}, 5) \Rightarrow (c)$	✓ (c)
CG-1 ②	HI 15 Ch 24	$AB = 3\sqrt{5}, BC = 5\sqrt{5}, CA = 2\sqrt{5} \Rightarrow \frac{10+6+15}{30\sqrt{5}} = \frac{31\sqrt{5}}{150}$	✓ (a)
CG-1 ③	HI 8 Ch 24	$\frac{1}{2} 0(3-1) + 2(1-2) + 3(2-3) = 2.5$	✓ (c)
CG-2 ①	EN 2 Ch 24	intercept = $6 \div \text{coefficient: } 2, -3/2, -3$	✓ (d)
CG-2 ①	HI 13 Ch 24	$D = 0 - 4 \cdot 9 \cdot 16 < 0 \Rightarrow x\text{-axis ko na kaate na chhue}$	✓ (d)
CG-2 ①	EN 37 Ch 24	$y = -4x + 2 \Rightarrow 4x + y = 2 \Rightarrow 2x + \frac{y}{2} = 1$	✓ (b)
GE-1 ④	EN 101 Ch 25	diagonals bisect: $y + 2 = 3y \Rightarrow y = 1; x + 3 = 2x + 1 \Rightarrow x = 2$	✓ (a)
GE-1 ③	HI 181 Ch 25	$180^\circ - \frac{360^\circ}{8} = 135^\circ$	✓ (d)
GE-1 ①	EN 249 Ch 25	alternate kon barabar $\Rightarrow 35^\circ$	✓ (a)
GE-2 ②	HI 98 Ch 25	$p = \sqrt{17^2 - 8^2} = 15$	✓ (a)
GE-2 ①	HI 57 Ch 25	chinh milao: $Q \leftrightarrow Q, R \leftrightarrow P, S \leftrightarrow T \Rightarrow \Delta QPT$	✓ (b)
GE-2 ①	HI 74 Ch 25	similar $\Rightarrow$ kon barabar $\Rightarrow 1 : 1$	✓ (b)
GE-3 ①	EN 122 Ch 25	$EA \cdot EB = EC \cdot ED: 12 \times 3 = 4 \times EC \Rightarrow EC = 9 \Rightarrow CD = 5$	✓ (c)
GE-3 ②	HI 167 Ch 25	tangent-chord kon = alternate segment ka kon $\Rightarrow 82^\circ$	✓ (d)
GE-3 ②	HI 170 Ch 25	chord = r $\Rightarrow$ centre par $60^\circ \Rightarrow$ tangent se kon $30^\circ = \pi/6$	✓ (a)
SP-1 ①	HI 7 Ch 26	$\sum fx = \bar{x} \sum f: 390 + 15X = 18(20 + X) \Rightarrow X = 10$	✓ (d)
SP-1 ④	HI 94 Ch 26	Mean = $\frac{3(35)-27}{2} = 39$	✓ (b)
SP-1 ④	EN 201 Ch 26	sort: 2,2,3,3,3,5,5,7,7,9 $\Rightarrow$ median 4, mode 3 $\Rightarrow 1$	✓ (b)
SP-2 ①	EN 209 Ch 26	$CV = 6 \div 24 \times 100 = 25\%$	✓ (b)
SP-2 ①	EN 228 Ch 26	$CV = 5 \div 11 \times 100 = 45.45\%$	✓ (a)
SP-2 ①	EN 235 Ch 26	$\sigma = \sqrt{324} = 18$	✓ (b)
SP-3 ①	EN 255 Ch 26	90 do-ank numbers, 21 prime $\Rightarrow 69/90 = 23/30$	✓ (b)
SP-3 ①	EN 249 Ch 26	circle mein do saath: $\frac{2}{n-1} = \frac{2}{8} = \frac{1}{4}$	✓ (c)

SP-3 ①	EN 264 Ch 26	$(3 + 5) \div 12 = 2/3$	✓ (a)
DI-1 ②	EN 57 Ch 27	2017: 731, 2021: 1071 $\Rightarrow 731 \div 1071 \times 100 = 68\%$	✓ (c)
DI-1 ③	EN 199 Ch 27	D1 men 430 $\rightarrow 240 \Rightarrow 190$ baante $\Rightarrow (250 + 350 + 190) : (440 + 180) = 79 : 62$ (graph par 450 chhapa, book ke hal mein 430) graph par District 1 men 450 chhapa hai, book ke hal mein 430; 430 se (d) 79 : 62 aata hai (450 se 81 : 62, option mein nahi)	✓ (d)
DI-1 ①	EN 22 Ch 28	$(10 + 20)\%$ of 360 = 108	✓ (a)
MS-1 ④	HI 27 Ch 28	3 9 1 9 0 9 1 6 8 $\Rightarrow$ aakhri 4: 9 + 1 + 6 + 8 = 24	✓ (c)
MS-1 ④	HI 40 Ch 28	2 4 6 2 4 2 0 $\Rightarrow 6 - 0 = 6$	✓ (b)
MS-1 ④	HI 35 Ch 28	5 5 7 3 1 9 $\Rightarrow 5 + 1 = 6$	✓ (a)
MN-1 ②	EN 190 Ch 21 (badle mein)	doosri side $\sqrt{7.5^2 - 4.5^2} = 6 \Rightarrow$ area $4.5 \times 6 = 27$	✓ (a)
MN-3 ①	EN 243 Ch 21 (badle mein)	lambai $\times 3 \Rightarrow$ volume $\times 3^3 = 27$ guna	✓ (d)

Chhode gaye (2) — inki jagah usi formula ka naya random sawaal liya

- **MN-1** — HI 259 (Ch 21): book mein options chhape hi nahi (khaali image box) — answer key se milaana sambhav nahi.
- **MN-3** — HI 159 (Ch 21): book mein options (a), (b), (d) khaali image box hain — answer key se milaana sambhav nahi.

Appendix B – Book Errata & Recovered Questions

B.1 Book ki galtiyan (answer key / question text)

Ch	Chapter	Q	Kya galat hai – sahi kya hai
10	Partnership	EN 1	English book mein A ki capital ₹50,000 chhapi hai (galat); Hindi book (HI2) mein ₹4,50,000 hai $\Rightarrow 450000 \times 12 : 9B = 3 : 2 \Rightarrow B = ₹4,00,000 = \text{option (c)}$.
12	Alligation / Mixture	EN 42	1.5% aur 2.5% ko milakar 3.5% ban hi nahi sakta (mean hamesha dono ke beech). Question galat; book ka 1 : 2 cross rule ko ulta lagane se aaya.
12	Alligation / Mixture	EN 45	Book ne ₹160 aur ₹400 (SP) ko hi rates maan kar mean ₹240 se alligation lagaya $\Rightarrow 2 : 1$. Loss/profit se asli CP (₹200, ₹333.3) lein to 7 : 3 aata hai — option mein nahi.
13	Pipe & Cistern	EN 119	Book ne 2 ghante ka net $\frac{1}{4}$ maan kar $4 \times 2 = 8$ h likha. Par 5th ghante mein A akela $\frac{1}{2}$ se poora bhar deta hai $\Rightarrow$ sahi $4 + \frac{3}{4} = 4.75$ h = option (a).
13	Pipe & Cistern	HI 79	EN119 jaisa: sahi uttar 4.75 घंटे (a); book ka (d) 8 घंटे aakhri ghante ka trap bhool gaya.
13	Pipe & Cistern	EN 59	Wording ulta chhapi hai. Sahi arth: doosra pehle se 5 h KAM, teesre se 4 h ZYADA leta $\Rightarrow 15, 10, 6 (1/15 + 1/10 = 1/6) \Rightarrow 15 \checkmark$.
18	Speed, Time & Distance	EN 59	Question mein “reduced his speed to 2 km/h” chhapa hai, par book ka answer (12:30 pm) “by 2 km/h” (yaani 4 km/h) se aata hai: $18/6 + 12/4 = 6$ h. “to 2 km/h” lene par 3:30 pm aata jo options mein nahi.
19	Train	EN 39	Answer key mein (c) 700 m chhapa hai, par book ki apni solution bhi $L = 800$ m nikalti hai (55 m/s ki train). Sahi option (a) 800 m — Hindi book (HI38) mein bhi (a).
23	Trigonometry	EN 60	$\sec \theta + \tan \theta = 2 - \sqrt{3}$ se $\tan \theta$ negative aata hai (strictly answer -1 , option mein nahi); book ne $\sec \theta = 2 \Rightarrow \theta = 60^\circ$ maan ke (b) 2 diya (Appendix A dekho)
27	Data Interpretation	EN 199	graph par District 1 men 450 chhapa hai, book ke hal mein 430; 430 se (d) 79 : 62 aata hai (450 se 81 : 62, option mein nahi) (Appendix A dekho)

B.2 Wapas nikaale gaye / numbering-gap sawaal

Ch	Chapter	Q	Kya hua
6	Percentage	EN 189	Page se wapas nikala: question number printed as '18.'; stem garbled in print as '110 of 100% of 10%', book solution uses 110 of 110% of 10%
7	Profit & Loss	HI 81	Page se wapas nikala: book prints this as a second '80.'
9	Ratio & Proportion	EN 136	Page se wapas nikala: book prints this as '137.'
9	Ratio & Proportion	EN 137	Page se wapas nikala: extraction mein chhoot gaya tha
15	Compound Interest	EN 78	Book mein ye number chhapa hi nahi (numbering gap) — koi sawaal gayab nahi.
16	Problems on Age	HI 54	Book mein ye number chhapa hi nahi (numbering gap) — koi sawaal gayab nahi.
18	Speed, Time & Distance	EN 35	Page se wapas nikala: book prints the number as '35' without a dot
22	Algebra	HI 262	Page se wapas nikala: page text of Q262 and Q263 was interleaved; repaired
22	Algebra	HI 263	Page se wapas nikala: page text of Q262 and Q263 was interleaved; repaired
25	Geometry	EN 161	Page se wapas nikala: book prints the number as '161' without a dot
27	Data Interpretation	HI 97	Page se wapas nikala: book prints this number as a second '96.'
27	Data Interpretation	HI 98	Page se wapas nikala: extraction mein chhoot gaya tha
27	Data Interpretation	HI 99	Page se wapas nikala: extraction mein chhoot gaya tha